dsPIC33F/PIC24H

dsPIC33F/PIC24H Flash Programming Specification

1.0 DEVICE OVERVIEW

This document defines the programming specification for the dsPIC33F 16-bit Digital Signal Controller (DSC) and PIC24H 16-bit Microcontroller (MCU) families. This programming specification is required only for those developing programming support for the dsPIC33F/PIC24H family. Customers only using one of these devices should use development tools that already provide support for device programming.

This document includes programming specifications for the following devices:

- dsPIC33FJ64GP206
- dsPIC33FJ64GP306
- dsPIC33FJ64GP310
- dsPIC33FJ64GP706
- dsPIC33FJ64GP708
- dsPIC33FJ64GP710
- dsPIC33FJ128GP206
- dsPIC33FJ128GP306
- dsPIC33FJ128GP310
- dsPIC33FJ128GP706
- dsPIC33FJ128GP708
- dsPIC33FJ128GP710
- dsPIC33FJ256GP506
- dsPIC33FJ256GP510
- dsPIC33FJ256GP710
- dsPIC33FJ64MC506
- dsPIC33FJ64MC508
- dsPIC33FJ64MC510
- dsPIC33FJ64MC706
- dsPIC33FJ64MC710
- dsPIC33FJ128MC506
- dsPIC33FJ128MC510
- dsPIC33FJ128MC706
- dsPIC33FJ128MC708
- dsPIC33FJ128MC710
- dsPIC33FJ256MC510
- dsPIC33FJ256MC710
- PIC24HJ64GP206
- PIC24HJ64GP210
- PIC24HJ64GP506
- PIC24HJ64GP510
- PIC24HJ128GP206
- PIC24HJ128GP210
- PIC24HJ128GP306
- PIC24HJ128GP310
- PIC24HJ128GP506
- PIC24HJ128GP510
- PIC24HJ256GP206
- PIC24HJ256GP210
- PIC24HJ256GP610
- dsPIC33FJ12GP201
- dsPIC33FJ12GP202
- dsPIC33FJ12MC201
- dsPIC33FJ12MC202
- PIC24HJ12GP201
- PIC24HJ12GP202

2.0 PROGRAMMING OVERVIEW OF THE dsPIC33F/PIC24H

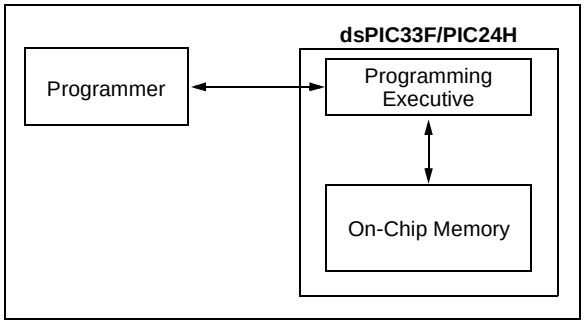
There are two methods of programming the dsPIC33F/PIC24H family of devices discussed in this programming specification. They are:

- In-Circuit Serial Programming™ (ICSP™) programming capability
- Enhanced In-Circuit Serial Programming

The ICSP programming method is the most direct method to program the device; however, it is also the slower of the two methods. It provides native, low-level programming capability to erase, program and verify the chip.

The Enhanced ICSP protocol uses a faster method that takes advantage of the programming executive, as illustrated in Figure 2-1. The programming executive provides all the necessary functionality to erase, program and verify the chip through a small command set. The command set allows the programmer to program the dsPIC33F/PIC24H Programming Specification devices without having to deal with the low-level programming protocols of the chip.

FIGURE 2-1: PROGRAMMING SYSTEM OVERVIEW FOR ENHANCED ICSP™



This specification is divided into major sections that describe the programming methods independently. **Section 3.0 "Device Programming – Enhanced ICSP"** describes the Enhanced ICSP method. **Section 5.0 "Device Programming – ICSP"** describes the ICSP method.

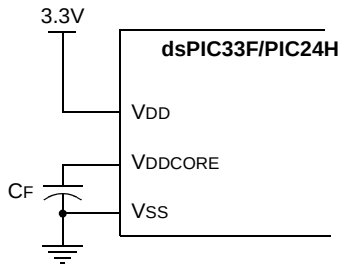
2.1 Power Requirements

All devices in the dsPIC33F/PIC24H family are dual voltage supply designs: one supply for the core and another for the peripherals and I/O pins. A regulator is provided on-chip to alleviate the need for two external voltage supplies.

All of the dsPIC33F/PIC24H devices power their core digital logic at a nominal 2.5V. To simplify system design, all devices in the dsPIC33F/PIC24H Programming Specification family incorporate an on-chip regulator that allows the device to run its core logic from VDD.

The regulator provides power to the core from the other VDD pins. A low-ESR capacitor (such as tantalum) must be connected to the VDDCORE pin (Figure 2-2). This helps to maintain the stability of the regulator. The specifications for core voltage and capacitance are listed in **Section TABLE 8-1: "AC/DC Characteristics and Timing Requirements"**.

FIGURE 2-2: CONNECTIONS FOR THE ON-CHIP REGULATOR



Note 1: These are typical operating voltages. Refer to **Section TABLE 8-1: "AC/DC Characteristics and Timing Requirements"** for the full operating ranges of VDD and

2.2 Program Memory Write/Erase Requirements

The program Flash memory on the dsPIC33F/PIC24H has a specific write/erase requirement that must be adhered to for proper device operation. The rule is that any given word in memory must not be written without first erasing the page in which it is located. Thus, the easiest way to conform to this rule is to write all the data in a programming block within one write cycle. The programming methods specified in this document comply with this requirement.

Note: A program memory word can be programmed twice before an erase, but only if (a) the same data is used in both program operations or (b) bits containing '1' are set to '0' but no '0' is set to '1'.

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2.3 Pin Diagrams

The pin diagrams for the dsPIC33F/PIC24H device family are shown in the following figures. The pins that are required for programming are listed in Table 2-1. The MCLR, PGC1, PGD1, PGC2, PGD2, PGC3 and PGD3 pins are shown in bold letters in the figures. Refer to the appropriate device data sheet for complete pin descriptions.

TABLE 2-1: PIN DESCRIPTIONS (PINS USED DURING PROGRAMMING)

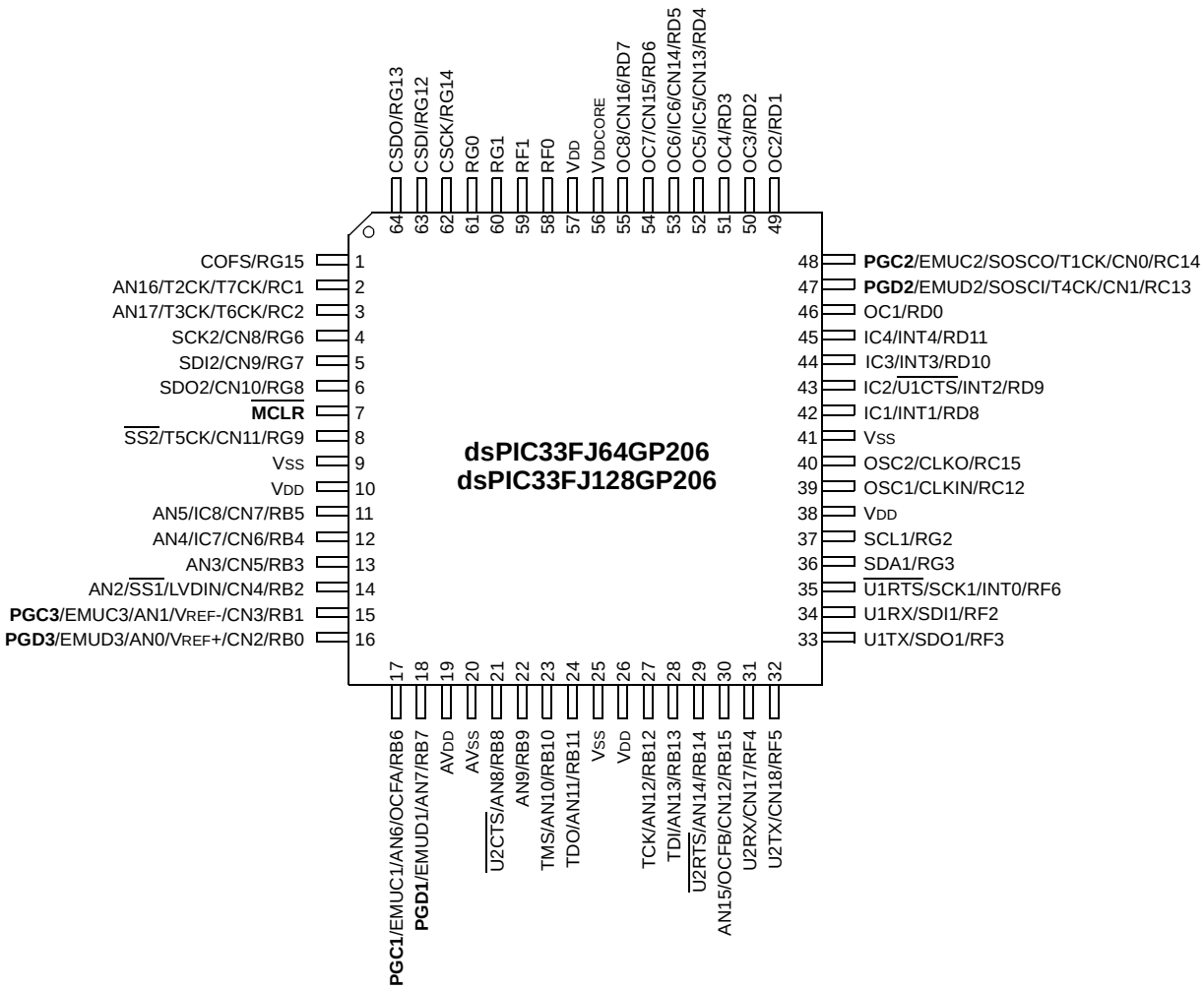
Pin Name	During Programming		
	Pin Name	Pin Type	Pin Description
MCLR	MCLR	P	Programming Enable
VDD and AVDD(1)	VDD	P	Power Supply
VSS and AVSS(1)	VSS	P	Ground
VDDCORE	VDDCORE	P	Regulated Power Supply for Core
PGC1	PGC1	I	Primary Programming Pin Pair: Serial Clock
PGD1	PGD1	I/O	Primary Programming Pin Pair: Serial Data
PGC2	PGC2	I	Secondary Programming Pin Pair: Serial Clock
PGD2	PGD2	I/O	Secondary Programming Pin Pair: Serial Data
PGC3	PGC3	I	Tertiary Programming Pin Pair: Serial Clock
PGD3	PGD3	I/O	Tertiary Programming Pin Pair: Serial Data

Legend: I = Input, O = Output, P = Power

Note 1: All power supply and ground pins must be connected, including analog supplies (AVDD) and ground (AVSS).

Pin Diagrams

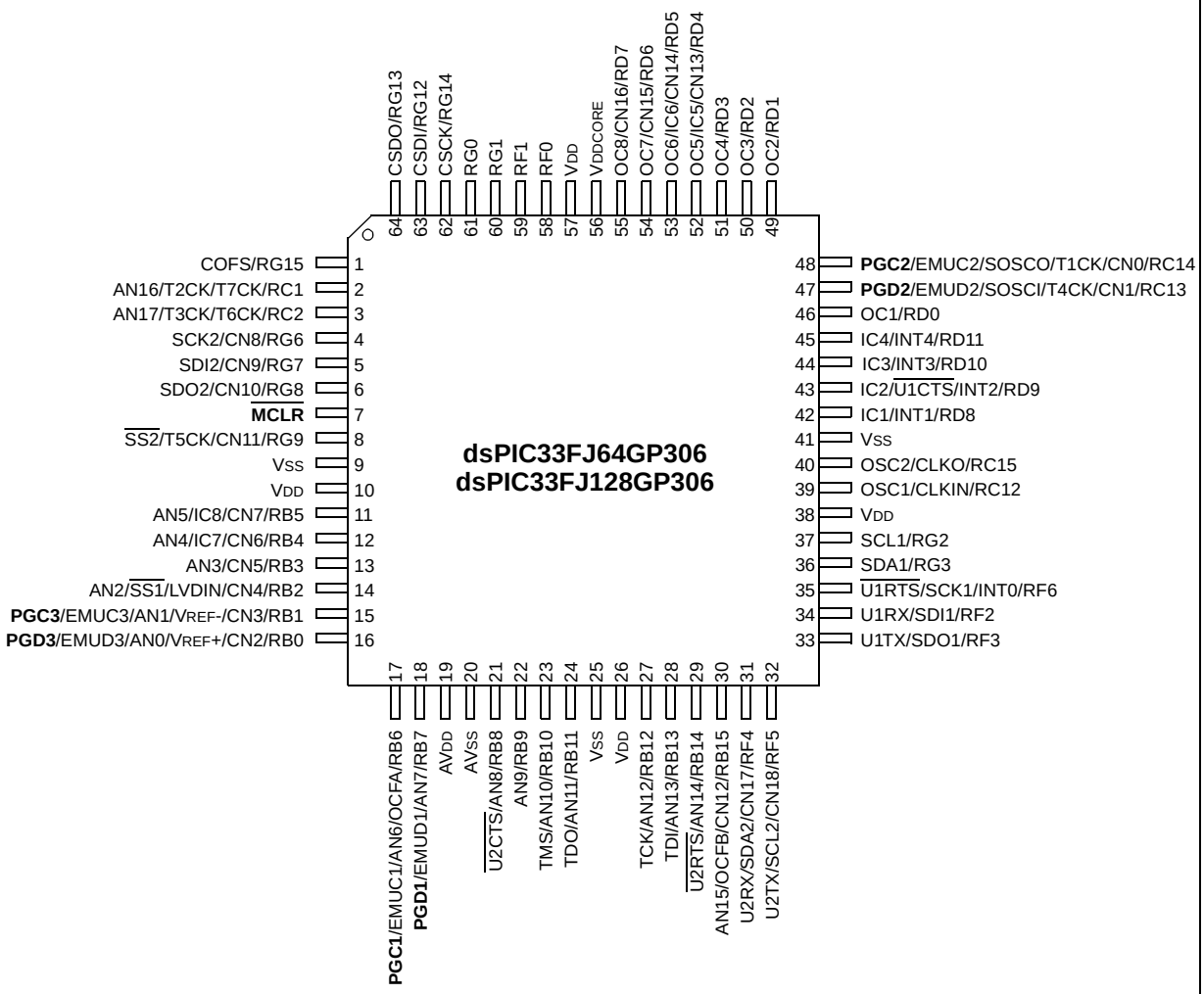
64-Pin TQFP



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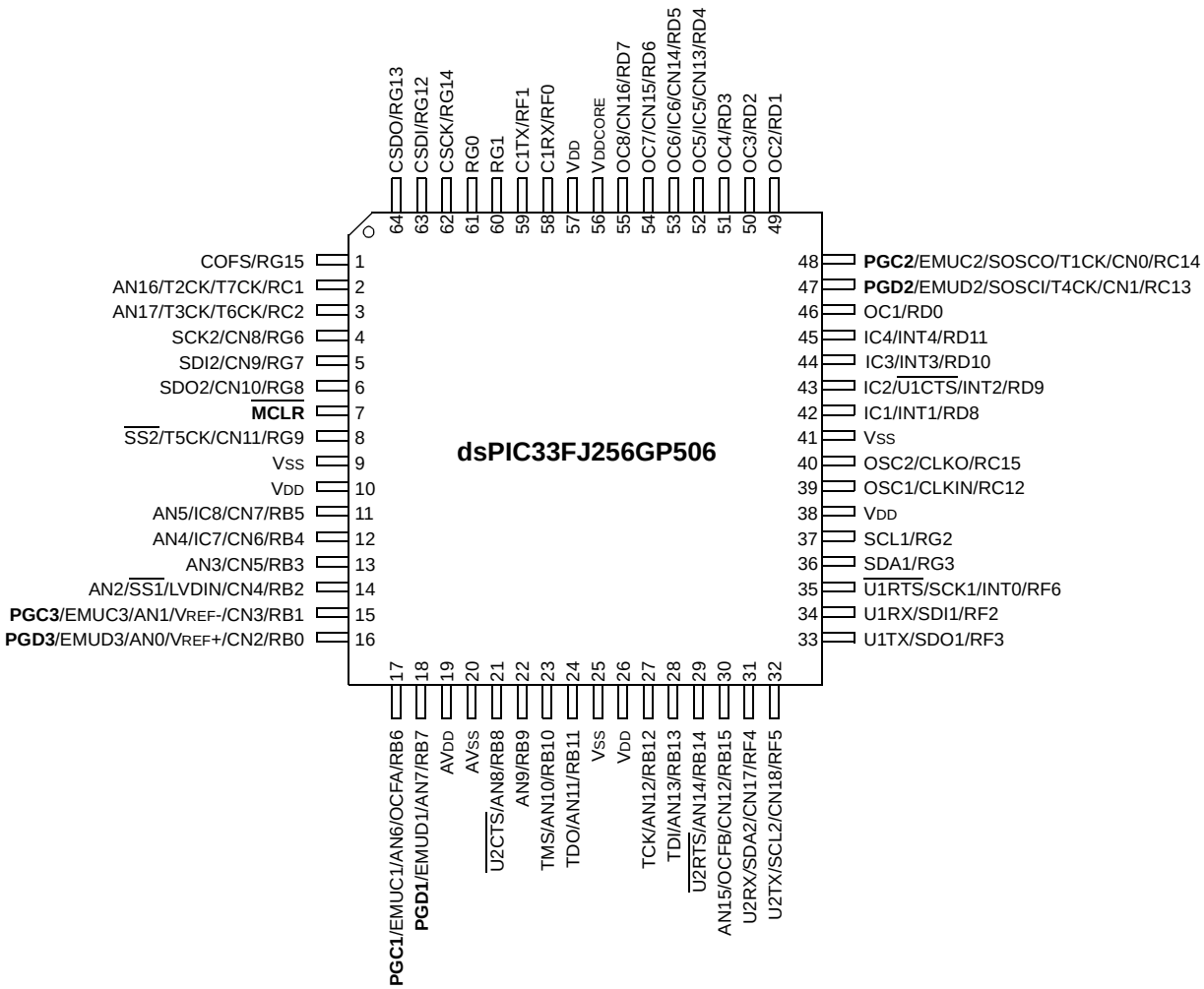
Pin Diagrams (Continued)

64-Pin TQFP



Pin Diagrams (Continued)

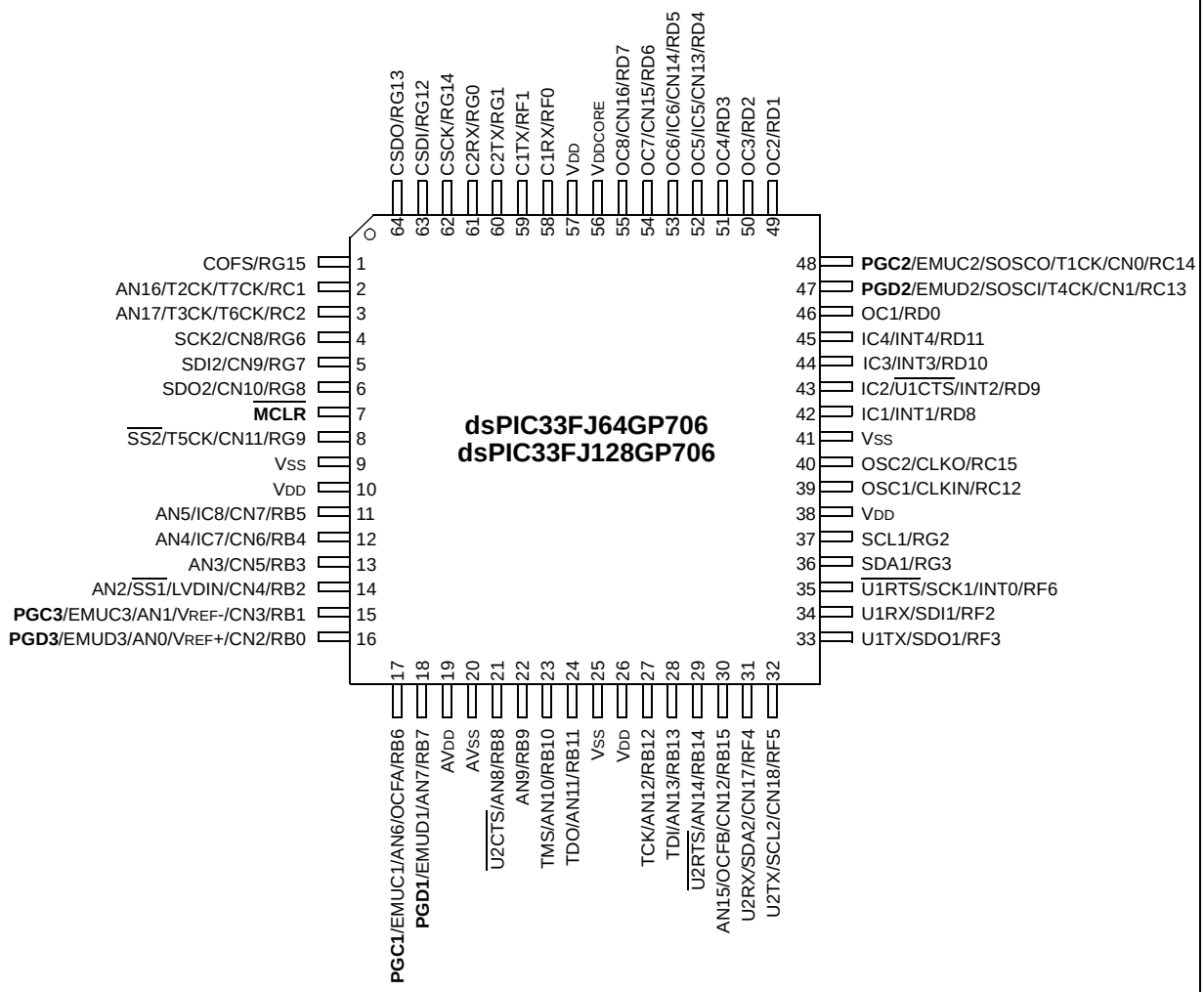
64-Pin TQFP



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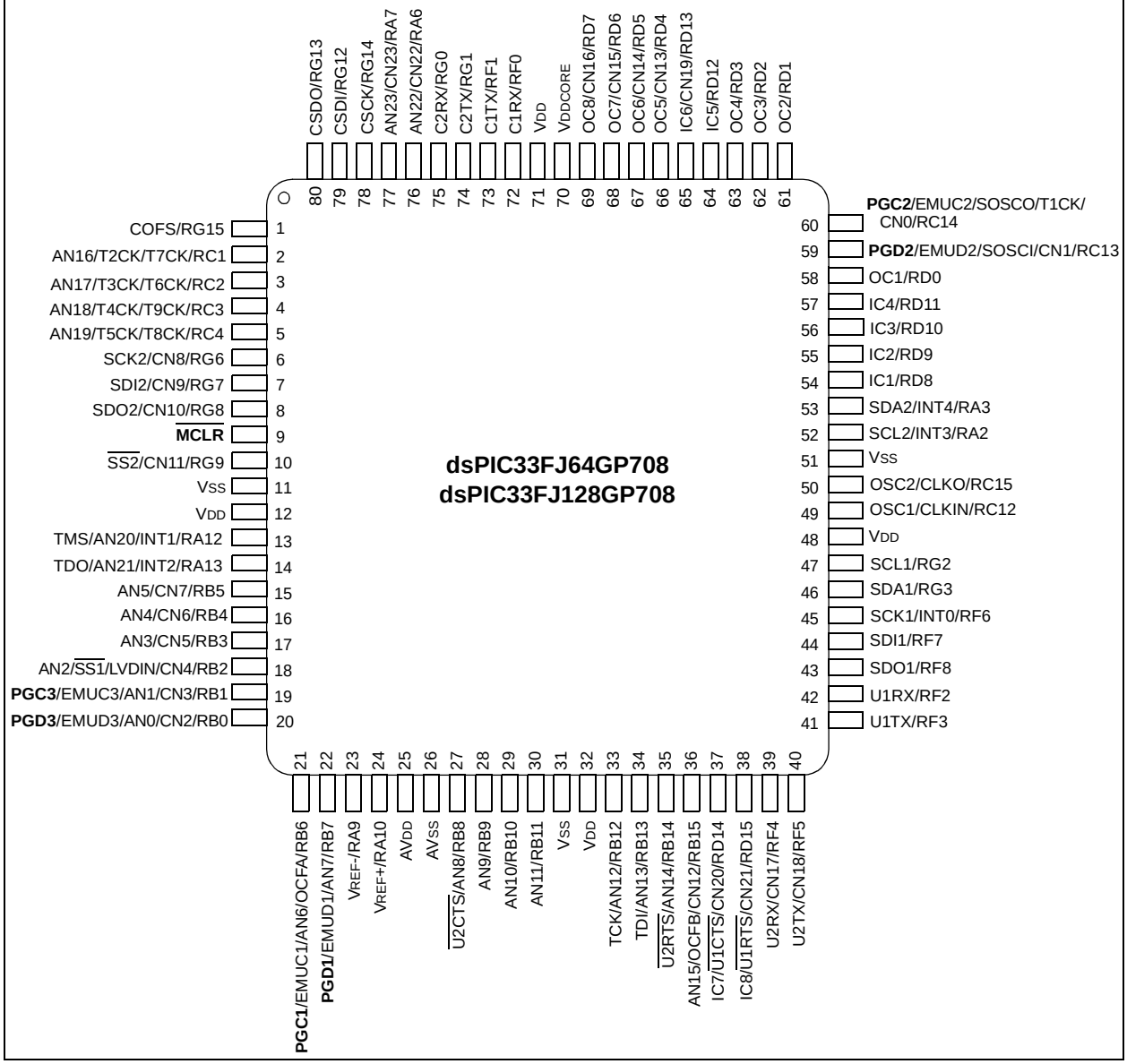
Pin Diagrams (Continued)

64-Pin TQFP



Pin Diagrams (Continued)

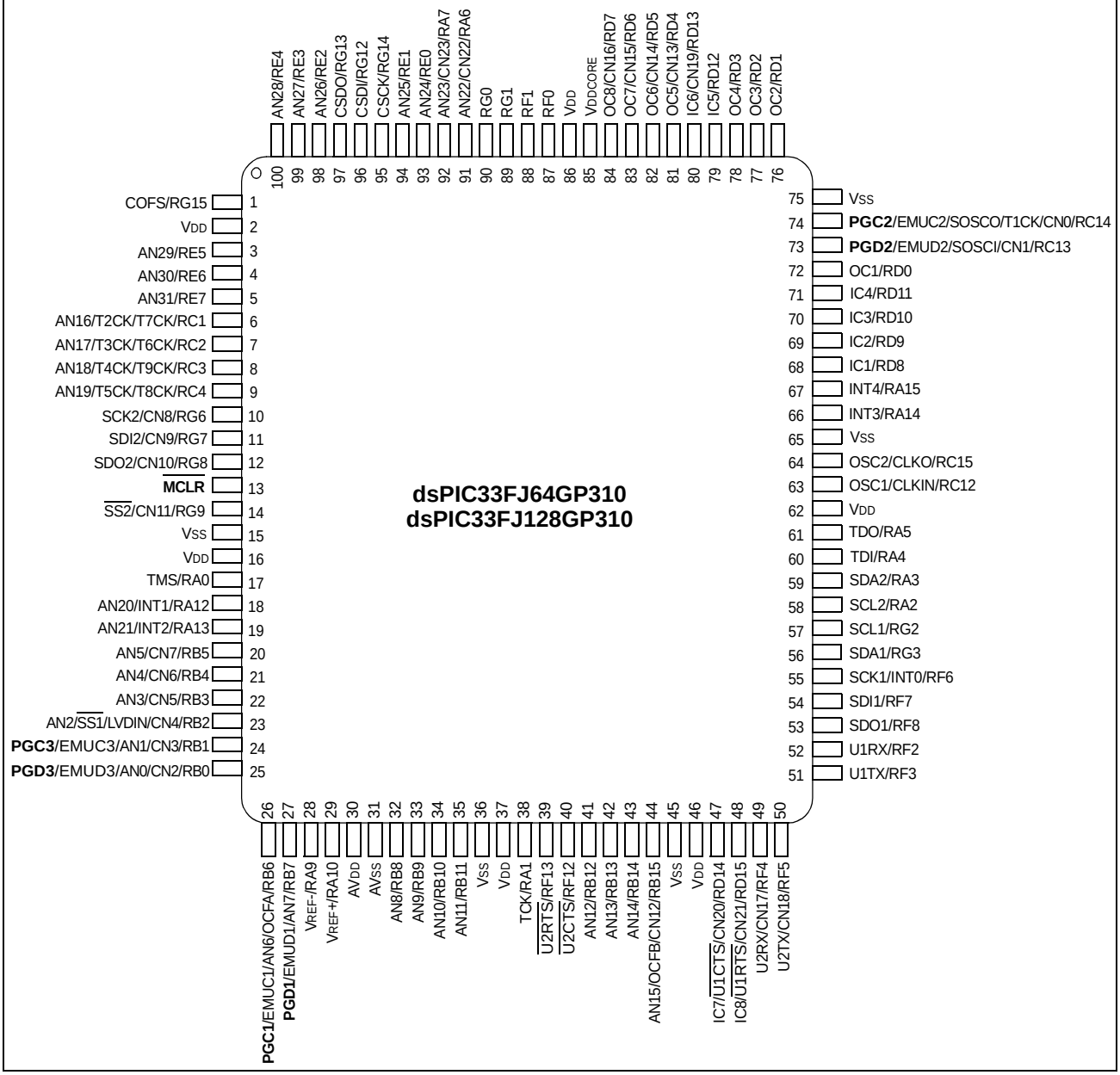
80-Pin TQFP



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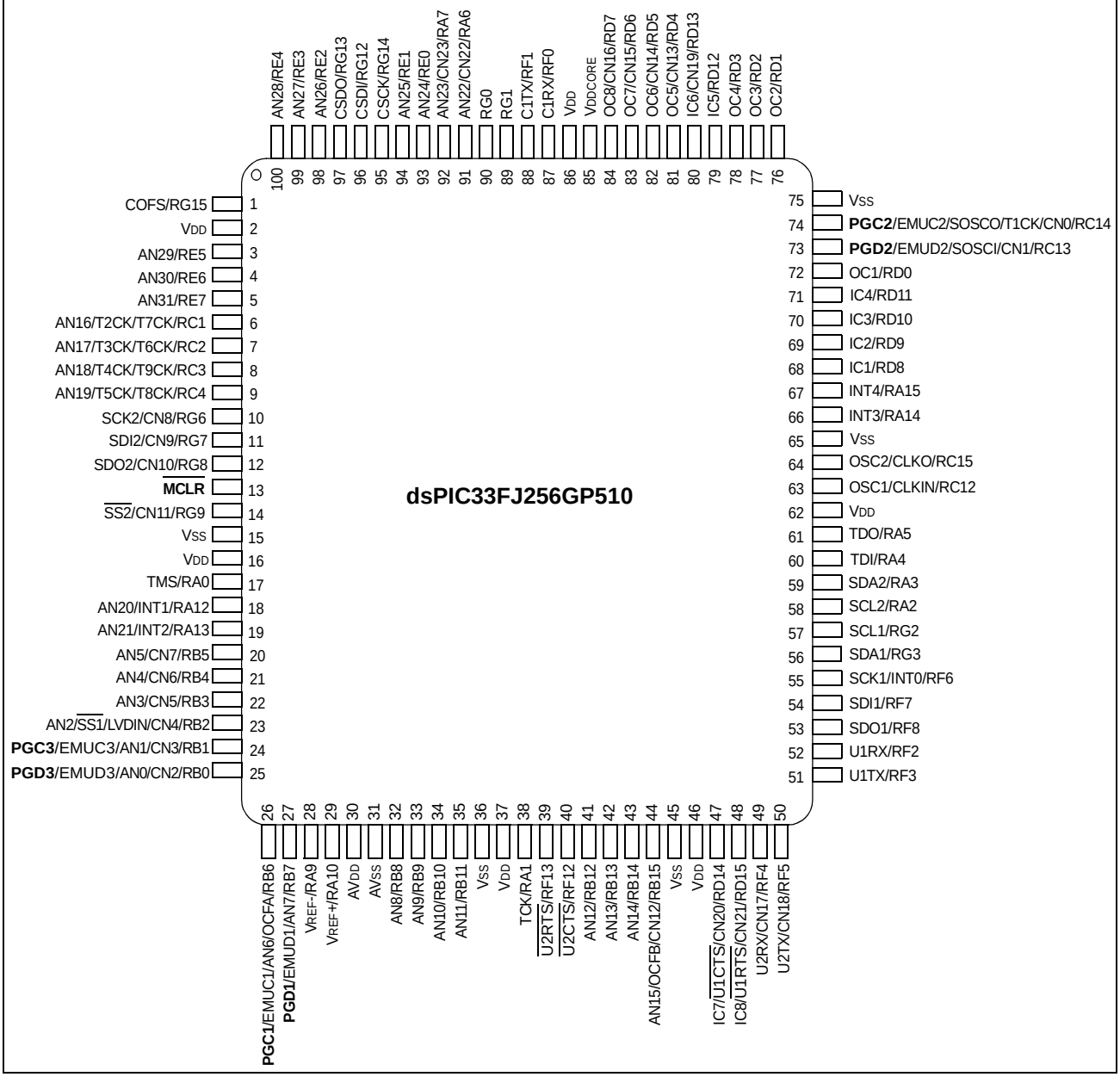
Pin Diagrams (Continued)

100-Pin TQFP



Pin Diagrams (Continued)

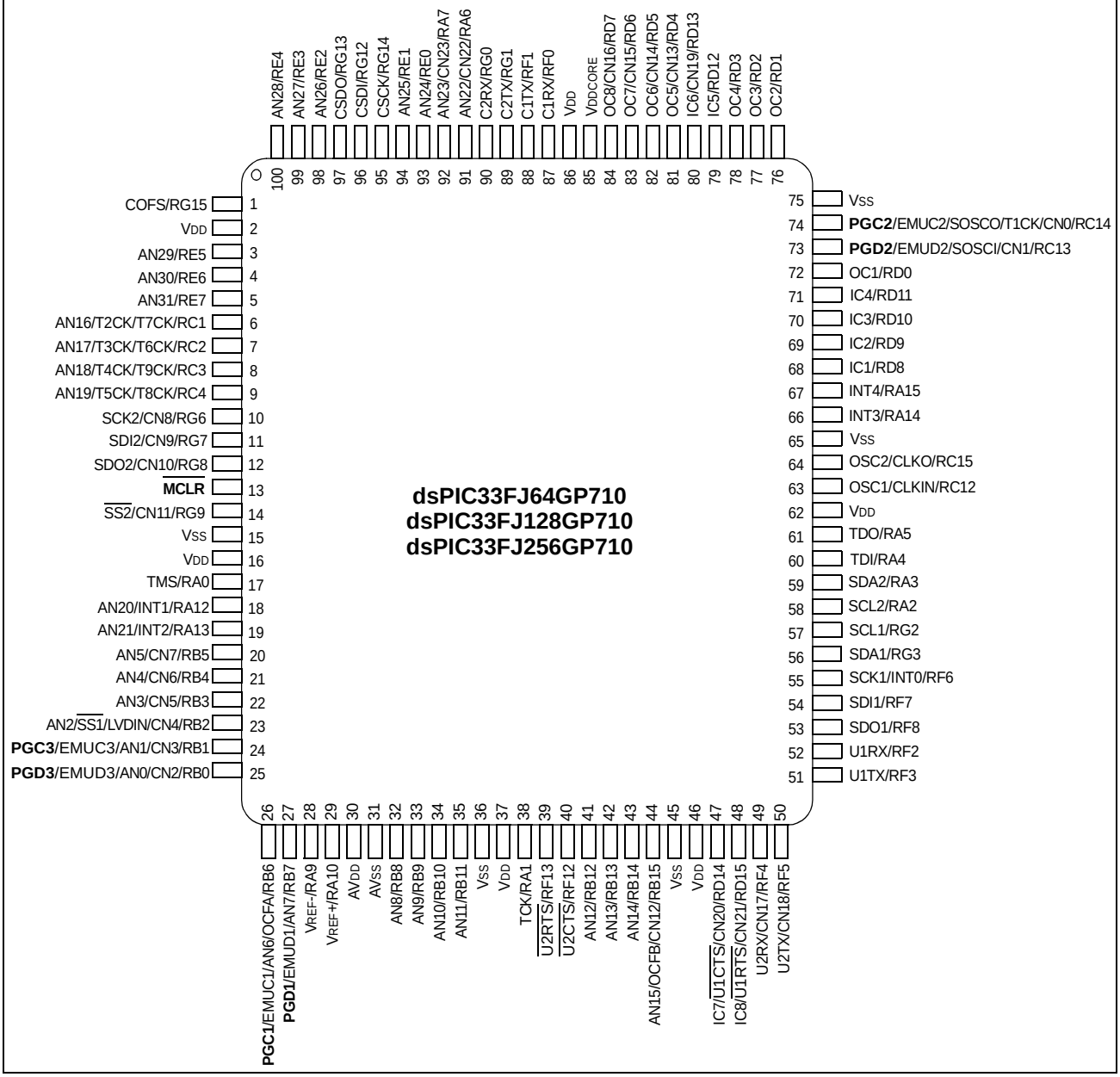
100-Pin TQFP



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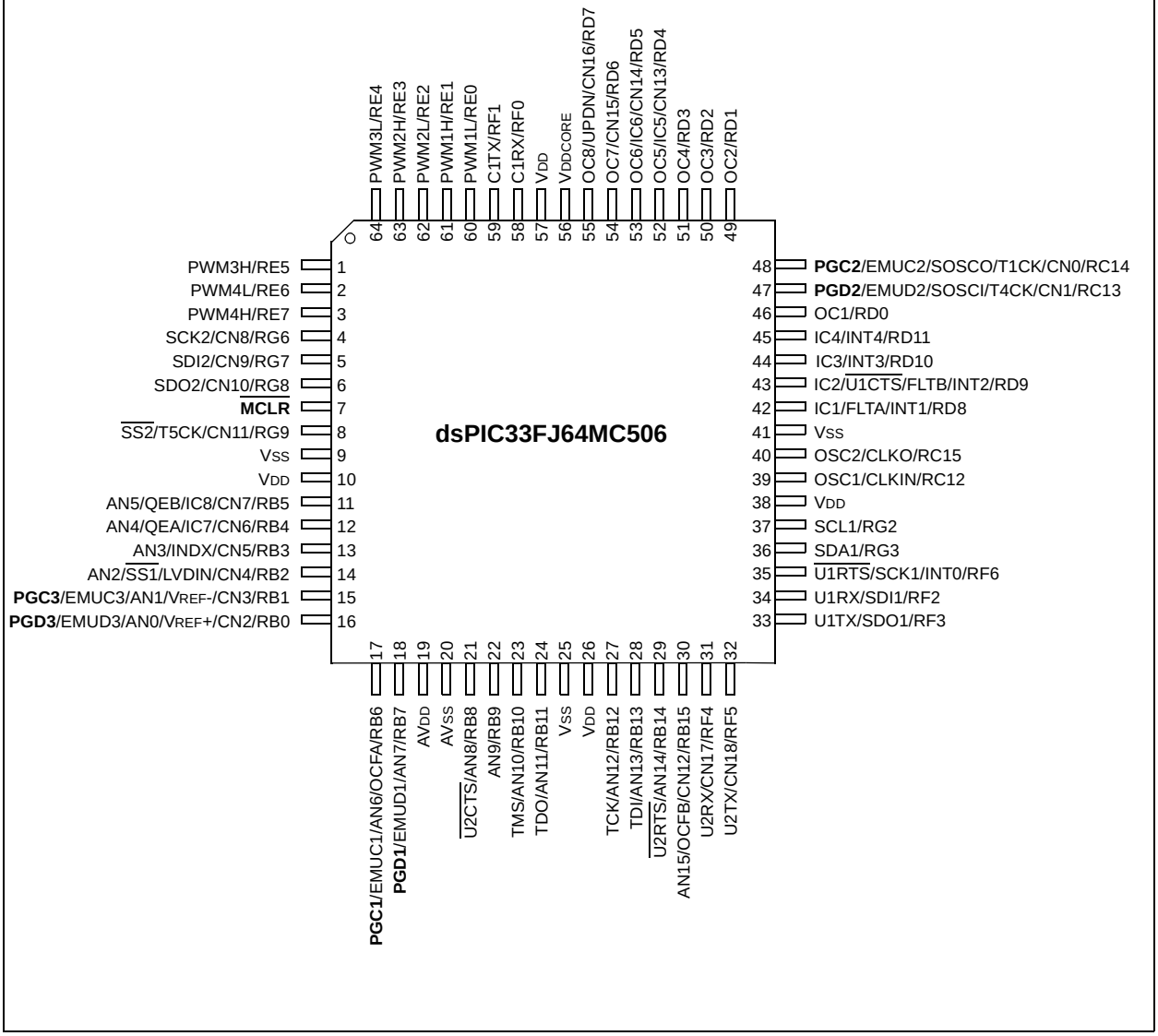
Pin Diagrams (Continued)

100-Pin TQFP



Pin Diagrams (Continued)

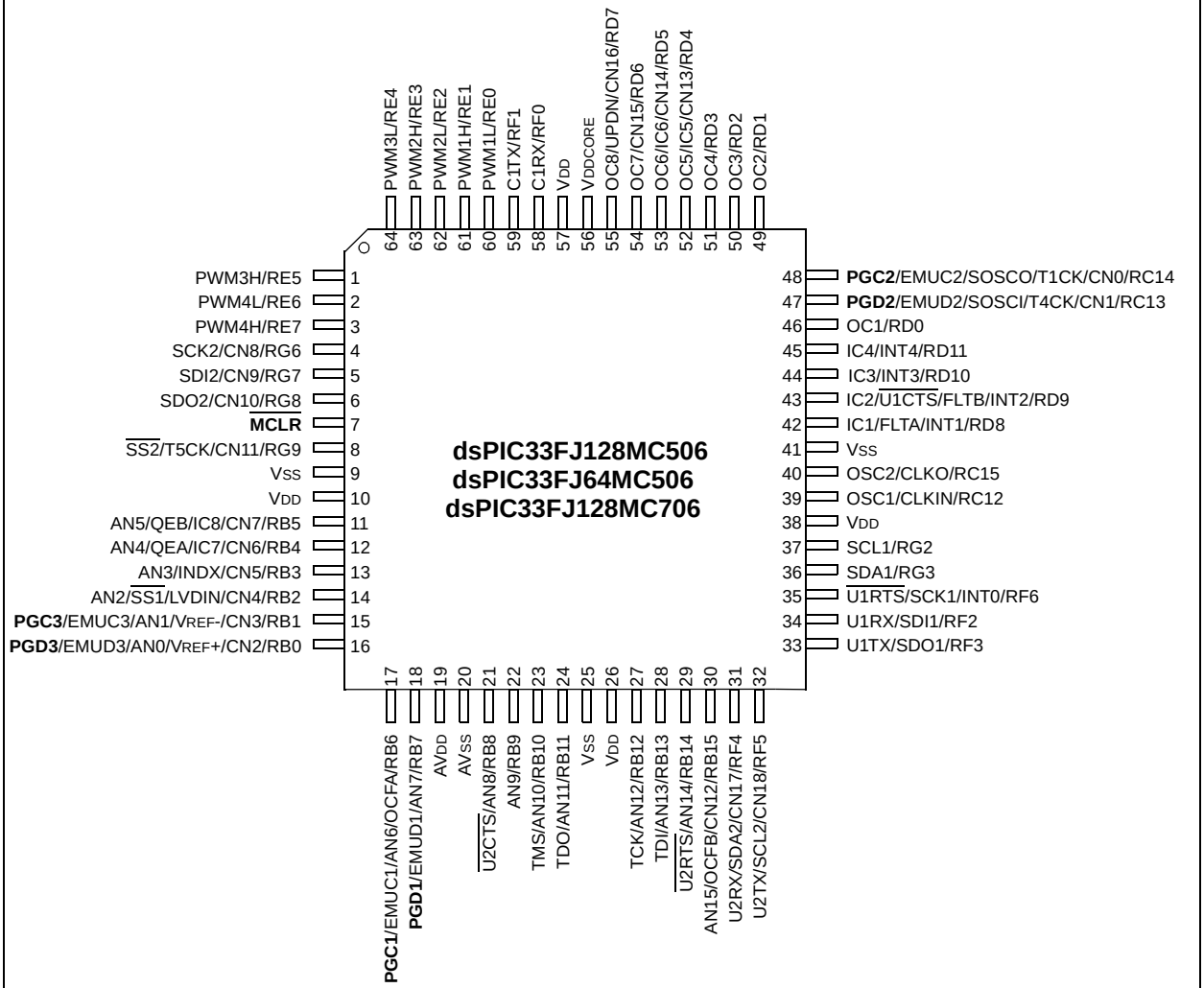
64-Pin TQFP



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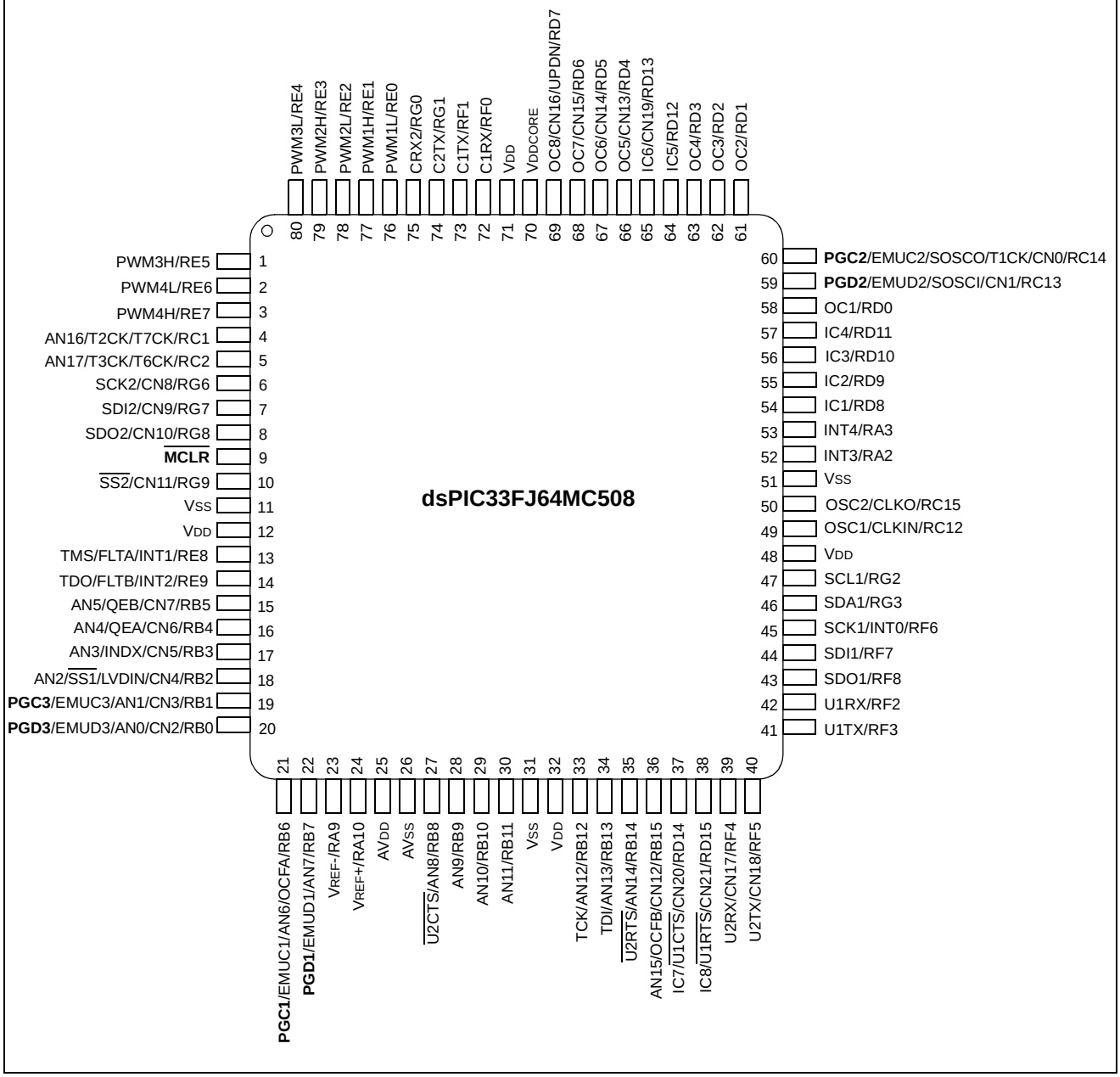
Pin Diagrams (Continued)

64-Pin TQFP



Pin Diagrams (Continued)

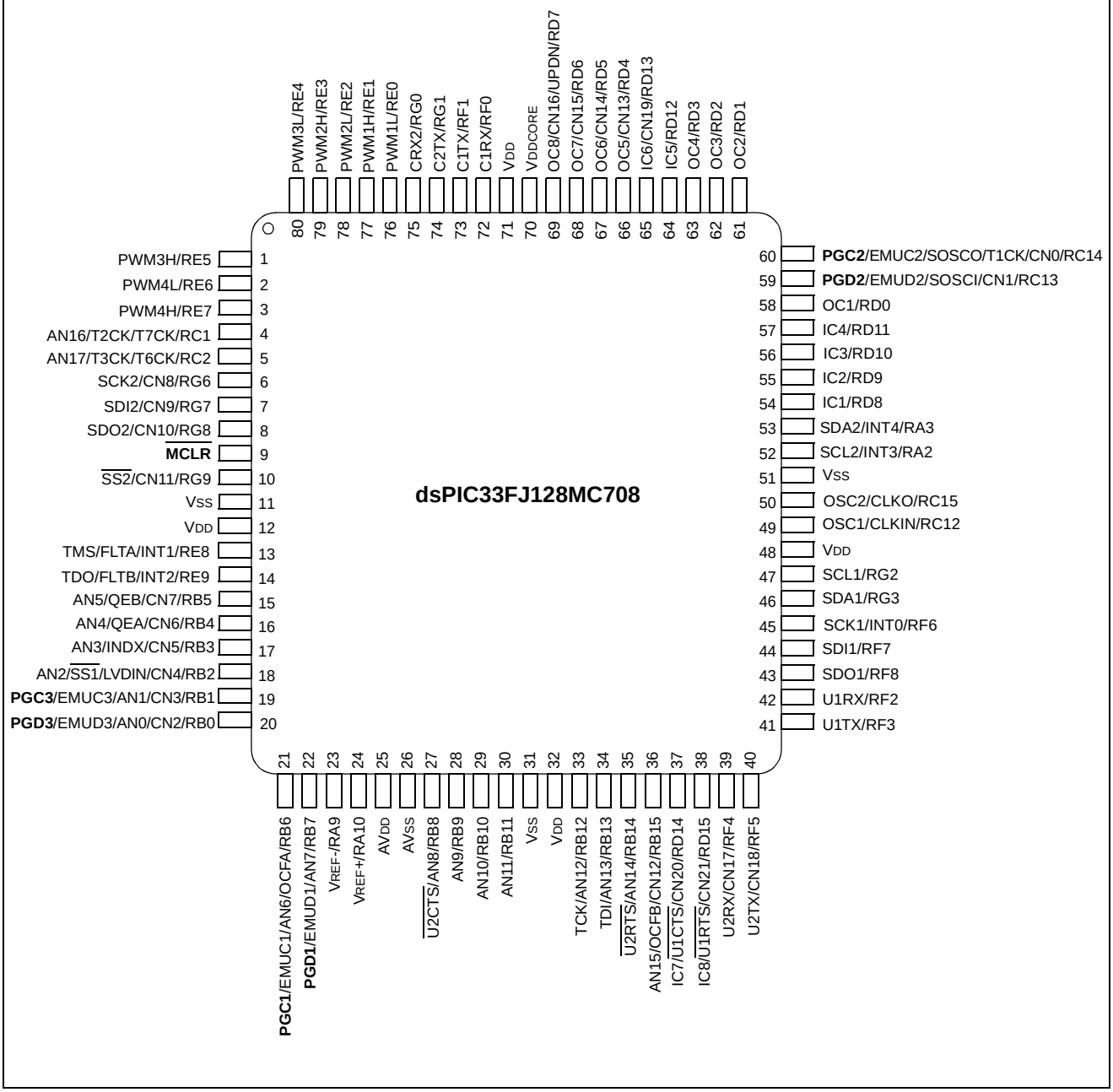
80-Pin TQFP



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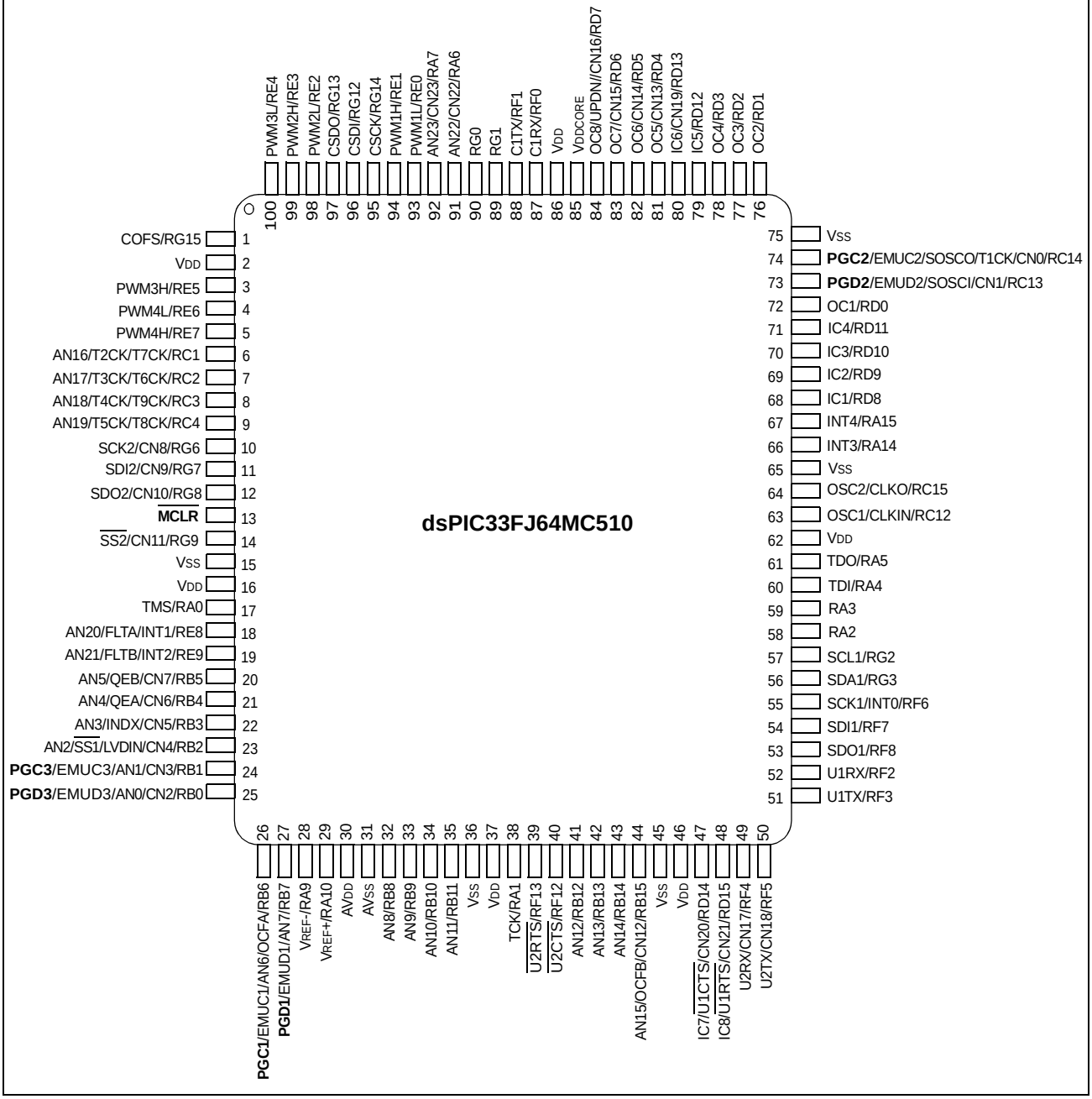
Pin Diagrams (Continued)

80-Pin TQFP



Pin Diagrams (Continued)

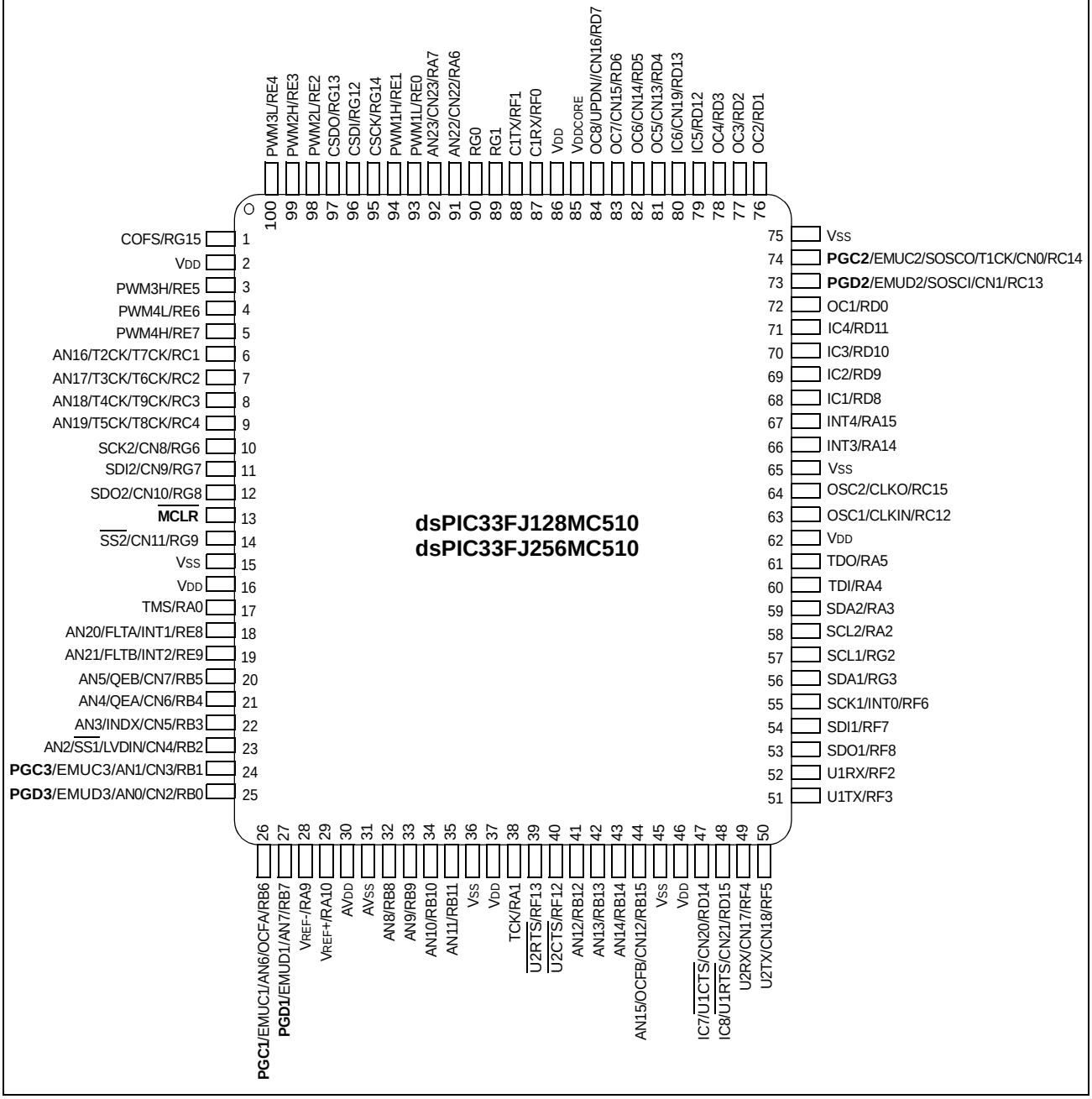
100-Pin TQFP



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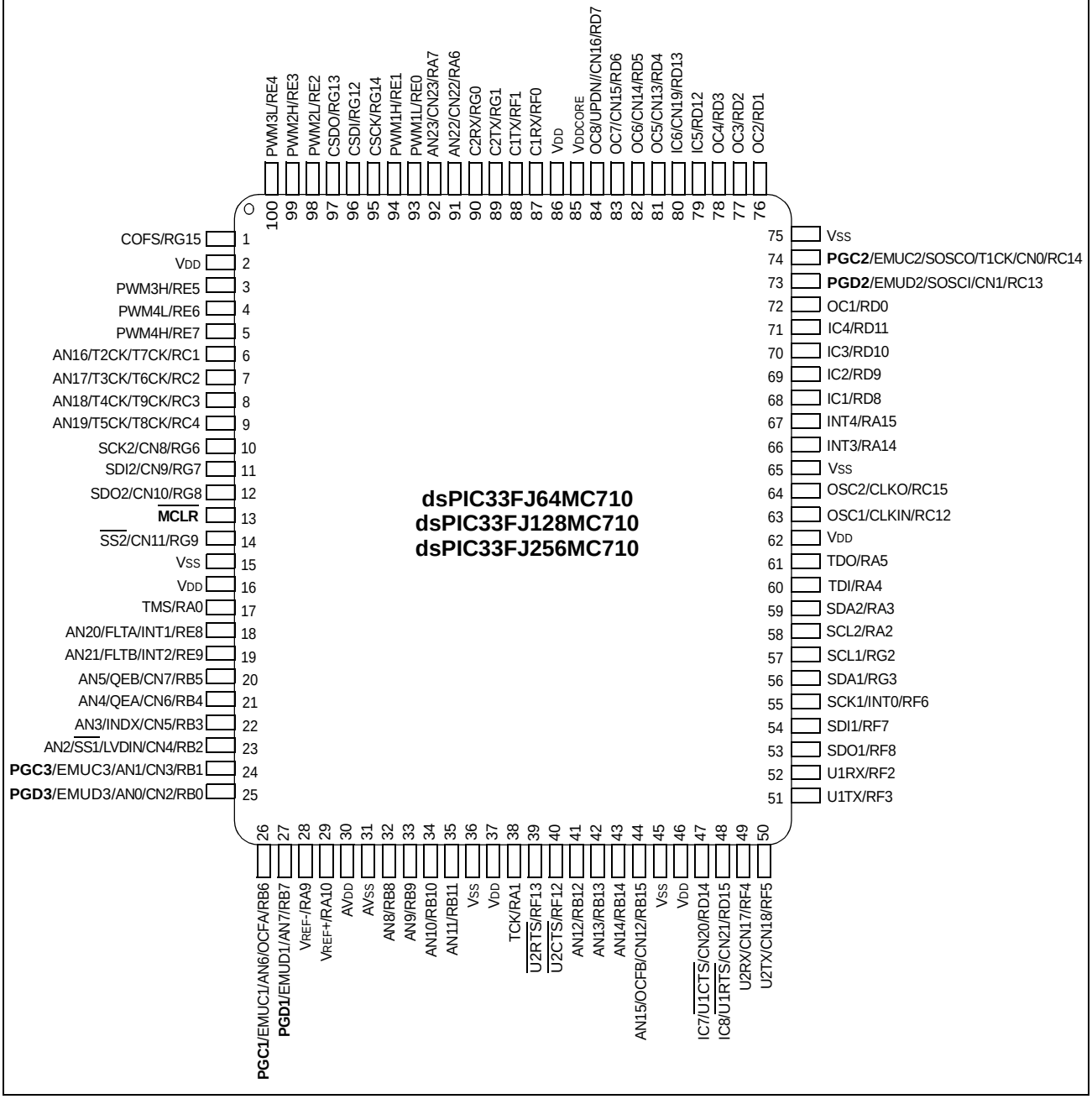
Pin Diagrams (Continued)

100-Pin TQFP



Pin Diagrams (Continued)

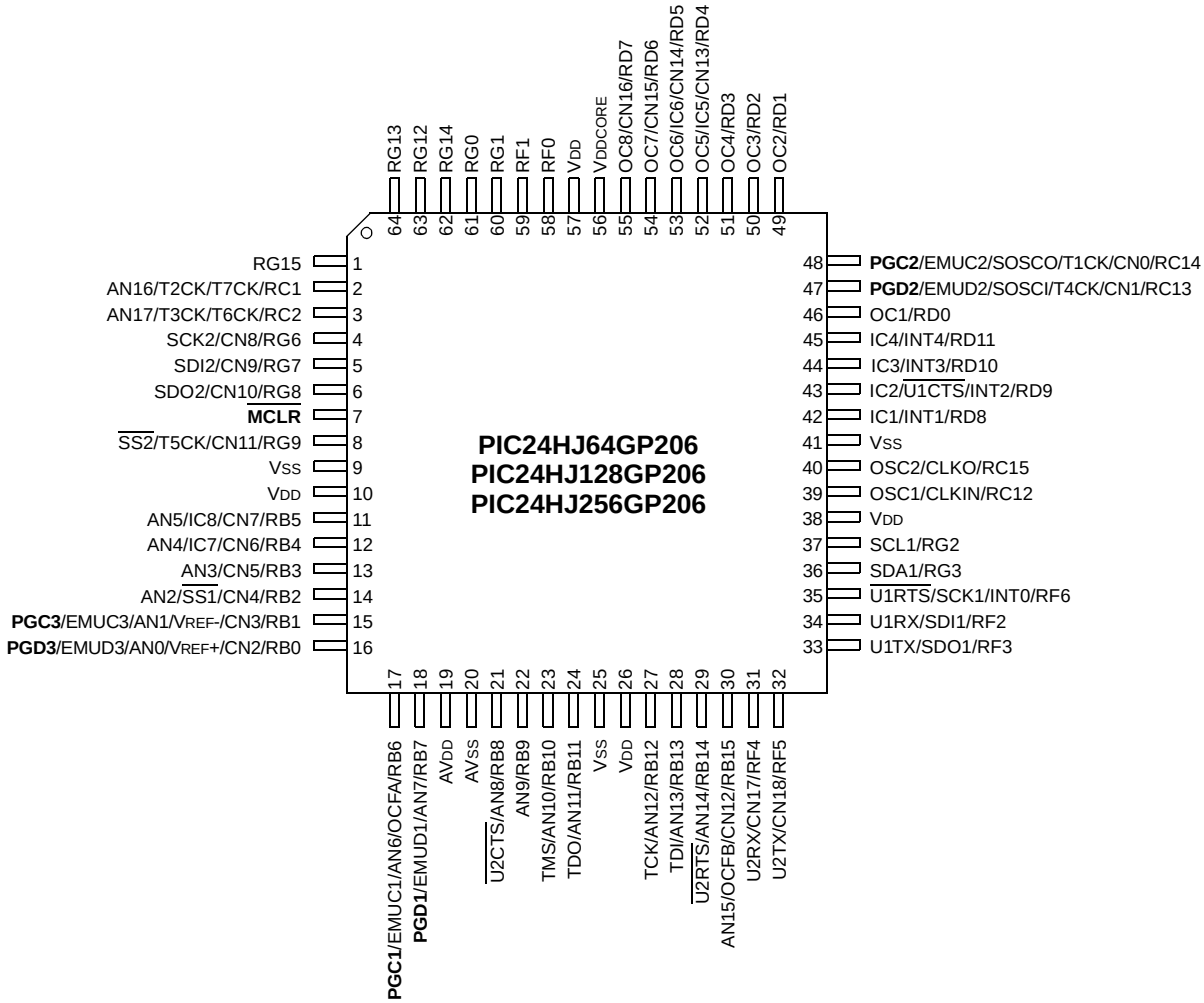
100-Pin TQFP



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Pin Diagrams (Continued)

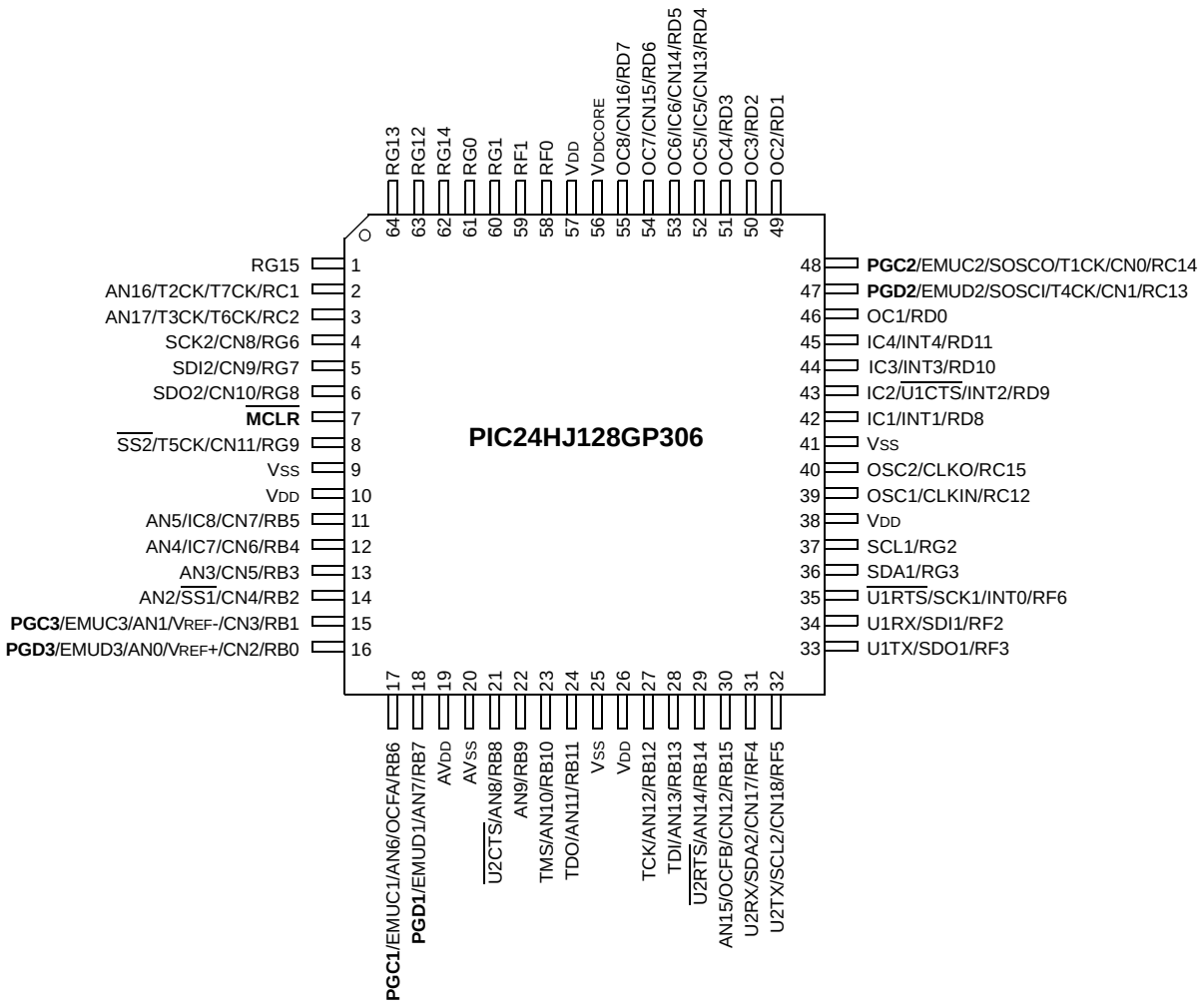
64-Pin TQFP



Note: The PIC24HJ64GP206 device does not have the SCL2 and SDA2 pins.

Pin Diagrams (Continued)

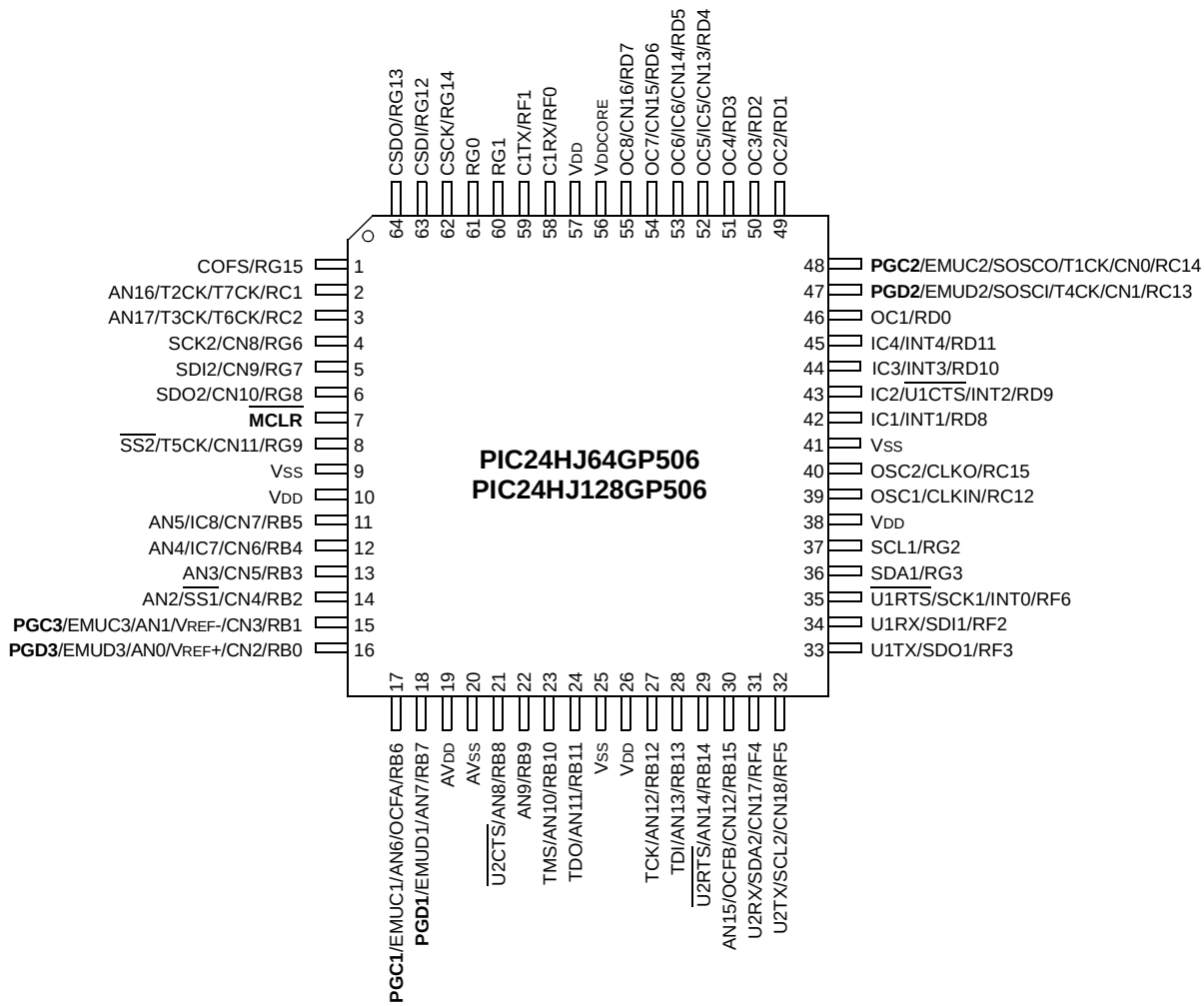
64-Pin TQFP



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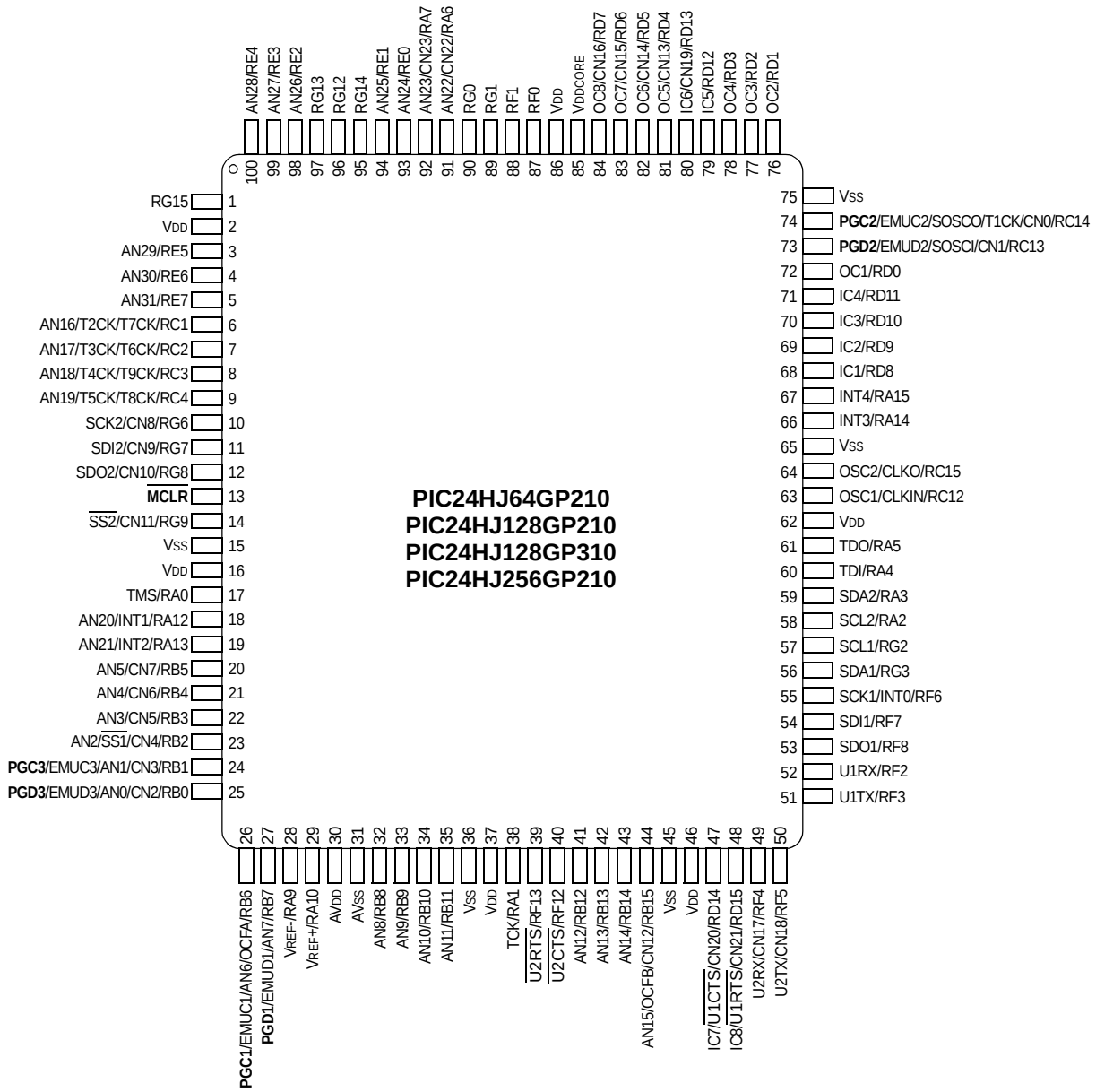
Pin Diagrams (Continued)

64-Pin TQFP



Pin Diagrams (Continued)

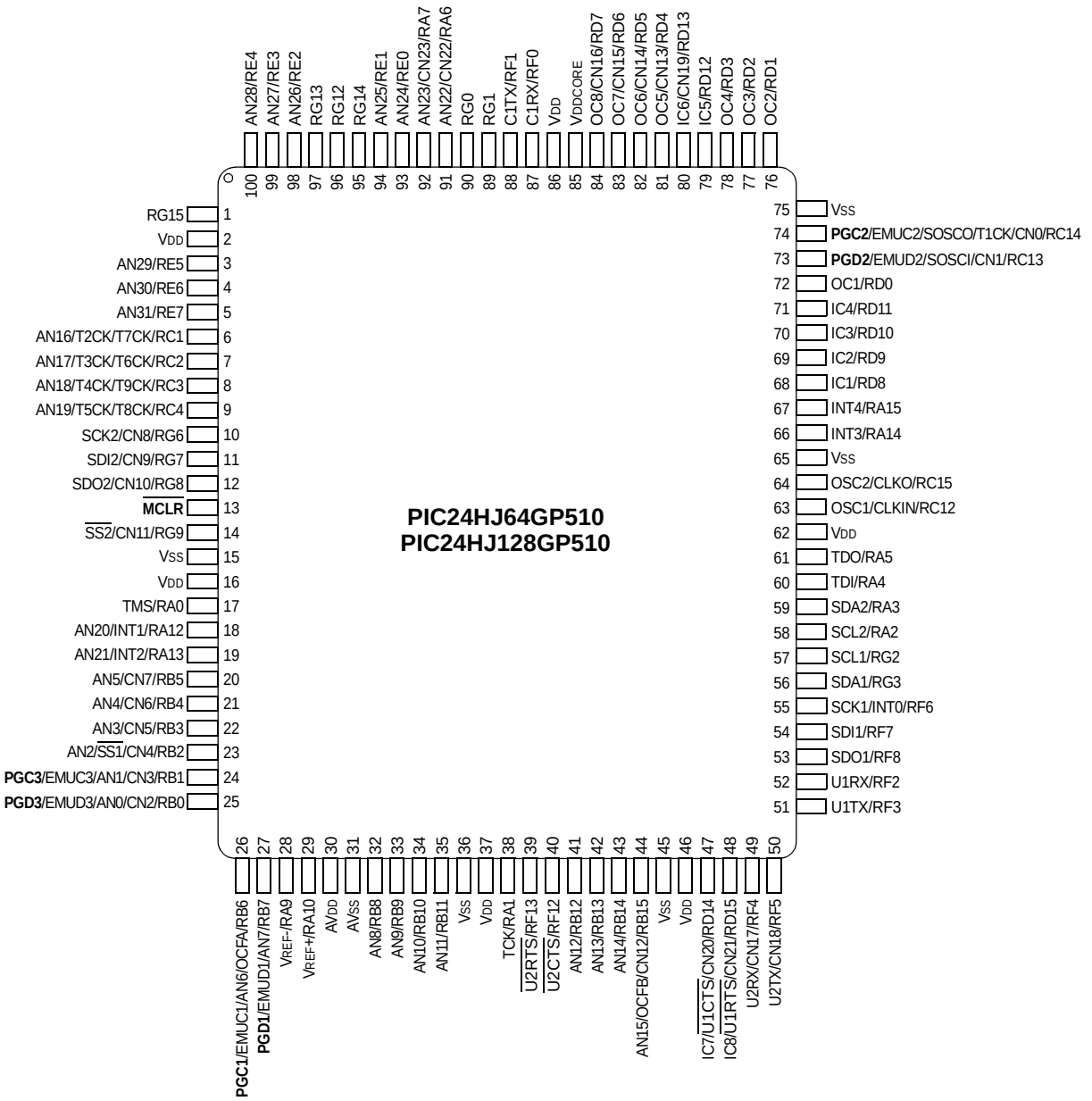
100-Pin TQFP



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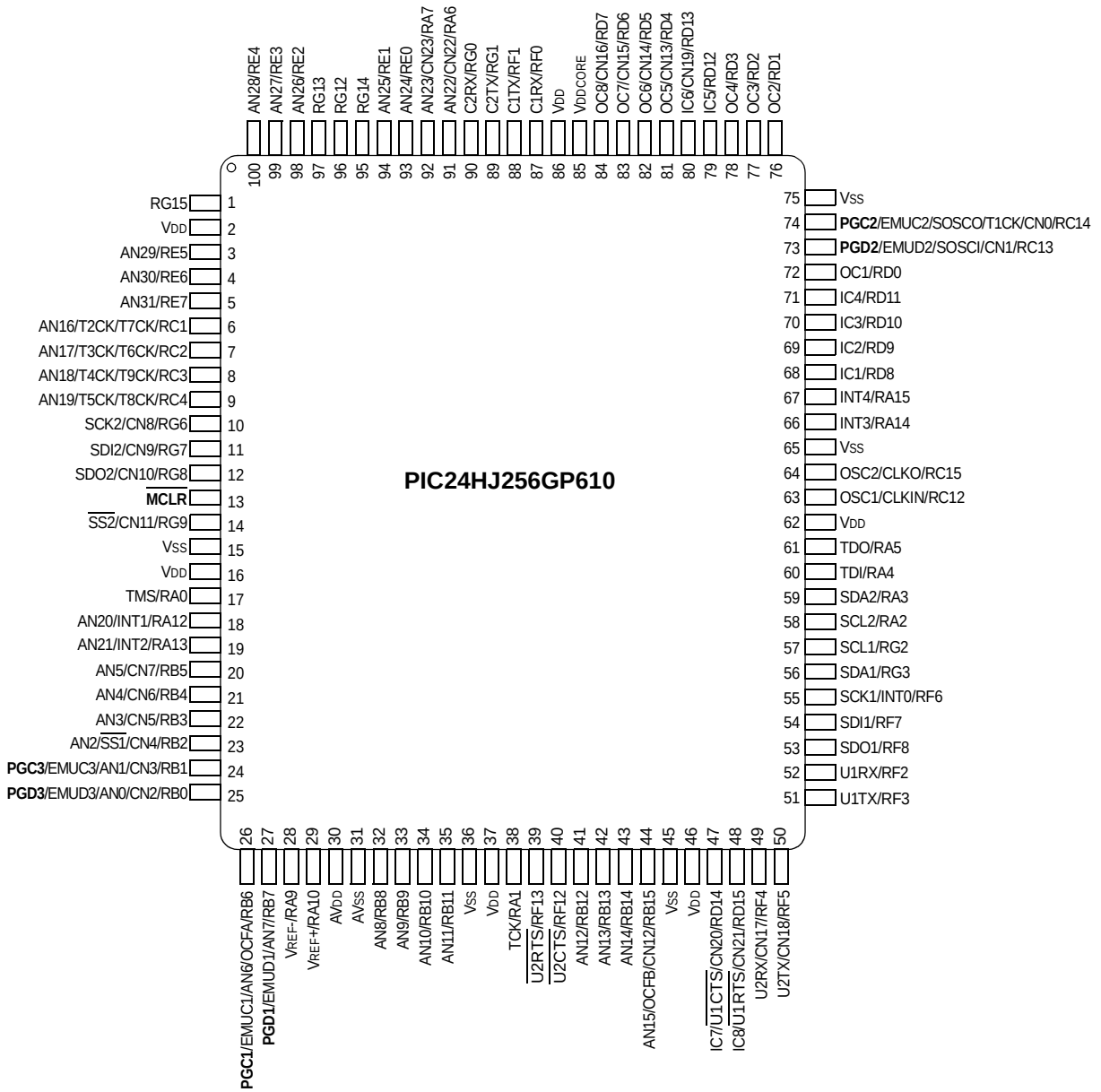
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100-Pin TQFP



Pin Diagrams (Continued)

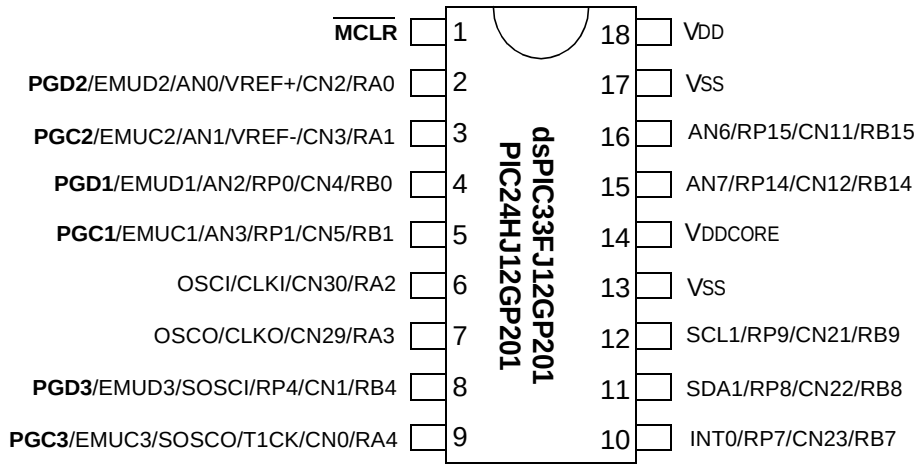
100-Pin TQFP



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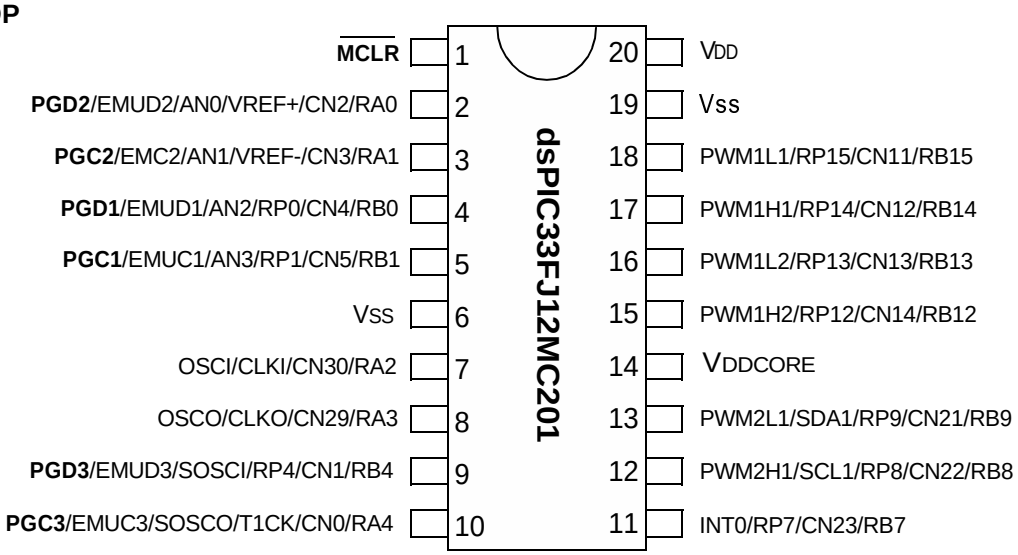
Pin Diagrams (Continued)

18-PIN SDIP
18-PIN SOIC



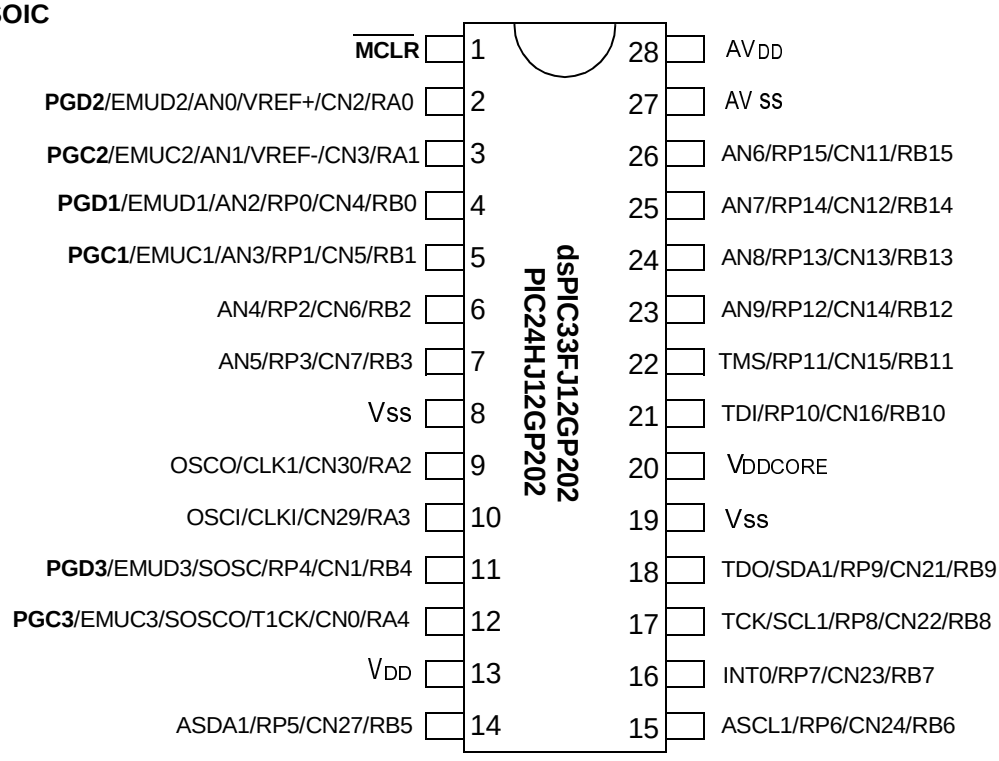
Pin Diagrams (Continued)

20-PIN SDIP
20-PIN SSOP



Pin Diagrams (Continued)

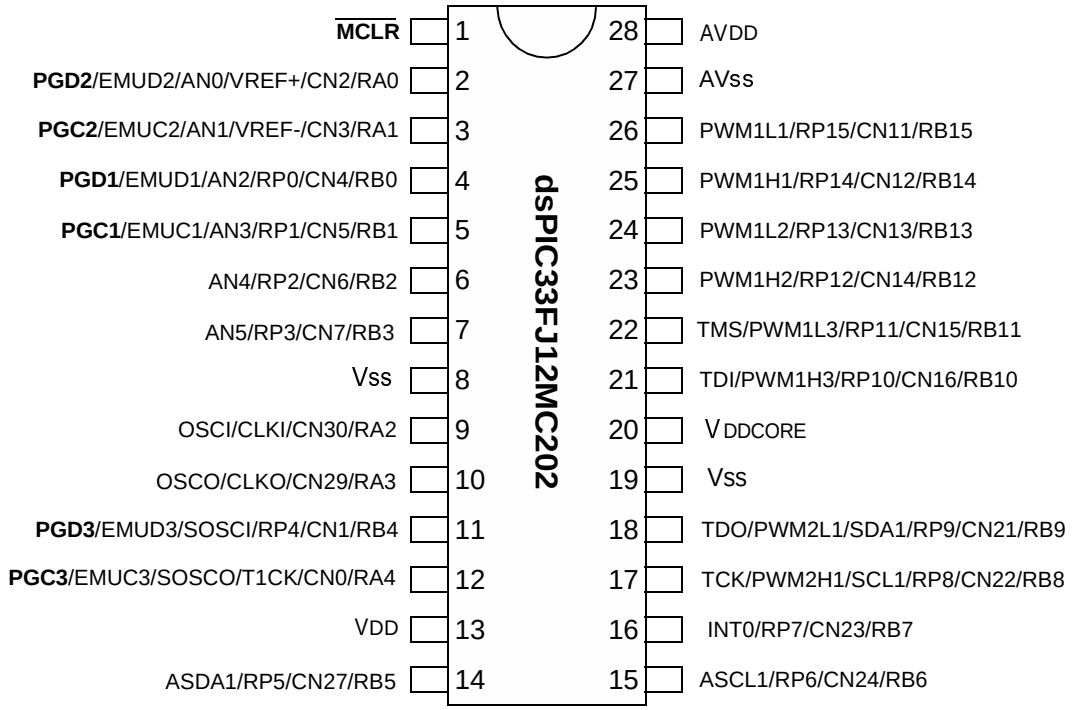
28-PIN SDIP
28-PIN SOIC



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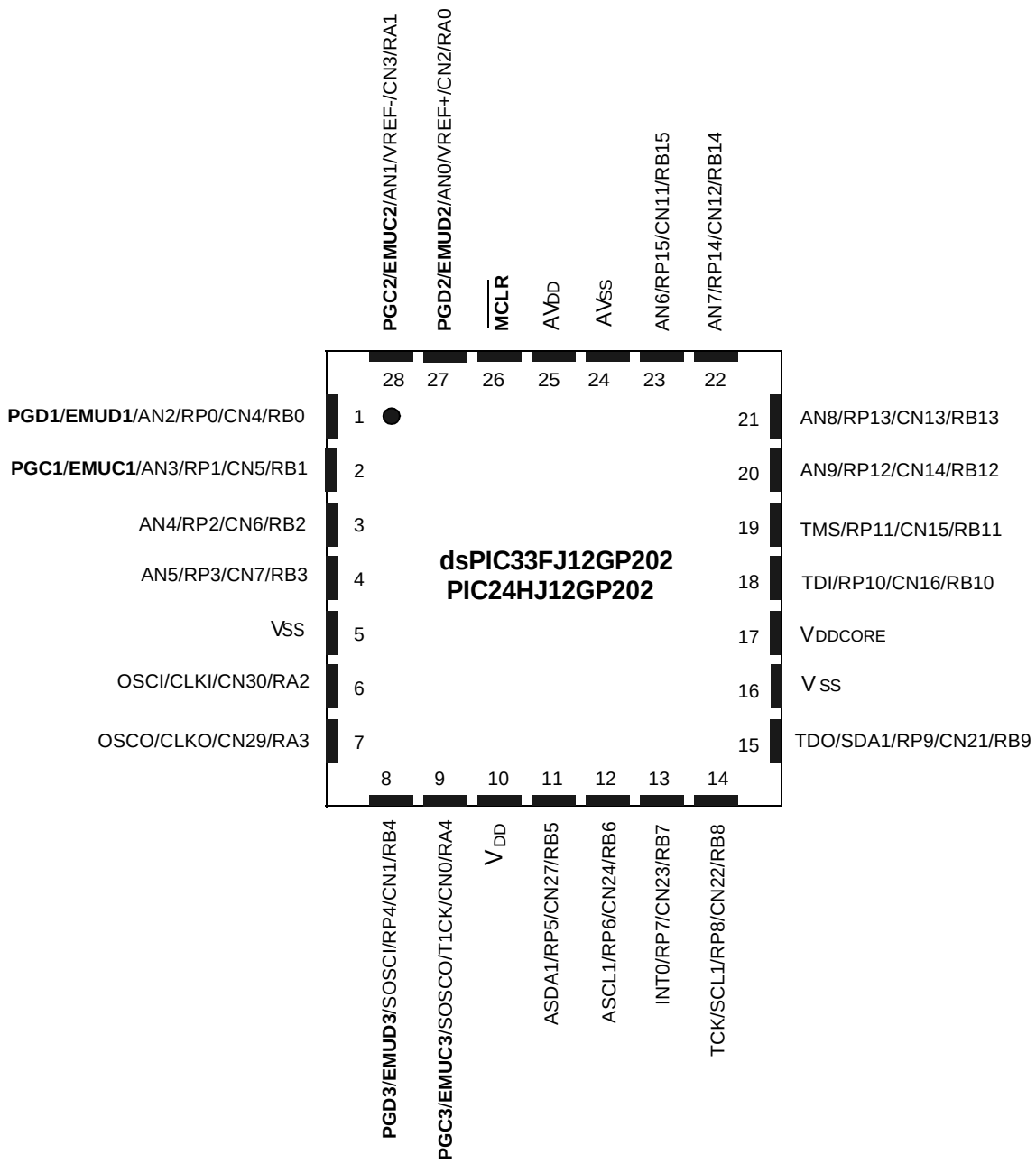
Pin Diagrams (Continued)

28-PIN SDIP
28-PIN SOIC



Pin Diagrams (Continued)

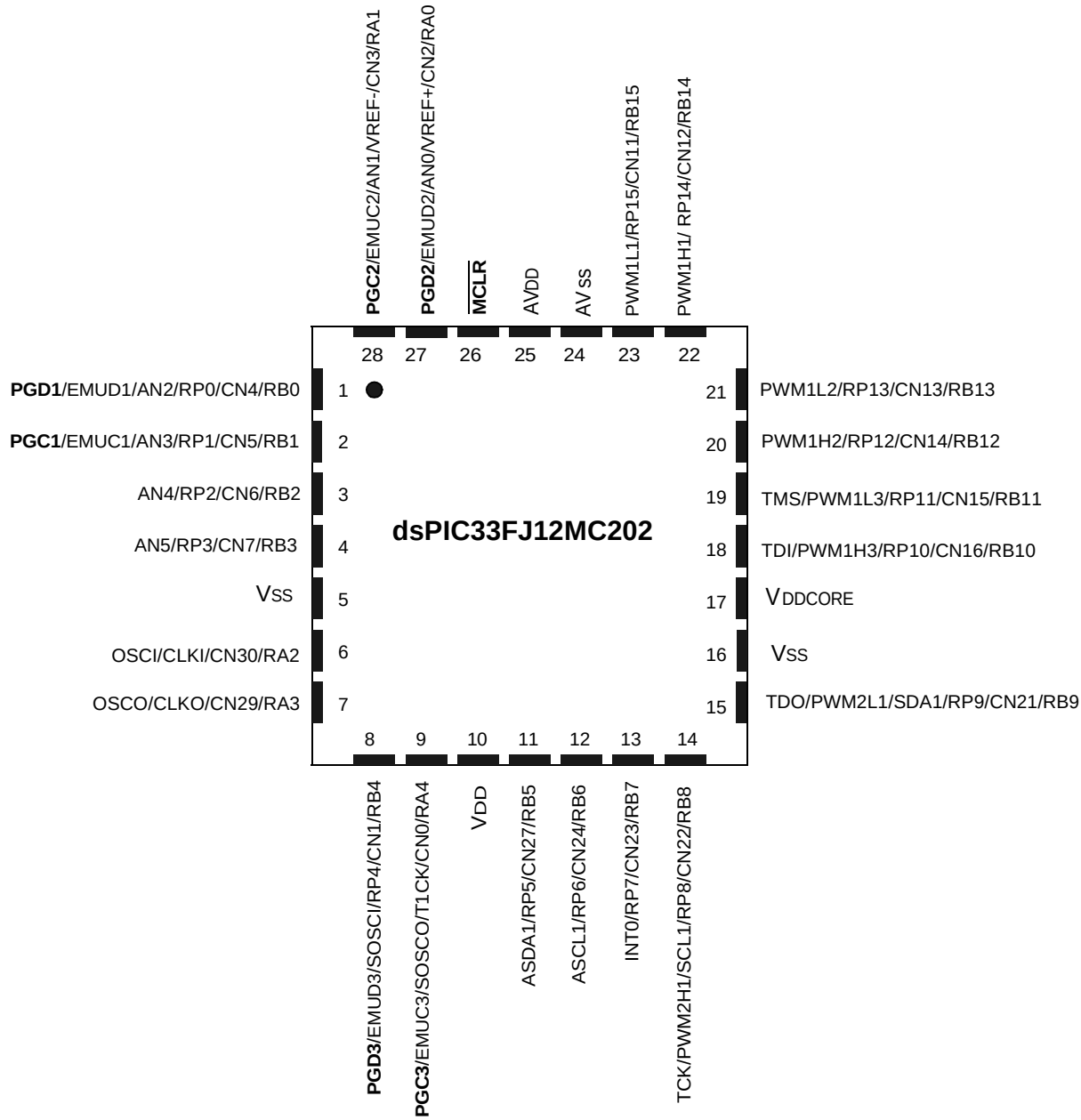
28-Pin QFN 6*6mm



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Pin Diagrams (Continued)

28-Pin QFN 6*6mm



2.4 Memory Map

The program memory map extends from 0x0 to 0xFFFFFE. Code storage is located at the base of the memory map and supports up to 88K instructions (about 256 Kbytes). Table 2-2 shows the program memory size and number of erase and program blocks present in each device variant. Each erase block, or page, contains 512 instructions and each program block, or row, contains 64 instructions.

Locations 0x800000 through 0x800FFE are reserved for executive code memory. This region stores the programming executive and the debugging executive. The programming executive is used for device programming and the debug executive is used for in-circuit debugging. This region of memory can not be used to store user code.

Locations 0xF80000 through 0xF80017 are reserved for the device Configuration registers.

Locations 0xFF0000 and 0xFF0002 are reserved for the Device ID Word registers. These bits can be used by the programmer to identify what device type is being programmed. They are described in **Section 7.0 "Device ID"**. The Device ID registers read out normally, even after code protection is applied.

Figure 2-3 shows the memory map for the dsPIC33F/PIC24H family variants.

TABLE 2-2: CODE MEMORY SIZE

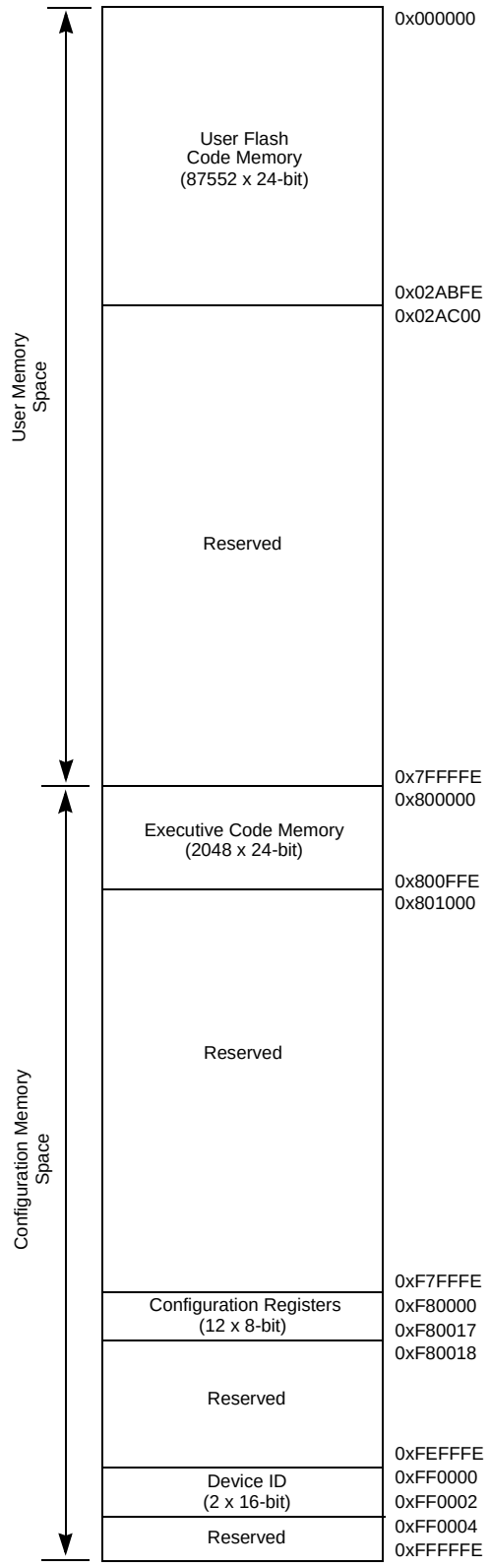
dsPIC33F/PIC24H Device	User Memory Address Limit (Instruction Words)	Write Blocks	Erase Blocks	Executive Memory Address Limit (Instruction Words)
dsPIC33FJ64GP206	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ64GP306	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ64GP310	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ64GP706	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ64GP708	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ64GP710	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ128GP206	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ128GP306	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ128GP310	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ128GP706	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ128GP708	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ128GP710	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ256GP506	0x02ABFE (88K)	1368	171	0x800FFE (2K)
dsPIC33FJ256GP510	0x02ABFE (88K)	1368	171	0x800FFE (2K)
dsPIC33FJ256GP710	0x02ABFE (88K)	1368	171	0x800FFE (2K)
dsPIC33FJ64MC506	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ64MC508	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ64MC510	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ64MC706	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ64MC710	0x00ABFE (22K)	344	43	0x800FFE (2K)
dsPIC33FJ128MC506	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ128MC510	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ128MC706	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ128MC708	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ128MC710	0x0157FE (44K)	688	86	0x800FFE (2K)
dsPIC33FJ256MC510	0x02ABFE (88K)	1368	171	0x800FFE (2K)
dsPIC33FJ256MC710	0x02ABFE (88K)	1368	171	0x800FFE (2K)
PIC24HJ64GP206	0x00ABFE (22K)	344	43	0x800FFE (2K)
PIC24HJ64GP210	0x00ABFE (22K)	344	43	0x800FFE (2K)
PIC24HJ64GP506	0x00ABFE (22K)	344	43	0x800FFE (2K)
PIC24HJ64GP510	0x00ABFE (22K)	344	43	0x800FFE (2K)
PIC24HJ128GP206	0x0157FE (44K)	688	86	0x800FFE (2K)
PIC24HJ128GP210	0x0157FE (44K)	688	86	0x800FFE (2K)

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TABLE 2-2: CODE MEMORY SIZE (CONTINUED)

PIC24HJ128GP306	0x0157FE (44K)	688	86	0x800FFE (2K)
PIC24HJ128GP310	0x0157FE (44K)	688	86	0x800FFE (2K)
PIC24HJ128GP506	0x0157FE (44K)	688	86	0x800FFE (2K)
PIC24HJ128GP510	0x0157FE (44K)	688	86	0x800FFE (2K)
PIC24HJ256GP206	0x02ABFE (88K)	1368	171	0x800FFE (2K)
PIC24HJ256GP210	0x02ABFE (88K)	1368	171	0x800FFE (2K)
PIC24HJ256GP610	0x02ABFE (88K)	1368	171	0x800FFE (2K)
dsPIC33FJ12GP201	0x001FFE (4K)	64	8	0x8007FE (1K)
dsPIC33FJ12GP202	0x001FFE (4K)	64	8	0x8007FE (1K)
dsPIC33FJ12MC201	0x001FFE (4K)	64	8	0x8007FE (1K)
dsPIC33FJ12MC202	0x001FFE (4K)	64	8	0x8007FE (1K)
PIC24HJ12GP201	0x001FFE (4K)	64	8	0x8007FE (1K)
PIC24HJ12GP202	0x001FFE (4K)	64	8	0x8007FE (1K)

FIGURE 2-3: PROGRAM MEMORY MAP



Note: The address boundaries for user Flash and Executive code memory are device dependent.

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3.0 DEVICE PROGRAMMING – ENHANCED ICSP

This section discusses programming the device through Enhanced ICSP and the programming executive. The programming executive resides in executive memory (separate from code memory) and is executed when Enhanced ICSP Programming mode is entered. The programming executive provides the mechanism for the programmer (host device) to program and verify the dsPIC33F/PIC24H Programming Specification family devices using a simple command set and communication protocol. There are several basic functions provided by the programming executive:

- Read Memory
- Erase Memory
- Program Memory
- Blank Check
- Read Executive Firmware Revision

The programming executive performs the low-level tasks required for erasing, programming and verifying a device. This allows the programmer to program the device by issuing the appropriate commands and data. Table 3-1 summarizes the commands. A detailed description for each command is provided in **Section 4.2 "Programming Executive Commands"**.

TABLE 3-1: COMMAND SET SUMMARY

Command	Description
SCHECK	Sanity check
READC	Read Configuration registers or Device ID registers
READP	Read code memory
PROGC	Program a Configuration register and verify
PROGP	Program one row of code memory and verify
PROGW	Program one word of code memory and verify
QBLANK	Query if the code memory is blank
QVER	Query the software version

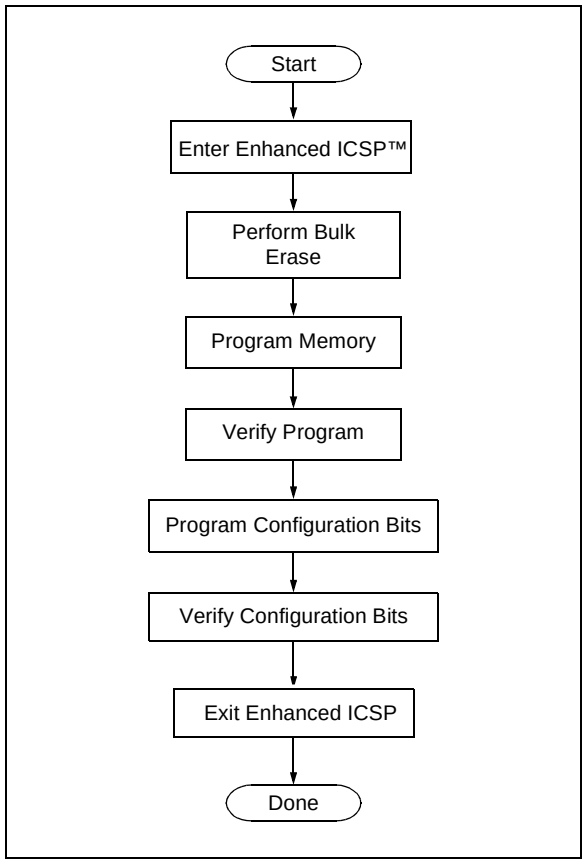
The programming executive uses the device's data RAM for variable storage and program execution. After the programming executive has run, no assumptions should be made about the contents of data RAM.

3.1 Overview of the Programming Process

Figure 3-1 shows the high-level overview of the programming process. After entering Enhanced ICSP mode, the programming executive is verified. Next, the device is erased. Then, the code memory is programmed, followed by the nonvolatile device Configuration registers. Code memory (including the Configuration registers) is then verified to ensure that programming was successful.

After the programming executive has been verified in memory (or loaded if not present), the dsPIC33F/PIC24H Programming Specification can be programmed using the command set shown in Table 3-1.

FIGURE 3-1: HIGH-LEVEL ENHANCED ICSP™ PROGRAMMING FLOW



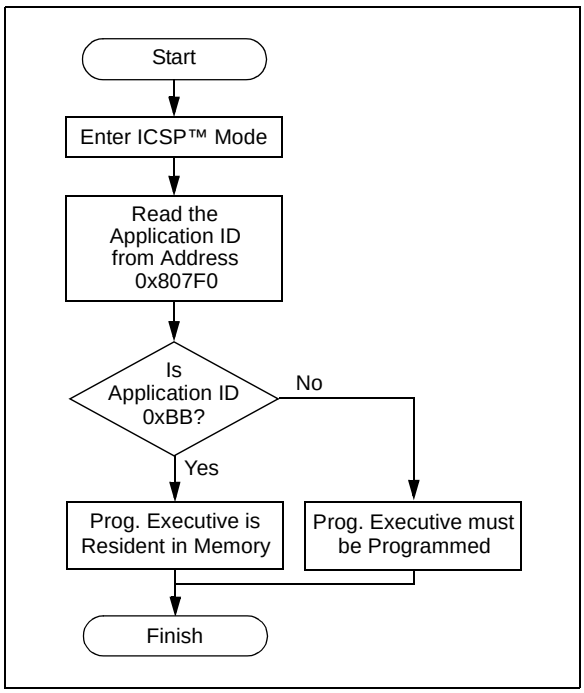
3.2 Confirming the Presence of the Programming Executive

Before programming can begin, the programmer must confirm that the programming executive is stored in executive memory. The procedure for this task is shown in Figure 3-2.

First, ICSP mode is entered. Then, the unique Application ID Word stored in executive memory is read. If the programming executive is resident, the Application ID Word is 0xBB, which means programming can resume as normal. However, if the Application ID Word is not 0xBB, the programming executive must be programmed to executive code memory using the method described in **Section 6.0 "Programming the Programming Executive to Memory"**.

Section 5.0 "Device Programming – ICSP" describes the ICSP programming method. **Section 5.11 "Reading the Application ID Word"** describes the procedure for reading the Application ID Word in ICSP mode.

FIGURE 3-2: CONFIRMING PRESENCE OF PROGRAMMING EXECUTIVE



3.3 Entering Enhanced ICSP Mode

As shown in Figure 3-3, entering Enhanced ICSP Program/Verify mode requires three steps:

1. The $\overline{\text{MCLR}}$ pin is briefly driven high then low.
2. A 32-bit key sequence is clocked into PGD.
3. $\overline{\text{MCLR}}$ is then driven high within a specified period of time and held.

The programming voltage applied to $\overline{\text{MCLR}}$ is V_{IH}, which is essentially V_{DD} in the case of dsPIC33F/PIC24H devices. There is no minimum time requirement for holding at V_{IH}. After V_{IH} is removed, an interval of at least P18 must elapse before presenting the key sequence on PGD.

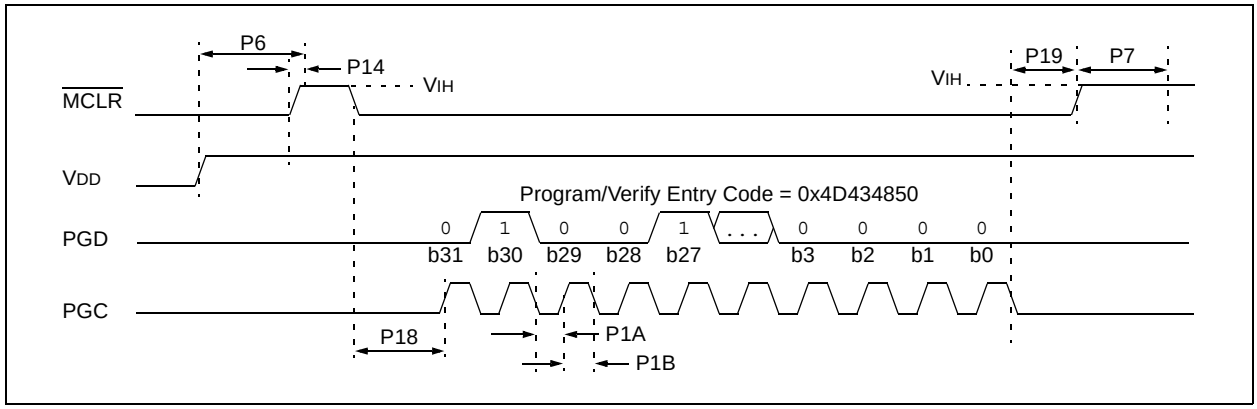
The key sequence is a specific 32-bit pattern, '0100 1101 0100 0011 0100 1000 0101 0000' (more easily remembered as 0x4D434850 in hexadecimal format). The device will enter Program/Verify mode only if the key sequence is valid. The Most Significant bit (MSb) of the most significant nibble must be shifted in first.

Once the key sequence is complete, V_{IH} must be applied to $\overline{\text{MCLR}}$ and held at that level for as long as Program/Verify mode is to be maintained. An interval time of at least P19 and P7 must elapse before presenting data on PGD. Signals appearing on PGD before P7 has elapsed will not be interpreted as valid.

On successful entry, the program memory can be accessed and programmed in serial fashion. While in the Program/Verify mode, all unused I/Os are placed in the high-impedance state.

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FIGURE 3-3: ENTERING ENHANCED ICSP™ MODE



3.4 Blank Check

The term “Blank Check” implies verifying that the device has been successfully erased and has no programmed memory locations. A blank or erased memory location is always read as a ‘1’.

The Device ID registers (0xFF0000:0xFF0002) can be ignored by the Blank Check since this region stores device information that cannot be erased. The device Configuration registers are also ignored by the Blank Check. Additionally, all unimplemented memory space should be ignored from the Blank Check.

The QBLANK command is used for the Blank Check. It determines if the code memory is erased by testing these memory regions. A ‘BLANK’ or ‘NOT BLANK’ response is returned. If it is determined that the device is not blank, it must be erased before attempting to program the chip.

3.5 Code Memory Programming

3.5.1 PROGRAMMING METHODOLOGY

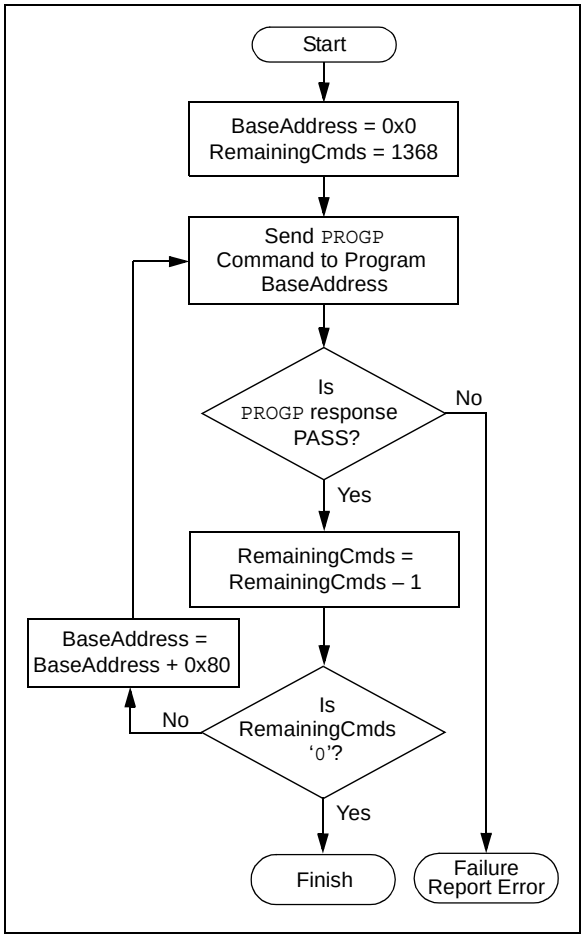
Code memory is programmed with the PROGP command. PROGP programs one row of code memory starting from the memory address specified in the command. The number of PROGP commands required to program a device depends on the number of write blocks that must be programmed in the device.

A flowchart for programming code memory is shown in Figure 3-4. In this example, all 88K instruction words of a dsPIC33F/PIC24H device are programmed. First, the number of commands to send (called ‘RemainingCmds’ in the flowchart) is set to 1368 and the destination address (called ‘BaseAddress’) is set to ‘0’. Next, one write block in the device is programmed with a PROGP command. Each PROGP command contains data for one row of code memory of the dsPIC33F/PIC24H. After the first command is processed successfully, ‘RemainingCmds’ is decremented by ‘1’ and compared with ‘0’. Since there are more PROGP commands to send, ‘BaseAddress’ is incremented by 0x80 to point to the next row of memory.

On the second PROGP command, the second row is programmed. This process is repeated until the entire device is programmed..

Note:	If a bootloader needs to be programmed, the bootloader code must not be programmed into the first page of code memory. For example, if a bootloader located at address 0x200 attempts to erase the first page, it would inadvertently erase itself. Instead, program the bootloader into the second page, e.g. 0x400.

FIGURE 3-4: FLOWCHART FOR PROGRAMMING CODE MEMORY



3.5.2 PROGRAMMING VERIFICATION

After code memory is programmed, the contents of memory can be verified to ensure that programming was successful. Verification requires code memory to be read back and compared against the copy held in the programmer's buffer.

The READP command can be used to read back all the programmed code memory.

Alternatively, you can have the programmer perform the verification after the entire device is programmed, using a checksum computation.

3.5.3 CHECKSUM COMPUTATION

Only the Configuration registers are included in the checksum computation. The Device ID and Unit ID are not included in the checksum computation.

Table 3-2 shows how this 16-bit computation can be made for each dsPIC33F and PIC24H device. Computations for read code protection are shown both enabled and disabled. The checksum values shown here assume that the Configuration registers are also erased. However, when code protection is enabled, the value of the FGS register is assumed to be 0x5.

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TABLE 3-2: CHECKSUM COMPUTATION

Device	Read Code Protection	Checksum Computation	Erased Value	Value with 0xAAAAAA at 0x0 and Last Code Address
dsPIC33FJ64GP206	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ64GP306	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ64GP310	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ64GP706	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ64GP708	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ64GP710	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128GP206	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128GP306	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128GP310	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128GP706	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128GP708	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128GP710	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ256GP506	Disabled	CFGB + SUM(0:02ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ256GP510	Disabled	CFGB + SUM(0:02ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ256GP710	Disabled	CFGB + SUM(0:02ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA

Item Description:

SUM(a:b) = Byte sum of locations a to b inclusive (all 3 bytes of code memory)

CFGB = **Configuration Block (masked)**

= Byte sum of ((FBS & 0xCF) + (FSS & 0xFF) + (FGS & 0x07) + (FOSCSEL & 0xA7) + (FOSC & 0xE7) + (FWDT & 0xDF) + (FPOR & 0xE7) + (FICD & 0xE3))

(for dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202)

= Byte sum of ((FBS & 0xCF) + (FSS & 0xCF) + (FGS & 0x07) + (FOSCSEL & 0xA7) + (FOSC & 0xC7) + (FWDT & 0xDF) + (FPOR & 0xE7) + (FICD & 0xE3))

(for all other devices)

TABLE 3-2: CHECKSUM COMPUTATION (CONTINUED)

Device	Read Code Protection	Checksum Computation	Erased Value	Value with 0xAAAAAA at 0x0 and Last Code Address
dsPIC33FJ64MC506	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ64MC508	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ64MC510	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ64MC706	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ64MC710	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128MC506	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128MC510	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128MC706	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128MC708	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ128MC710	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ256MC510	Disabled	CFGB + SUM(0:02ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ256MC710	Disabled	CFGB + SUM(0:02ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ64GP206	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ64GP210	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ64GP506	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA

Item Description:

SUM(a:b) = Byte sum of locations a to b inclusive (all 3 bytes of code memory)

CFGB = **Configuration Block (masked)**

= Byte sum of ((FBS & 0xCF) + (FSS & 0xFF) + (FGS & 0x07) + (FOSCSEL & 0xA7) + (FOSC & 0xE7) + (FWDT & 0xDF) + (FPOR & 0xE7) + (FICD & 0xE3))

(for dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202)

= Byte sum of ((FBS & 0xCF) + (FSS & 0xCF) + (FGS & 0x07) + (FOSCSEL & 0xA7) + (FOSC & 0xC7) + (FWDT & 0xDF) + (FPOR & 0xE7) + (FICD & 0xE3))

(for all other devices)

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TABLE 3-2: CHECKSUM COMPUTATION (CONTINUED)

Device	Read Code Protection	Checksum Computation	Erased Value	Value with 0xAAAAAA at 0x0 and Last Code Address
PIC24HJ64GP510	Disabled	CFGB + SUM(0:00ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ128GP206	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ128GP210	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ128GP306	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ128GP310	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ128GP506	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ128GP510	Disabled	CFGB + SUM(0:0157FF)	0x01BC	0xFFBE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ256GP206	Disabled	CFGB + SUM(0:02ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ256GP210	Disabled	CFGB + SUM(0:02ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
PIC24HJ256GP610	Disabled	CFGB + SUM(0:02ABFF)	0x03BC	0x01BE
	Enabled	CFGB	0x05BA	0x05BA
dsPIC33FJ12GP201	Disabled	CFGB + SUM(0:001FFF)	0xD60C	0xD40E
	Enabled	CFGB	0x060A	0x060A
dsPIC33FJ12GP202	Disabled	CFGB + SUM(0:001FFF)	0xD60C	0xD40E
	Enabled	CFGB	0x060A	0x060A
dsPIC33FJ12MC201	Disabled	CFGB + SUM(0:001FFF)	0xD60C	0xD40E
	Enabled	CFGB	0x060A	0x060A
dsPIC33FJ12MC202	Disabled	CFGB + SUM(0:001FFF)	0xD60C	0xD40E
	Enabled	CFGB	0x060A	0x060A
PIC24HJ12GP201	Disabled	CFGB + SUM(0:001FFF)	0xD60C	0xD40E
	Enabled	CFGB	0x060A	0x060A
PIC24HJ12GP202	Disabled	CFGB + SUM(0:001FFF)	0xD60C	0xD40E
	Enabled	CFGB	0x060A	0x060A

Item Description:

SUM(a:b) = Byte sum of locations a to b inclusive (all 3 bytes of code memory)

CFGB = **Configuration Block (masked)**

= Byte sum of ((FBS & 0xCF) + (FSS & 0xFF) + (FGS & 0x07) + (FOSCSEL & 0xA7) + (FOSC & 0xE7) + (FWDT & 0xDF) + (FPOR & 0xE7) + (FICD & 0xE3))

(for dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202)

= Byte sum of ((FBS & 0xCF) + (FSS & 0xCF) + (FGS & 0x07) + (FOSCSEL & 0xA7) + (FOSC & 0xC7) + (FWDT & 0xDF) + (FPOR & 0xE7) + (FICD & 0xE3))

(for all other devices)

3.6 Configuration Bits Programming

3.6.1 OVERVIEW

The dsPIC33F/PIC24H has Configuration bits stored in twelve 8-bit Configuration registers, aligned on even configuration memory address boundaries. These bits can be set or cleared to select various device configurations. There are three types of Configuration bits: system operation bits, code-protect bits and unit ID bits. The system operation bits determine the power-on settings for system level components, such as oscillator and Watchdog Timer. The code-protect bits prevent program memory from being read and written.

The register descriptions for the FBS, FSS, FGS, FOSCSEL, FOSC, FWDT, FPOR and FICD Configuration registers are shown in Table 3-3.

The Configuration register map is shown in Table 3-4..

Note:	If any of the code-protect bits in FBS, FSS or FGS is clear, then the entire device must be erased before it can be reprogrammed.

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TABLE 3-3: dsPIC33F/PIC24H CONFIGURATION BITS DESCRIPTION

Bit Field	Register	Description
RBS<1:0>	FBS	**Boot Segment Data RAM Code Protection** 11 = No RAM is reserved for Boot Segment 10 = Small-sized Boot RAM [128 bytes of RAM are reserved for Boot Segment] 01 = Medium-sized Boot RAM [256 bytes of RAM are reserved for Boot Segment] 00 = Large-sized Boot RAM [1024 bytes of RAM are reserved for Boot Segment] [**Note:** This bit is Reserved in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202.]
BSS<2:0>	FBS	**Boot Segment Program Memory Code Protection** 111 = No Boot Segment 110 = Standard security, Small-sized Boot Program Flash [Boot Segment ends at 0x0003FF in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202. Boot Segment ends at 0x0007FF in other all other devices.] 101 = Standard security, Medium-sized Boot Program Flash [Boot Segment ends at 0x0007FF in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202. Boot Segment ends at 0x001FFF in all other devices.] 100 = Standard security, Large-sized Boot Program Flash [Boot Segment ends at 0x000FFF in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202. Boot Segment ends at 0x003FFF in all other devices.] 011 = No Boot Segment 010 = High security, Small-sized Boot Program Flash [Reserved in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202. Boot Segment ends at 0x0007FF in all other devices.] 001 = High security, Medium-sized Boot Program Flash [Reserved in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202. Boot Segment ends at 0x001FFF in all other devices.] 000 = High security, Large-sized Boot Program Flash [Reserved in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202. Boot Segment ends at 0x003FFF in all other devices.]
BWRP	FBS	**Boot Segment Program Memory Write Protection** 1 = Boot Segment program memory is not write-protected 0 = Boot program memory is write-protected

TABLE 3-3: dsPIC33F/PIC24H CONFIGURATION BITS DESCRIPTION (CONTINUED)

Bit Field	Register	Description
RSS<1:0>	FSS	**Secure Segment Data RAM Code Protection** 11 = No Data RAM is reserved for Secure Segment 10 = Small-sized Secure RAM [(256 – N) bytes of RAM are reserved for Secure Segment in all other devices.] 01 = Medium-sized Secure RAM [(2048 – N) bytes of RAM are reserved for Secure Segment in all other devices.] 00 = Large-sized Secure RAM [(4096 – N) bytes of RAM are reserved for Secure Segment in all other devices.] where N = Number of bytes of RAM reserved for Boot Sector. **Note 1:** This bit is Reserved in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202.] **2:** If the defined Boot Segment size is greater than or equal to the defined Secure Segment, then the Secure Segment size selection has no effect and the Secure Segment is disabled.
SSS<2:0>	FSS	**Secure Segment Program Memory Code Protection** 111 = No Secure Segment 110 = Standard security, Small-sized Secure Program Flash [Secure Segment ends at 0x001FFF for dsPIC33FJ64GPxxx/ dsPIC33FJ64MCxxx/PIC24HJ64GPxxx devices, and at 0x003FFF in other devices.] 101 = Standard security, Medium-sized Secure Program Flash [Secure Segment ends at 0x003FFF for dsPIC33FJ64GPxxx/ dsPIC33FJ64MCxxx/PIC24HJ64GPxxx devices, and at 0x007FFF in other devices.] 100 = Standard security, Large-sized Secure Program Flash [Secure Segment ends at 0x007FFF for dsPIC33FJ64GPxxx/ dsPIC33FJ64MCxxx/PIC24HJ64GPxxx devices, and at 0x00FFFF in other devices.] 011 = No Secure Segment 010 = High security, Small-sized Secure Program Flash [Secure Segment ends at 0x001FFF for dsPIC33FJ64GPxxx/ dsPIC33FJ64MCxxx/PIC24HJ64GPxxx devices, and at 0x003FFF in other devices.] 001 = High security, Medium-sized Secure Program Flash [Secure Segment ends at 0x003FFF for dsPIC33FJ64GPxxx/ dsPIC33FJ64MCxxx/PIC24HJ64GPxxx devices, and at 0x007FFF in other devices.] 000 = High security, Large-sized Secure Program Flash [Secure Segment ends at 0x007FFF for dsPIC33FJ64GPxxx/ dsPIC33FJ64MCxxx/PIC24HJ64GPxxx devices, and at 0x00FFFF in other devices.] [**Note:** This bit is Reserved in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202.]
SWRP	FSS	**Secure Segment Program Memory Write Protection** 1 = Secure Segment program memory is not write-protected 0 = Secure program memory is write-protected [**Note:** This bit is Reserved in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202.]

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TABLE 3-3: dsPIC33F/PIC24H CONFIGURATION BITS DESCRIPTION (CONTINUED)

Bit Field	Register	Description
GSS<1:0>	FGS	**General Segment Code-Protect bit** 11 = Code protection is disabled 10 = Standard security code protection is enabled 0x = Reserved in dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202. In all other devices, high security code protection is enabled.
GWRP	FGS	**General Segment Write-Protect bit** 1 = General Segment program memory is not write-protected 0 = General Segment program memory is write-protected
IESO	FOSCSEL	**Two-speed Oscillator Start-Up Enable bit** 1 = Start up device with FRC, then automatically switch to the user-selected oscillator source when ready 0 = Start up device with user-selected oscillator source
TEMP	FOSCSEL	**Temperature Protection Enable bit** 1 = Temperature protection disabled 0 = Temperature protection enabled
FNOSC<2:0>	FOSCSEL	**Initial Oscillator Source Selection bits** 111 = Internal Fast RC (FRC) oscillator 110 = Reserved 101 = LPRC oscillator 100 = Secondary (LP) oscillator 011 = Primary (XT, HS, EC) oscillator with PLL 010 = Primary (XT, HS, EC) oscillator 001 = Internal Fast RC (FRC) oscillator with PLL 000 = Reserved
FCKSM<1:0>	FOSC	**Clock Switching Mode bits** 1x = Clock switching is disabled, Fail-Safe Clock Monitor is disabled 01 = Clock switching is enabled, Fail-Safe Clock Monitor is disabled 00 = Clock switching is enabled, Fail-Safe Clock Monitor is enabled
IOL1WAY	FOSC	**Peripheral Pin Select Configuration** 1 = Allow only one reconfiguration 0 = Allow multiple reconfigurations [**Note:** This bit is only present in the dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202 devices.]
OSCIOFNC	FOSC	**OSC2 Pin Function bit (except in XT and HS modes)** 1 = OSC2 is clock output 0 = OSC2 is general purpose digital I/O pin
POSCMD<1:0>	FOSC	**Primary Oscillator Mode Select bits** 11 = Primary oscillator disabled 10 = HS crystal oscillator mode 01 = XT crystal oscillator mode 00 = EC (external clock) mode
FWDTEN	FWDT	**Watchdog Enable bit** 1 = Watchdog always enabled (LPRC oscillator cannot be disabled. Clearing the SWDTEN bit in the RCON register will have no effect) 0 = Watchdog enabled/disabled by user software (LPRC can be disabled by clearing the SWDTEN bit in the RCON register)
WINDIS	FWDT	**Watchdog Timer Window Enable bit** 1 = Watchdog Timer in Non-Window mode 0 = Watchdog Timer in Window mode
WDTPRE	FWDT	**Watchdog Timer Prescaler bit** 1 = 1:128 0 = 1:32

TABLE 3-3: dsPIC33F/PIC24H CONFIGURATION BITS DESCRIPTION (CONTINUED)

Bit Field	Register	Description
WDTPOST	FWDT	**Watchdog Timer Postscaler bits** 1111 = 1:32,768 1110 = 1:16,384 . . . 0001 = 1:2 0000 = 1:1
PWMPIN	FPOR	**Motor Control PWM Module Pin mode** 1 = PWM module pins controlled by PORT register at device Reset (tri-stated) 0 = PWM module pins controlled by PWM module at device Reset (configured as output pins)
HPOL	FPOR	**Motor Control PWM High-side Polarity bit** 1 = PWM module high-side output pins have active-high output polarity 0 = PWM module high-side output pins have active-low output polarity
LPOL	FPOR	**Motor Control PWM Low-side Polarity bit** 1 = PWM module low-side output pins have active-high output polarity 0 = PWM module low-side output pins have active-low output polarity
ALTI2C	FPOR	**Alternate I^2C™ pins** 1 = I^2C mapped to SDA1/SCL1 pins 0 = I^2C mapped to ASDA1/SACL1 pins [**Note:** This bit is only present in the dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202 devices.]
FPWRT<2:0>	FPOR	**Power-on Reset Timer Value Select bits** 111 = PWRT = 128 ms 110 = PWRT = 64 ms 101 = PWRT = 32 ms 100 = PWRT = 16 ms 011 = PWRT = 8 ms 010 = PWRT = 4 ms 001 = PWRT = 2 ms 000 = PWRT Disabled
BKBUG	FICD	**Background Debug Enable bit** 1 = Device will reset in User mode 0 = Device will reset in Debug mode
COE	FICD	**Debugger/Emulator Enable bit** 1 = Device will reset in Operational mode 0 = Device will reset in Clip-On Emulation mode
JTAGEN	FICD	**JTAG Enable bit** 1 = JTAG enabled 0 = JTAG disabled
ICS<1:0>	FICD	**ICD Communication Channel Select bits** 11 = Communicate on PGC1/EMUC1 and PGD1/EMUD1 10 = Communicate on PGC2/EMUC2 and PGD2/EMUD2 01 = Communicate on PGC3/EMUC3 and PGD3/EMUD3 00 = Reserved, do not use
—	All	Unimplemented (read as '0', write as '0')

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TABLE 3-4: dsPIC33F/PIC24H DEVICE CONFIGURATION REGISTER MAP

Address	Name	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1	Bit 0
0xF80000	FBS	RBS<1:0>(3)		—		BSS<2:0>			BWRP
0xF80002	FSS	RSS<1:0>(3)		—		SSS<2:0>(3)			SWRP(3)
0xF80004	FGS	—					GSS<1:0>		GWRP
0xF80006	FOSCSEL	IESO	—	TEMP	—		FNOSC<2:0>		
0xF80008	FOSC	FCKSM<1:0>		IOL1WAY(2)	—		OSCIOFNC	POSCMD<1:0>	
0xF8000A	FWDT	FWDTEN	WINDIS	-	WDTPRE	WDTPOST<3:0>			
0xF8000C	FPOR	PWMPIN(1)	HPOL(1)	LPOL(1)	ALTI2C(2)	—	FPWRT<2:0>		
0xF8000E	FICD	BKBUG	COE	JTAGEN	—			ICS<1:0>	
0xF80010	FUID0	User Unit ID Byte 0							
0xF80012	FUID1	User Unit ID Byte 1							
0xF80014	FUID2	User Unit ID Byte 2							
0xF80016	FUID3	User Unit ID Byte 3							

Note 1: On the dsPIC33F General Purpose Family devices (dsPIC33FJXXXGPXXX) and PIC24H devices, these bits are reserved (read as '1' and must be programmed as '1').

2: These bits are only present in the dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202 devices. In all other devices, they are unimplemented (read as '0').

3: In the dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202 devices, these bits are reserved (read as '1' and must be programmed as '1').

3.6.2 PROGRAMMING METHODOLOGY

Configuration bits may be programmed a single byte at a time using the PROGC command. This command specifies the configuration data and Configuration register address. When Configuration bits are programmed, any unimplemented bits must be programmed with a '0' and any reserved bits must be programmed with a '1'.

Twelve PROGC commands are required to program all the Configuration bits. A flowchart for Configuration bit programming is shown in Figure 3-5.

Note:	If the General Code Segment Code-Protect bit (GCP) is programmed to '0', code memory is code-protected and can not be read. Code memory must be verified before enabling read protection. See **Section 3.6.4 "CodeGuard Security Configuration Bits"** for more information about code-protect Configuration bits.

3.6.3 PROGRAMMING VERIFICATION

After the Configuration bits are programmed, the contents of memory should be verified to ensure that the programming was successful. Verification requires the Configuration bits to be read back and compared against the copy held in the programmer's buffer. The READC command reads back the programmed Configuration bits and verifies that the programming was successful.

Any unimplemented Configuration bits are read-only and read as '0'. The reserved bits are read-only and read as '1'.

3.6.4 CODEGUARD SECURITY CONFIGURATION BITS

The FBS, FSS and FGS Configuration registers are special Configuration registers that control the size and level of code protection for the Boot Segment, Secure Segment and General Segment, respectively. For each segment, two main forms of code protection are provided. One form prevents code memory from being written (write protection), while the other prevents code memory from being read (read protection).

BWRP, SWRP and GWRP bits control write protection and BSS<2:0>, SSS<2:0> and GSS<1:0> bits controls read protection. The Chip Erase ERASEB command sets all the code protection bits to '1', which allows the device to be programmed.

When write protection is enabled, any programming operation to code memory will fail. When read protection is enabled, any read from code memory will cause a '0x0' to be read, regardless of the actual contents of code memory. Since the programming executive always verifies what it programs, attempting to program code memory with read protection enabled will also result in failure.

It is imperative that all code protection bits are '1' while the device is being programmed and verified. Only after the device is programmed and verified should any of the above bits be programmed to '0'.

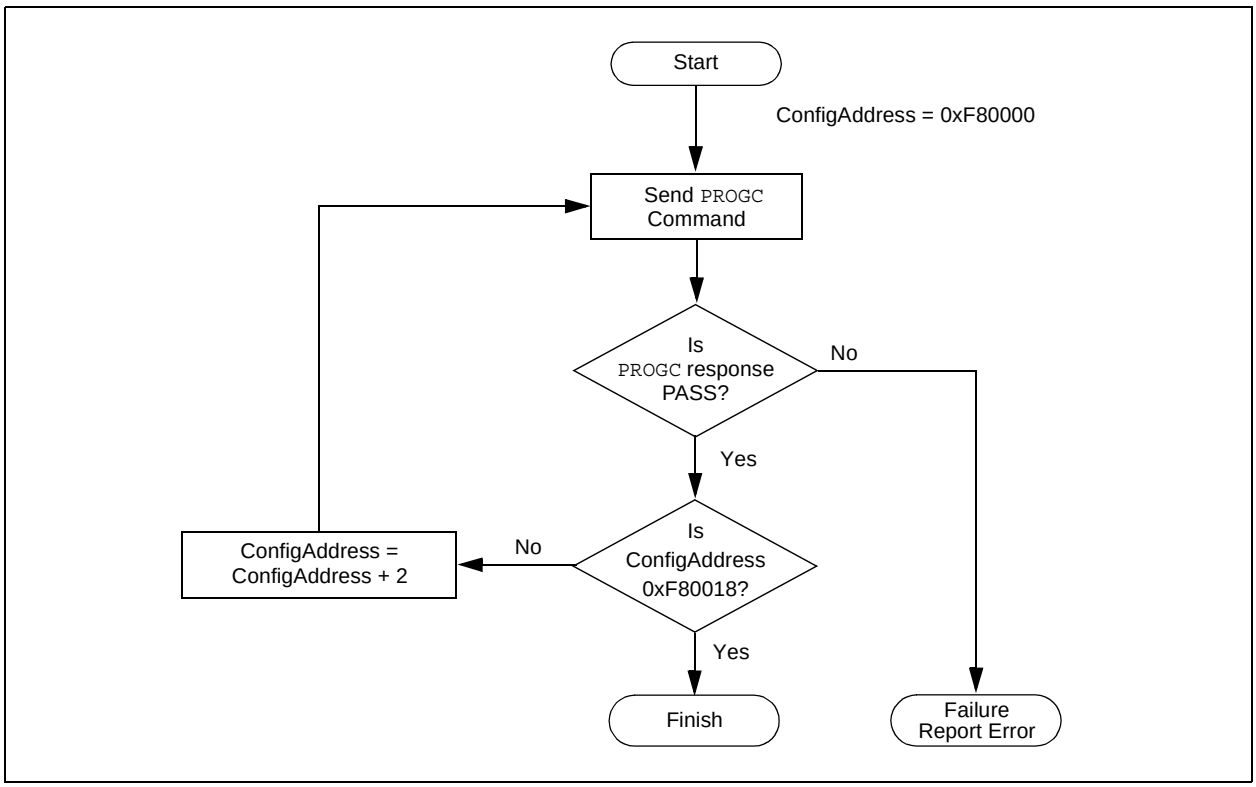
In addition to code memory protection, a part of Data RAM can be configured to be accessible only by code resident in the Boot Segment and/or Secure Segment. The sizes of these “reserved” sections are user-configurable, using the RBS<1:0> and RSS<1:0> bits.

Note:	All bits in the FBS, FSS and FGS Configuration registers can only be programmed to a value of ‘0’. the ERASEB command is the only way to reprogram code-protect bits from ON (‘0’) to OFF (‘1’).

3.6.5 USER UNIT ID

The dsPIC33F/PIC24H devices provide four 8-bit Configuration registers (FUID0 through FUID3) for the user to store product-specific information, such as unit serial numbers and other product manufacturing data.

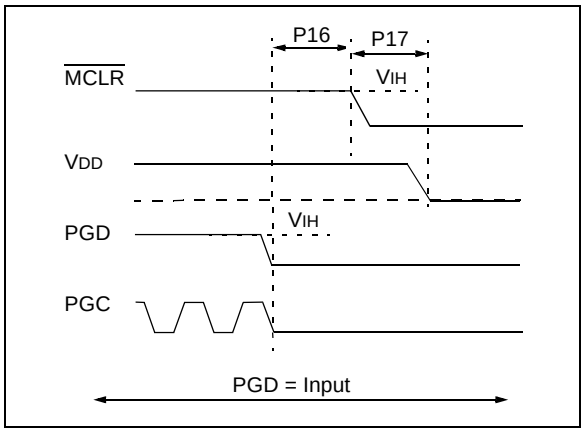
FIGURE 3-5: CONFIGURATION BIT PROGRAMMING FLOW



3.7 Exiting Enhanced ICSP Mode

Exiting Program/Verify mode is done by removing VIH from MCLR, as shown in Figure 3-6. The only requirement for exit is that an interval P16 should elapse between the last clock and program signals on PGC and PGD before removing VIH.

FIGURE 3-6: EXITING ENHANCED ICSP™ MODE



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4.0 THE PROGRAMMING EXECUTIVE

4.1 Programming Executive Communication

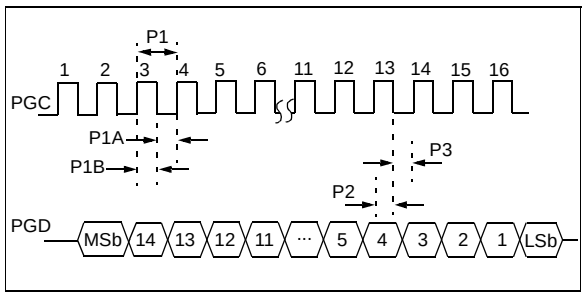
The programmer and programming executive have a master-slave relationship, where the programmer is the master programming device and the programming executive is the slave.

All communication is initiated by the programmer in the form of a command. Only one command at a time can be sent to the programming executive. In turn, the programming executive only sends one response to the programmer after receiving and processing a command. The programming executive command set is described in **Section 4.2 "Programming Executive Commands"**. The response set is described in **Section 4.3 "Programming Executive Responses"**.

4.1.1 COMMUNICATION INTERFACE AND PROTOCOL

The ICSP/Enhanced ICSP interface is a 2 wire SPI implemented using the PGC and PGD pins. The PGC pin is used as a clock input pin and the clock source must be provided by the programmer. The PGD pin is used for sending command data to and receiving response data from the programming executive. All serial data is transmitted on the falling edge of PGC and latched on the rising edge of PGC. All data transmissions are sent to the Most Significant bit (MSb) first using 16-bit mode (see Figure 4-1).

FIGURE 4-1: PROGRAMMING EXECUTIVE SERIAL TIMING



Since a 2 wire SPI is used, and data transmissions are bidirectional, a simple protocol is used to control the direction of PGD. When the programmer completes a command transmission, it releases the PGD line and allows the programming executive to drive this line high. The programming executive keeps the PGD line high to indicate that it is processing the command.

After the programming executive has processed the command, it brings PGD low for 15 µsec to indicate to the programmer that the response is available to be clocked out. The programmer can begin to clock out the response 23 µsec after PGD is brought low and it must provide the necessary amount of clock pulses to receive the entire response from the programming executive.

After the entire response is clocked out, the programmer should terminate the clock on PGC until it is time to send another command to the programming executive. This protocol is shown in Figure 4-2.

4.1.2 SPI RATE

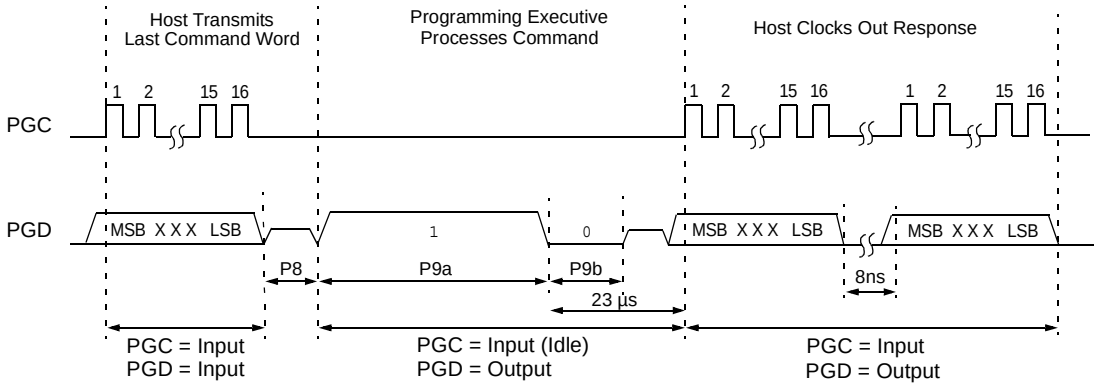
In Enhanced ICSP mode, the dsPIC33F/PIC24H family devices operate from the Fast Internal RC oscillator, which has a nominal frequency of 7.3728 MHz. This oscillator frequency yields an effective system clock frequency of 1.8432 MHz. To ensure that the programmer does not clock too fast, it is recommended that a 7.35 MHz clock be provided by the programmer.

4.1.3 TIME OUTS

The programming executive uses no Watchdog or time out for transmitting responses to the programmer. If the programmer does not follow the flow control mechanism using PGC as described in **Section 4.1.1 "Communication Interface and Protocol"**, it is possible that the programming executive will behave unexpectedly while trying to send a response to the programmer. Since the programming executive has no time out, it is imperative that the programmer correctly follow the described communication protocol.

As a safety measure, the programmer should use the command time outs identified in Table 4-1. If the command time out expires, the programmer should reset the programming executive and start programming the device again.

FIGURE 4-2: PROGRAMMING EXECUTIVE – PROGRAMMER COMMUNICATION PROTOCOL



4.2 Programming Executive Commands

The programming executive command set is shown in Table 4-1. This table contains the opcode, mnemonic, length, time out and description for each command. Functional details on each command are provided in the command descriptions (**Section 4.2.4 "Command Descriptions"**).

4.2.1 COMMAND FORMAT

All programming executive commands have a general format consisting of a 16-bit header and any required data for the command (see Figure 4-3). The 16-bit header consists of a 4-bit opcode field, which is used to identify the command, followed by a 12-bit command length field.

FIGURE 4-3: COMMAND FORMAT

15 12	11 0
Opcode	Length
Command Data First Word (if required)	
•	
•	
Command Data Last Word (if required)	

The command opcode must match one of those in the command set. Any command that is received which does not match the list in Table 4-1 will return a "NACK" response (see **Section 5.3.1.1 "Opcode Field"**).

The command length is represented in 16-bit words since the SPI operates in 16-bit mode. The programming executive uses the command length field to determine the number of words to read from the SPI port. If the value of this field is incorrect, the command will not be properly received by the programming executive.

4.2.2 PACKED DATA FORMAT

When 24-bit instruction words are transferred across the 16-bit SPI interface, they are packed to conserve space using the format shown in Figure 4-4. This format minimizes traffic over the SPI and provides the programming executive with data that is properly aligned for performing table write operations.

FIGURE 4-4: PACKED INSTRUCTION WORD FORMAT

15 8	7 0
LSW1	
MSB2	MSB1
LSW2	

LSWx: Least Significant 16 bits of instruction word
MSBx: Most Significant Byte of instruction word

Note: When the number of instruction words transferred is odd, MSB2 is zero and LSW2 can not be transmitted.

4.2.3 PROGRAMMING EXECUTIVE ERROR HANDLING

The programming executive will "NACK" all unsupported commands. Additionally, due to the memory constraints of the programming executive, no checking is performed on the data contained in the programmer command. It is the responsibility of the programmer to command the programming executive with valid command arguments or the programming operation may fail. Additional information on error handling is provided in **Section 5.3.1.3 "QE_Code Field"**.

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TABLE 4-1: PROGRAMMING EXECUTIVE COMMAND SET

Opcode	Mnemonic	Length (16-bit words)	Time Out	Description
0x0	SCHECK	1	1 msec	Sanity check.
0x1	READC	3	1 msec	Read an 8-bit word from the specified Configuration register or Device ID register.
0x2	READP	4	1 msec/row	Read 'N' 24-bit instruction words of code memory starting from the specified address.
0x3	RESERVED	N/A	N/A	This command is reserved. It will return a NACK.
0x4	PROGC	4	5 msec	Write an 8-bit word to the specified Configuration register.
0x5	PROGP	99	5 msec	Program one row of code memory at the specified address, then verify.
0x6	PROGW	5	5 msec	Program one instruction word of code memory at the specified address, then verify.
0x7	RESERVED	N/A	N/A	This command is reserved. It will return a NACK.
0x8	RESERVED	N/A	N/A	This command is reserved. It will return a NACK.
0x9	RESERVED	N/A	N/A	This command is reserved. It will return a NACK.
0xA	QBLANK	2	TBD	Query if the code memory is blank.
0xB	QVER	1	1 msec	Query the programming executive software version.
0xC	RESERVED	N/A	N/A	This command is reserved. It will return a NACK.
0xD	RESERVED	N/A	N/A	This command is reserved. It will return a NACK.

Legend: TBD = To Be Determined

Note: One row of code memory consists of (64) 24-bit words. Refer to Table 2-2 for device-specific information.

4.2.4 COMMAND DESCRIPTIONS

All commands supported by the programming executive are described in **Section 5.2.5 "SCHECK Command"** through **Section 4.2.12 "QVER Command"**.

4.2.5 SCHECK COMMAND

15 12	11 0
Opcode	Length

Field	Description
Opcode	0x0
Length	0x1

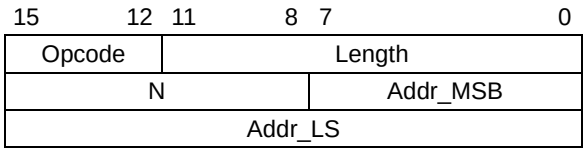
The SCHECK command instructs the programming executive to do nothing but generate a response. This command is used as a "Sanity Check" to verify that the programming executive is operational.

Expected Response (2 words):

0x1000
0x0002

Note:	This instruction is not required for programming, but is provided for development purposes only.

4.2.6 READC COMMAND



Field	Description
Opcode	0x1
Length	0x3
N	Number of 8-bit Configuration registers or Device ID registers to read (max of 256)
Addr_MSB	MSB of 24-bit source address
Addr_LS	Least Significant 16 bits of 24-bit source address

The READC command instructs the programming executive to read N Configuration registers or Device ID registers, starting from the 24-bit address specified by Addr_MSB and Addr_LS. This command can only be used to read 8-bit or 16-bit data.

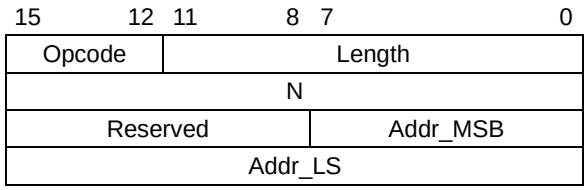
When this command is used to read Configuration registers, the upper byte in every data word returned by the programming executive is 0x00 and the lower byte contains the Configuration register value.

Expected Response (4 + 3 * (N – 1)/2 words for N odd):

0x1100
2 + N
Configuration register or Device ID Register 1
...
Configuration register or Device ID Register N

Note:	Reading unimplemented memory will cause the programming executive to reset. Please ensure that only memory locations present on a particular device are accessed.

4.2.7 READP COMMAND



Field	Description
Opcode	0x2
Length	0x4
N	Number of 24-bit instructions to read (max of 32768)
Reserved	0x0
Addr_MSB	MSB of 24-bit source address
Addr_LS	Least Significant 16 bits of 24-bit source address

The READP command instructs the programming executive to read N 24-bit words of code memory, starting from the 24-bit address specified by Addr_MSB and Addr_LS. This command can only be used to read 24-bit data. All data returned in the response to this command uses the packed data format described in **Section 4.2.2 “Packed Data Format”**.

Expected Response (2 + 3 * N/2 words for N even):

0x1200
2 + 3 * N/2
Least significant program memory word 1
...
Least significant data word N

Expected Response (4 + 3 * (N – 1)/2 words for N odd):

0x1200
4 + 3 * (N – 1)/2
Least significant program memory word 1
...
MSB of program memory word N (zero padded)

Note:	Reading unimplemented memory will cause the programming executive to reset. Please ensure that only memory locations present on a particular device are accessed.

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4.2.8 PROGC COMMAND

15 12	11 8	7 0
Opcode	Length	
Reserved		Addr_MSB
Addr_LS		
Data		

Field	Description
Opcode	0x4
Length	0x4
Reserved	0x0
Addr_MSB	MSB of 24-bit destination address
Addr_LS	Least Significant 16 bits of 24-bit destination address
Data	8-bit data word

The PROGC command instructs the programming executive to program a single Configuration register, located at the specified memory address.

After the specified data word has been programmed to code memory, the programming executive verifies the programmed data against the data in the command.

Expected Response (2 words):
0x1400
0x0002

4.2.9 PROGP COMMAND

15 12	11 8	7 0
Opcode	Length	
Reserved		Addr_MSB
Addr_LS		
D_1		
D_2		
...		
D_N		

Field	Description
Opcode	0x5
Length	0x63
Reserved	0x0
Addr_MSB	MSB of 24-bit destination address
Addr_LS	Least Significant 16 bits of 24-bit destination address
D_1	16-bit data word 1
D_2	16-bit data word 2
...	16-bit data word 3 through 95
D_96	16-bit data word 96

The PROGP command instructs the programming executive to program one row of code memory (64 instruction words) to the specified memory address. Programming begins with the row address specified in the command. The destination address should be a multiple of 0x80.

The data to program to memory, located in command words D_1 through D_96, must be arranged using the packed instruction word format shown in Figure 4-4.

After all data has been programmed to code memory, the programming executive verifies the programmed data against the data in the command.

Expected Response (2 words):
0x1500
0x0002

Note: Refer to Table 2-2 for code memory size information.

4.2.10 PROGW COMMAND

15 12	11 8	7 0
Opcode	Length	
Reserved		Addr_MSB
Addr_LS		
Data_LS		
Reserved		Data_MSB

Field	Description
Opcode	0x6
Length	0x5
Reserved	0x0
Addr_MSB	MSB of 24-bit destination address
Addr_LS	Least Significant 16 bits of 24-bit destination address
Data_MSB	MSB of 24-bit data
Data_LS	Least Significant 16 bits of 24-bit data

The PROGW command instructs the programming executive to program one word of code memory (3 bytes) to the specified memory address.

After the word has been programmed to code memory, the programming executive verifies the programmed data against the data in the command.

Expected Response (2 words):

0x1600
0x0002

4.2.11 QBLANK COMMAND

15 12	11 0
Opcode	Length
PSize	

Field	Description
Opcode	0xA
Length	0x2
PSize	Length of program memory to check (in 24-bit words) +1, up to a max of 49152

The QBLANK command queries the programming executive to determine if the contents of code memory are blank (contains all '1's). The size of code memory to check must be specified in the command.

The Blank Check for code memory begins at 0x0 and advances toward larger addresses for the specified number of instruction words.

QBLANK returns a QE_Code of 0xF0 if the specified code memory and code-protect bits are blank; otherwise, QBLANK returns a QE_Code of 0x0F.

Expected Response (2 words for blank device):

0x1AF0
0x0002

Expected Response (2 words for non-blank device):

0x1A0F
0x0002

Note:	The QBLANK command does not check the system operation Configuration bits since these bits are not set to '1' when a Chip Erase is performed.

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4.2.12 QVER COMMAND

15 12	11 0
Opcode	Length

Field	Description
Opcode	0xB
Length	0x1

The QVER command queries the version of the programming executive software stored in test memory. The “version.revision” information is returned in the response’s QE_Code using a single byte with the following format: main version in upper nibble and revision in the lower nibble (i.e., 0x23 means version 2.3 of programming executive software).

Expected Response (2 words):

0x1BMN (where “MN” stands for version M.N)
0x0002

4.3 Programming Executive Responses

The programming executive sends a response to the programmer for each command that it receives. The response indicates if the command was processed correctly. It includes any required response data or error data.

The programming executive response set is shown in Table 4-2. This table contains the opcode, mnemonic and description for each response. The response format is described in **Section 4.3.1 “Response Format”**.

TABLE 4-2: PROGRAMMING EXECUTIVE RESPONSE OPCODES

Opcode	Mnemonic	Description
0x1	PASS	Command successfully processed.
0x2	FAIL	Command unsuccessfully processed.
0x3	NACK	Command not known.

4.3.1 RESPONSE FORMAT

All programming executive responses have a general format consisting of a two-word header and any required data for the command.

15 12	11 8	7 0
Opcode	Last_Cmd	QE_Code
Length		
D_1 (if applicable)		
...		
D_N (if applicable)		

Field	Description
Opcode	Response opcode.
Last_Cmd	Programmer command that generated the response.
QE_Code	Query code or error code.
Length	Response length in 16-bit words (includes 2 header words).
D_1	First 16-bit data word (if applicable).
D_N	Last 16-bit data word (if applicable).

4.3.1.1 Opcode Field

The opcode is a 4-bit field in the first word of the response. The opcode indicates how the command was processed (see Table 4-2). If the command was processed successfully, the response opcode is PASS. If there was an error in processing the command, the response opcode is FAIL and the QE_Code indicates the reason for the failure. If the command sent to the programming executive is not identified, the programming executive returns a NACK response.

4.3.1.2 Last_Cmd Field

The Last_Cmd is a 4-bit field in the first word of the response and indicates the command that the programming executive processed. Since the programming executive can only process one command at a time, this field is technically not required. However, it can be used to verify that the programming executive correctly received the command that the programmer transmitted.

4.3.1.3 QE_Code Field

The QE_Code is a byte in the first word of the response. This byte is used to return data for query commands and error codes for all other commands.

When the programming executive processes one of the two query commands (QBLANK or QVER), the returned opcode is always PASS and the QE_Code holds the query response data. The format of the QE_Code for both queries is shown in Table 4-3.

TABLE 4-3: QE_Code FOR QUERIES

Query	QE_Code
QBLANK	0x0F = Code memory is NOT blank 0xF0 = Code memory is blank
QVER	0xMN, where programming executive software version = M.N (i.e., 0x32 means software version 3.2).

When the programming executive processes any command other than a Query, the QE_Code represents an error code. Supported error codes are shown in Table 4-4. If a command is successfully processed, the returned QE_Code is set to 0x0, which indicates that there was no error in the command processing. If the verify of the programming for the PROGP or PROGC command fails, the QE_Code is set to 0x1. For all other programming executive errors, the QE_Code is 0x2.

TABLE 4-4: QE_Code FOR NON-QUERY COMMANDS

QE_Code	Description
0x0	No error.
0x1	Verify failed.
0x2	Other error.

4.3.1.4 Response Length

The response length indicates the length of the programming executive's response in 16-bit words. This field includes the 2 words of the response header.

With the exception of the response for the READP command, the length of each response is only 2 words.

The response to the READP command uses the packed instruction word format described in **Section 4.2.2 "Packed Data Format"**. When reading an odd number of program memory words (N odd), the response to the READP command is (3 * (N + 1) / 2 + 2) words. When reading an even number of program memory words (N even), the response to the READP command is (3 * N / 2 + 2) words.

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5.0 DEVICE PROGRAMMING – ICSP

ICSP mode is a special programming protocol that allows you to read and write to dsPIC33F/PIC24H device family memory. The ICSP mode is the most direct method used to program the device; note, however, that Enhanced ICSP is faster. ICSP mode also has the ability to read the contents of executive memory to determine if the programming executive is present. This capability is accomplished by applying control codes and instructions serially to the device using pins PGC and PGD.

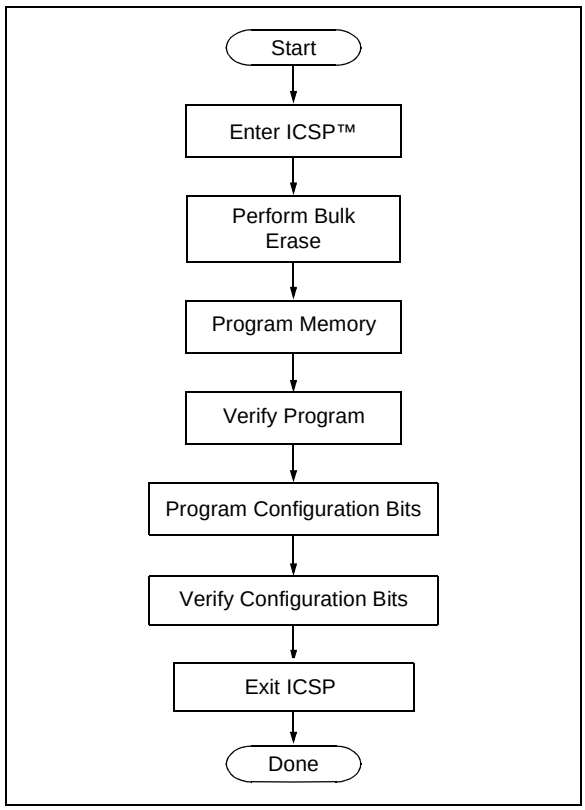
In ICSP mode, the system clock is taken from the PGC pin, regardless of the device's oscillator Configuration bits. All instructions are shifted serially into an internal buffer, then loaded into the instruction register and executed. No program fetching occurs from internal memory. Instructions are fed in 24 bits at a time. PGD is used to shift data in, and PGC is used as both the serial shift clock and the CPU execution clock.

Note:	During ICSP operation, the operating frequency of PGC must not exceed 5 MHz.

5.1 Overview of the Programming Process

Figure 5-1 shows the high-level overview of the programming process. After entering ICSP mode, the first action is to Bulk Erase the device. Next, the code memory is programmed, followed by the device Configuration registers. Code memory (including the Configuration registers) is then verified to ensure that programming was successful. Then, program the code-protect Configuration bits, if required.

FIGURE 5-1: HIGH-LEVEL ICSP™ PROGRAMMING FLOW



5.2 ICSP Operation

Upon entry into ICSP mode, the CPU is Idle. Execution of the CPU is governed by an internal state machine. A 4-bit control code is clocked in using PGC and PGD and this control code is used to command the CPU (see Table 5-1).

The SIX control code is used to send instructions to the CPU for execution and the REGOUT control code is used to read data out of the device via the VISI register.

TABLE 5-1: CPU CONTROL CODES IN ICSP™ MODE

4-Bit Control Code	Mnemonic	Description
0000b	SIX	Shift in 24-bit instruction and execute.
0001b	REGOUT	Shift out the VISI register.
0010b-1111b	N/A	Reserved.

5.2.1 SIX SERIAL INSTRUCTION EXECUTION

The SIX control code allows execution of dsPIC33F/PIC24H Programming Specification assembly instructions. When the SIX code is received, the CPU is suspended for 24 clock cycles, as the instruction is then clocked into the internal buffer. Once the instruction is shifted in, the state machine allows it to be executed over the next four clock cycles. While the received instruction is executed, the state machine simultaneously shifts in the next 4-bit command (see Figure 5-2).

Note 1:	Coming out of Reset, the first 4-bit control code is always forced to SIX and a forced `NOP` instruction is executed by the CPU. Five additional PGC clocks are needed on start-up, thereby resulting in a 9-bit SIX command instead of the normal 4-bit SIX command. After the forced SIX is clocked in, ICSP operation resumes as normal (the next 24 clock cycles load the first instruction word to the CPU).
2:	`TBLRDH`, `TBLRDL`, `TBLWTH` and `TBLWTL` instructions must be followed by a `NOP` instruction.

5.2.2 REGOUT SERIAL INSTRUCTION EXECUTION

The REGOUT control code allows for data to be extracted from the device in ICSP mode. It is used to clock the contents of the VISI register out of the device over the PGD pin. After the REGOUT control code is received, the CPU is held Idle for 8 cycles. After these eight cycles, an additional 16 cycles are required to clock the data out (see Figure 5-3).

The REGOUT code is unique because the PGD pin is an input when the control code is transmitted to the device. However, after the control code is processed, the PGD pin becomes an output as the VISI register is shifted out.

Note:	Data is transmitted on the falling edge and latched on the rising edge of PGC. For all data transmissions, the Least Significant bit (LSb) is transmitted first.

FIGURE 5-2: SIX SERIAL EXECUTION

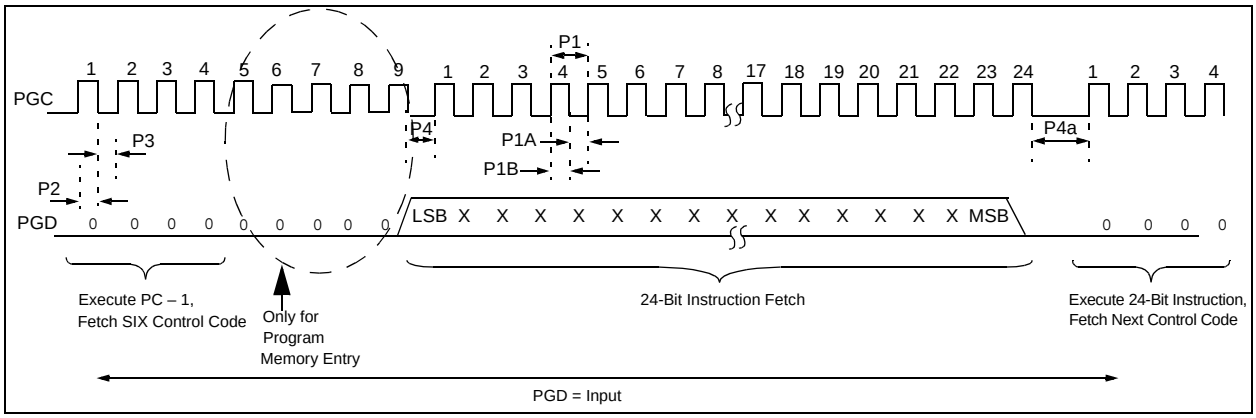
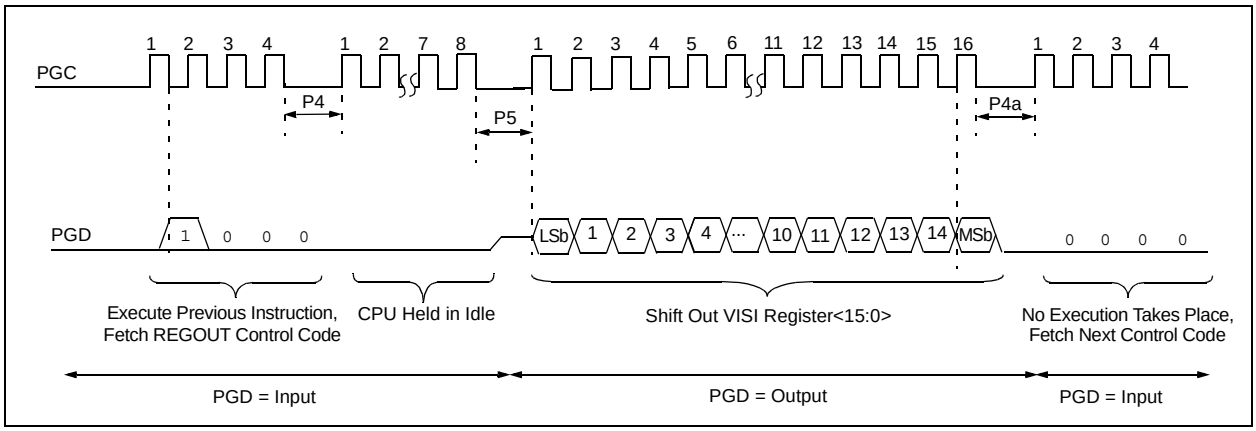


FIGURE 5-3: REGOUT SERIAL EXECUTION



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5.3 Entering ICSP Mode

As shown in Figure 5-4, entering ICSP Program/Verify mode requires three steps:

1. $\overline{\text{MCLR}}$ is briefly driven high then low.
2. A 32-bit key sequence is clocked into PGD.
3. $\overline{\text{MCLR}}$ is then driven high within a specified period of time and held.

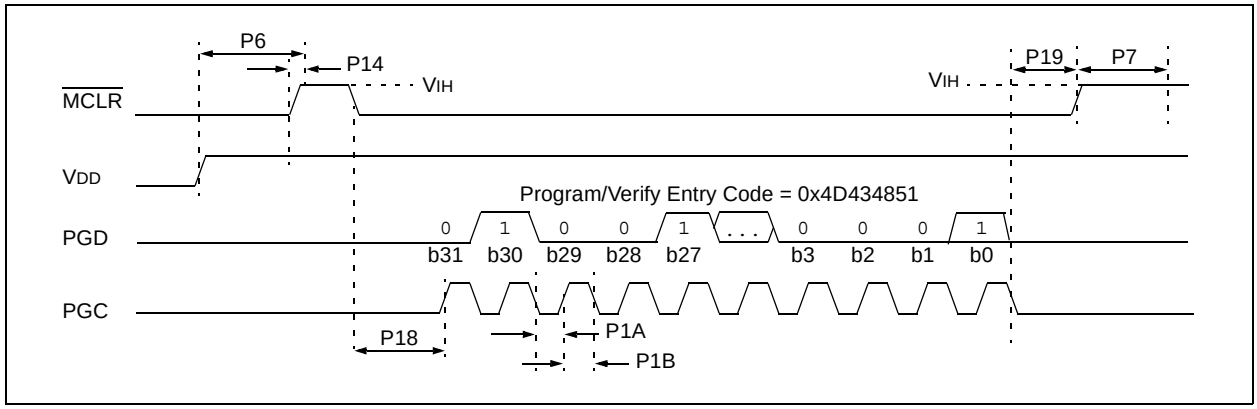
The programming voltage applied to $\overline{\text{MCLR}}$ is V_{IH}, which is essentially V_{DD} in the case of dsPIC33F/PIC24H devices. There is no minimum time requirement for holding at V_{IH}. After V_{IH} is removed, an interval of at least P18 must elapse before presenting the key sequence on PGD.

The key sequence is a specific 32-bit pattern, '0100 1101 0100 0011 0100 1000 0101 0001' (more easily remembered as 0x4D434851 in hexadecimal). The device will enter Program/Verify mode only if the sequence is valid. The Most Significant bit (MSb) of the most significant nibble must be shifted in first.

Once the key sequence is complete, V_{IH} must be applied to $\overline{\text{MCLR}}$ and held at that level for as long as Program/Verify mode is to be maintained. An interval of at least time P19 and P7 must elapse before presenting data on PGD. Signals appearing on PGD before P7 has elapsed will not be interpreted as valid.

On successful entry, the program memory can be accessed and programmed in serial fashion. While in ICSP mode, all unused I/Os are placed in the high-impedance state.

FIGURE 5-4: ENTERING ICSP™ MODE



5.4 Flash Memory Programming in ICSP Mode

5.4.1 PROGRAMMING OPERATIONS

Flash memory write and erase operations are controlled by the NVMCON register. Programming is performed by setting NVMCON to select the type of erase operation (Table 5-2) or write operation (Table 5-3) and initiating the programming by setting the WR control bit (NVMCON<15>).

In ICSP mode, all programming operations are self-timed. There is an internal delay between the user setting the WR control bit and the automatic clearing of the WR control bit when the programming operation is complete. Please refer to **Section TABLE 8-1: "AC/DC Characteristics and Timing Requirements"** for information about the delays associated with various programming operations.

TABLE 5-2: NVMCON ERASE OPERATIONS

NVMCON Value	Erase Operation
0x404F	Erase all code memory, executive memory and Configuration registers (does not erase Unit ID or Device ID registers).
0x404D	Erase General Segment and FGS Configuration register.
0x404C	Erase Secure Segment and FSS Configuration register. This operation will also erase the General Segment and FGS Configuration register.
0x4042	Erase a page of code memory or executive memory.
0x4040	Erase a Configuration register byte.

TABLE 5-3: NVMCON WRITE OPERATIONS

NVMCON Value	Write Operation
0x4001	Program 1 row (64 instruction words) of code memory or executive memory.
0x4000	Write a Configuration register byte.
0x4003	Program a code memory word.

5.4.2 STARTING AND STOPPING A PROGRAMMING CYCLE

The WR bit (NVMCON<15>) is used to start an erase or write cycle. Setting the WR bit initiates the programming cycle.

All erase and write cycles are self-timed. The WR bit should be polled to determine if the erase or write cycle has been completed. Starting a programming cycle is performed as follows:

```
BSET    NVMCON, #WR
```

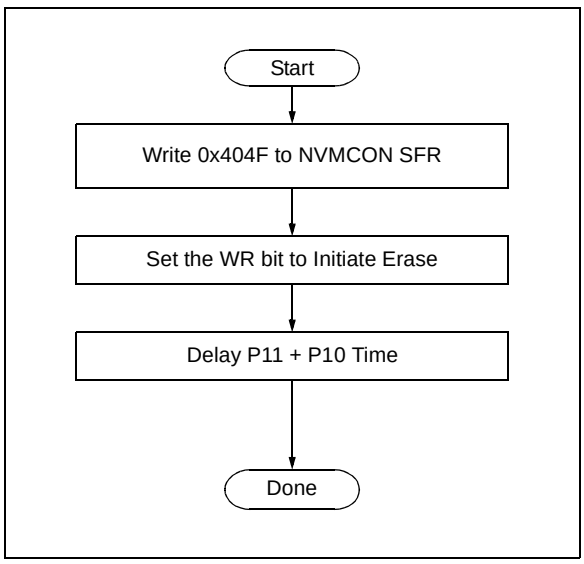
5.5 Erasing Program Memory

The procedure for erasing program memory (all of code memory, data memory, executive memory and code-protect bits) consists of setting NVMCON to 0x404F and then executing the programming cycle. For segment erase operations, the NVMCON value should be modified suitably, according to Table 5-2.

Figure 5-5 shows the ICSP programming process for Bulk Erasing program memory. This process includes the ICSP command code, which must be transmitted (for each instruction) Least Significant bit first, using the PGC and PGD pins (see Figure 5-2).

Note:	Program memory must be erased before writing any data to program memory.

FIGURE 5-5: BULK ERASE FLOW



If a Segment Erase operation is required, Step 3 must be modified with the appropriate NVMCON value as per Table 5-2.

The ability to individually erase various segments is a critical component of the CodeGuard™ Security features on dsPIC33F/PIC24H devices. An individual code segment may be erased without affecting other segments. In addition, the Configuration register corresponding to the erased code segment also gets erased. For example, the user might want to erase the code in the General Segment without erasing a Boot Loader located in Boot Segment.

The Secure Segment Erase command is used to erase the Secure Segment and the FSS Configuration register. The General Segment Erase command is used to erase the General Segment and the FGS Configuration register. This command is only effective if a Boot Segment or Secure Segment has been enabled.

Note 1:	The Boot Segment and FBS Configuration register can only be erased using a Bulk Erase.
2:	A Secure Segment Erase operation also erases the General Segment and FGS Configuration register. This is true even if Secure Segment is present on a device but not enabled.

Before performing any segment erase operation, the programmer must first determine if the dsPIC33F/PIC24H device has defined a Boot Segment or Secure Segment, and ensure that a segment does not get overwritten by operations on any other segment. Also, a Bulk Erase should not be performed if a Boot Segment or Secure Segment has been defined.

The BSS bit field in the FBS configuration register can be read to determine whether a Boot Segment has been defined. If a Boot Segment has already been defined (and probably already been programmed), the user must be warned about this fact. Similarly, the SSS bit field in the FSS configuration register can be read to determine whether a Secure Segment has been defined. If a Secure Segment has already been defined (and probably already been programmed), the user must be warned about this fact.

A Bulk Erase operation is the recommended mechanism to allow a user to overwrite the Boot Segment (if one chooses to do so).

In general, the segments and CodeGuard Security-related configuration registers should be programmed in the following order:

- FBS and Boot Segment
- FSS and Secure Segment
- FGS and General Segment

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TABLE 5-4: SERIAL INSTRUCTION EXECUTION FOR BULK ERASING CODE MEMORY

Command (Binary)	Data (Hex)	Description
Step 1: Exit the Reset vector.		
0000	000000	NOP
0000	000000	NOP
0000	040200	GOTO 0x200
0000	000000	NOP
Step 2: Set the NVMCON to erase all program memory.		
0000	2404FA	MOV #0x404F, W10
0000	883B0A	MOV W10, NVMCON
Step 3: Initiate the erase cycle.		
0000	A8E761	BSET NVMCON, #WR
0000	000000	NOP
0000	000000	NOP
Step 4: Wait for Bulk Erase operation to complete and make sure WR bit is clear.		
-	-	Externally time 'P11' msec (see **Section TABLE 8-1: "AC/DC Characteristics and Timing Requirements"**) to allow sufficient time for the Bulk Erase operation to complete.
0000	807600	MOV NVMCON, W0
0000	887840	MOV W0, VISI
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register. Repeat until the WR bit is clear.

5.6 Writing Code Memory

The procedure for writing code memory is similar to the procedure for writing the Configuration registers, except that 64 instruction words are programmed at a time. To facilitate this operation, working registers, W0:W5, are used as temporary holding registers for the data to be programmed.

Table 5-5 shows the ICSP programming details, including the serial pattern with the ICSP command code, which must be transmitted Least Significant bit first using the PGC and PGD pins (see Figure 5-2). In Step 1, the Reset vector is exited. In Step 2, the NVMCON register is initialized for programming of code memory. In Step 3, the 24-bit starting destination address for programming is loaded into the TBLPAG register and W7 register. The upper byte of the starting destination address is stored in TBLPAG and the lower 16 bits of the destination address are stored in W7.

To minimize the programming time, the same packed instruction format that the programming executive uses is utilized (Figure 4-4). In Step 4, four packed instruction words are stored in working registers, W0:W5, using the MOV instruction and the read pointer, W6, is initialized. The contents of W0:W5 holding the packed instruction word data are shown in Figure 5-6. In Step 5, eight TBLWT instructions are used to copy the data from W0:W5 to the write latches of code memory. Since code memory is programmed 64 instruction words at a time, Steps 4 and 5 are repeated 16 times to load all the write latches (Step 6).

After the write latches are loaded, programming is initiated by writing to the NVMCON register in Steps 7 and 8. In Step 9, the internal PC is reset to 0x200. This is a precautionary measure to prevent the PC from incrementing into unimplemented memory when large devices are being programmed. Lastly, in Step 10, Steps 3-9 are repeated until all of code memory is programmed.

FIGURE 5-6: PACKED INSTRUCTION WORDS IN W0:W5

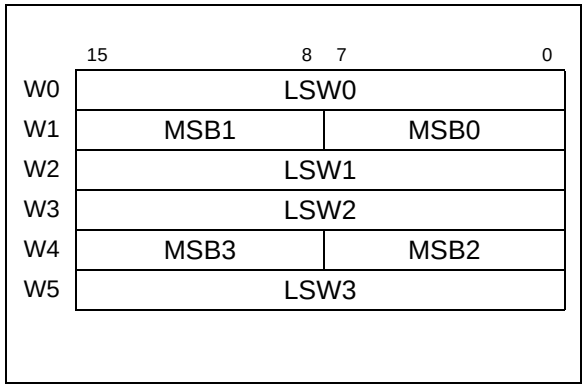


TABLE 5-5: SERIAL INSTRUCTION EXECUTION FOR WRITING CODE MEMORY

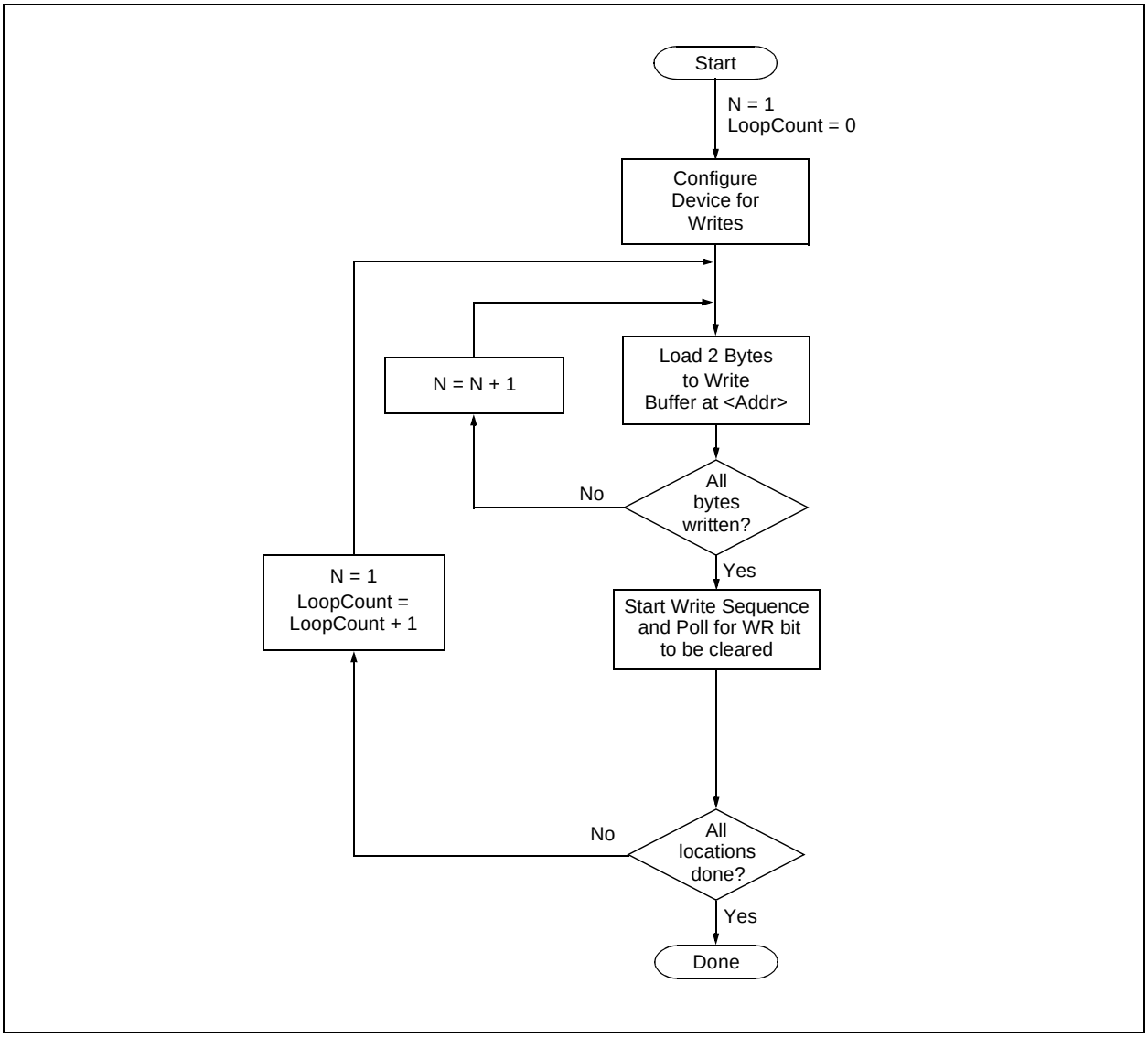
Command (Binary)	Data (Hex)	Description
Step 1: Exit the Reset vector.		
0000	000000	NOP
0000	000000	NOP
0000	040200	GOTO 0x200
0000	000000	NOP
Step 2: Set the NVMCON to program 64 instruction words.		
0000	24001A	MOV #0x4001, W10
0000	883B0A	MOV W10, NVMCON
Step 3: Initialize the write pointer (W7) for TBLWT instruction.		
0000	200xx0	MOV #<DestinationAddress23:16>, W0
0000	880190	MOV W0, TBLPAG
0000	2xxxx7	MOV #<DestinationAddress15:0>, W7
Step 4: Initialize the read pointer (W6) and load W0:W5 with the next 4 instruction words to program.		
0000	2xxxx0	MOV #<LSW0>, W0
0000	2xxxx1	MOV #<MSB1:MSB0>, W1
0000	2xxxx2	MOV #<LSW1>, W2
0000	2xxxx3	MOV #<LSW2>, W3
0000	2xxxx4	MOV #<MSB3:MSB2>, W4
0000	2xxxx5	MOV #<LSW3>, W5
Step 5: Set the read pointer (W6) and load the (next set of) write latches.		
0000	EB0300	CLR W6
0000	000000	NOP
0000	BB0BB6	TBLWTL [W6++], [W7]
0000	000000	NOP
0000	000000	NOP
0000	BBDBB6	TBLWTH.B [W6++], [W7++]
0000	000000	NOP
0000	000000	NOP
0000	BBEBB6	TBLWTH.B [W6++], [++W7]
0000	000000	NOP
0000	000000	NOP
0000	BB1BB6	TBLWTL [W6++], [W7++]
0000	000000	NOP
0000	000000	NOP
0000	BB0BB6	TBLWTL [W6++], [W7]
0000	000000	NOP
0000	000000	NOP
0000	BBDBB6	TBLWTH.B [W6++], [W7++]
0000	000000	NOP
0000	000000	NOP
0000	BBEBB6	TBLWTH.B [W6++], [++W7]
0000	000000	NOP
0000	000000	NOP
0000	BB1BB6	TBLWTL [W6++], [W7++]
0000	000000	NOP
0000	000000	NOP
Step 6: Repeat steps 4-5 sixteen times to load the write latches for 64 instructions.		
Step 7: Initiate the write cycle.		
0000	A8E761	BSET NVMCON, #WR
0000	000000	NOP
0000	000000	NOP
Step 8: Wait for Row Program operation to complete and make sure WR bit is clear.		

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TABLE 5-5: SERIAL INSTRUCTION EXECUTION FOR WRITING CODE MEMORY (CONTINUED)

Command (Binary)	Data (Hex)	Description
-	-	Externally time 'P13' msec (see **Section TABLE 8-1: "AC/DC Characteristics and Timing Requirements"**) to allow sufficient time for the Row Program operation to complete.
0000	807600	MOV NVMCON, W0
0000	887840	MOV W0, VISI
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register. Repeat until the WR bit is clear.
Step 9: Reset device internal PC.		
0000	040200	GOTO 0x200
0000	000000	NOP
Step 10: Repeat steps 3-9 until all code memory is programmed.		

FIGURE 5-7: PROGRAM CODE MEMORY FLOW



5.7 Writing Configuration Memory

The 8-bit Configuration registers are programmable, one register at a time. The default programming values recommended for the Configuration registers are shown in Table 5-6 and Table 5-7. The recommended default FOSCSEL value is 0x07, which selects the FRC clock oscillator setting.

The FBS, FSS and FGS Configuration registers are special since they enable code protection for the device. For security purposes, once any bit in these registers is programmed to '0' (to enable code protection), it can only be set back to '1' by performing a Bulk Erase as described in **Section 5.5 "Erasing Program Memory"**. Programming any of these bits from a '0' to '1' is not possible, but they may be programmed from a '1' to a '0' to enable code protection.

Table 5-8 shows the ICSP programming details for clearing the Configuration registers. In Step 1, the Reset vector is exited. In Step 2, the write pointer (W7) is loaded with 0x0000, which is the original destination address (in TBLPAG, 0xF8 of program memory). In Step 3, the NVMCON is set to program one Configuration register. In Step 4, the TBLPAG register is initialized to 0xF8 for writing to the Configuration registers. In Step 5, the value to write to each Configuration register is loaded to W0. In Step 6, the Configuration register data is written to the write latch using the TBLWTL instruction. In Steps 7 and 8, the programming cycle is initiated. In Step 9, the internal PC is set to 0x200 as a safety measure to prevent the PC from incrementing into unimplemented memory. Lastly, Steps 4-9 are repeated until all twelve Configuration registers are written.

TABLE 5-6: DEFAULT CONFIGURATION REGISTER VALUES FOR dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 AND PIC24HJ12GP201/202

Address	Name	Default Value
0xF80000	FBS	0xCF
0xF80002	FSS	0xFF
0xF80004	FGS	0x07
0xF80006	FOSCSEL	0xA7
0xF80008	FOSC	0xE7
0xF8000A	FWDT	0xDF
0xF8000C	FPOR	0xF7
0xF8000E	FICD	0xE3
0xF80010	FUID0	0xFF
0xF80012	FUID1	0xFF
0xF80014	FUID2	0xFF
0xF80016	FUID3	0xFF

TABLE 5-7: DEFAULT CONFIGURATION REGISTER VALUES FOR ALL OTHER DEVICES

Address	Name	Default Value
0xF80000	FBS	0xCF
0xF80002	FSS	0xCF
0xF80004	FGS	0x07
0xF80006	FOSCSEL	0xA7
0xF80008	FOSC	0xC7
0xF8000A	FWDT	0xDF
0xF8000C	FPOR	0xE7
0xF8000E	FICD	0xE3
0xF80010	FUID0	0xFF
0xF80012	FUID1	0xFF
0xF80014	FUID2	0xFF
0xF80016	FUID3	0xFF

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TABLE 5-8: SERIAL INSTRUCTION EXECUTION FOR WRITING CONFIGURATION REGISTERS

Command (Binary)	Data (Hex)	Description
Step 1: Exit the Reset vector.		
0000	000000	NOP
0000	000000	NOP
0000	040200	GOTO 0x200
0000	000000	NOP
Step 2: Initialize the write pointer (W7) for the TBLWT instruction.		
0000	200007	MOV #0x0000, W7
Step 3: Set the NVMCON register to program one Configuration register.		
0000	24000A	MOV #0x4000, W10
0000	883B0A	MOV W10, NVMCON
Step 4: Initialize the TBLPAG register.		
0000	200F80	MOV #0xF8, W0
0000	880190	MOV W0, TBLPAG
Step 5: Load the Configuration register data to W6.		
0000	2xxxx0	MOV #<CONFIG_VALUE>, W0
Step 6: Write the Configuration register data to the write latch and increment the write pointer.		
0000	BB1B96	TBLWTL W0, [W7++]
0000	000000	NOP
0000	000000	NOP
Step 7: Initiate the write cycle.		
0000	A8E761	BSET NVMCON, #WR
0000	000000	NOP
0000	000000	NOP
Step 8: Wait for the Configuration Register Write operation to complete and make sure WR bit is clear.		
-	-	Externally time 'P20' msec (see **Section TABLE 8-1: "AC/DC Characteristics and Timing Requirements"**) to allow sufficient time for the Configuration Register Write operation to complete.
0000	807600	MOV, NVMCON, W0
0000	887840	MOV W0, VISI
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register. Repeat until the WR bit is clear.
Step 9: Reset device internal PC.		
0000	040200	GOTO 0x200
0000	000000	NOP
Step 10: Repeat steps 5-9 until all twelve Configuration registers are written.		

5.8 Reading Code Memory

Reading from code memory is performed by executing a series of TBLRD instructions and clocking out the data using the REGOUT command.

Table 5-9 shows the ICSP programming details for reading code memory. In Step 1, the Reset vector is exited. In Step 2, the 24-bit starting source address for reading is loaded into the TBLPAG register and W6 register. The upper byte of the starting source address is stored in TBLPAG and the lower 16 bits of the source address are stored in W6.

To minimize the reading time, the packed instruction word format that was utilized for writing is also used for reading (see Figure 5-6). In Step 3, the write pointer, W7, is initialized. In Step 4, two instruction words are read from code memory and clocked out of the device, through the VISI register, using the REGOUT command. Step 4 is repeated until the desired amount of code memory is read.

TABLE 5-9: SERIAL INSTRUCTION EXECUTION FOR READING CODE MEMORY

Command (Binary)	Data (Hex)	Description
Step 1: Exit Reset vector.		
0000	000000	NOP
0000	000000	NOP
0000	040200	GOTO 0x200
0000	000000	NOP
Step 2: Initialize TBLPAG and the read pointer (W6) for TBLRD instruction.		
0000	200xx0	MOV #<SourceAddress23:16>, W0
0000	880190	MOV W0, TBLPAG
0000	2xxxx6	MOV #<SourceAddress15:0>, W6
Step 3: Initialize the write pointer (W7) to point to the VISI register.		
0000	207847	MOV #VISI, W7
0000	000000	NOP
Step 4: Read and clock out the contents of the next two locations of code memory, through the VISI register, using the REGOUT command.		
0000	BA1B96	TBLRDL [W6], [W7]
0000	000000	NOP
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register
0000	BA9BB6	TBLRDH [W6++], [W7]
0000	000000	NOP
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register
Step 5: Repeat step 4 until all desired code memory is read.		
Step 6: Reset device internal PC.		
0000	040200	GOTO 0x200
0000	000000	NOP

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5.9 Reading Configuration Memory

The procedure for reading configuration memory is similar to the procedure for reading code memory, except that 16-bit data words are read (with the upper byte read being all '0's) instead of 24-bit words. Since there are twelve Configuration registers, they are read one register at a time.

Table 5-10 shows the ICSP programming details for reading all of configuration memory. Note that the TBLPAG register is hard coded to 0xF8 (the upper byte address of configuration memory) and the read pointer, W6, is initialized to 0x0000.

TABLE 5-10: SERIAL INSTRUCTION EXECUTION FOR READING ALL CONFIGURATION MEMORY

Command (Binary)	Data (Hex)	Description
Step 1: Exit Reset vector.		
0000	000000	NOP
0000	000000	NOP
0000	040200	GOTO 0x200
0000	000000	NOP
Step 2: Initialize TBLPAG, the read pointer (W6) and the write pointer (W7) for TBLRD instruction.		
0000	200F80	MOV #0xF8, W0
0000	880190	MOV W0, TBLPAG
0000	EB0300	CLR W6
0000	207847	MOV #VISI, W7
0000	000000	NOP
Step 3: Read the Configuration register and write it to the VISI register (located at 0x784) and clock out the VISI register using the REGOUT command.		
0000	BA0BB6	TBLRDL [W6++], [W7]
0000	000000	NOP
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register
Step 4: Repeat step 3 twelve times to read all the Configuration registers.		
Step 5: Reset device internal PC.		
0000	040200	GOTO 0x200
0000	000000	NOP

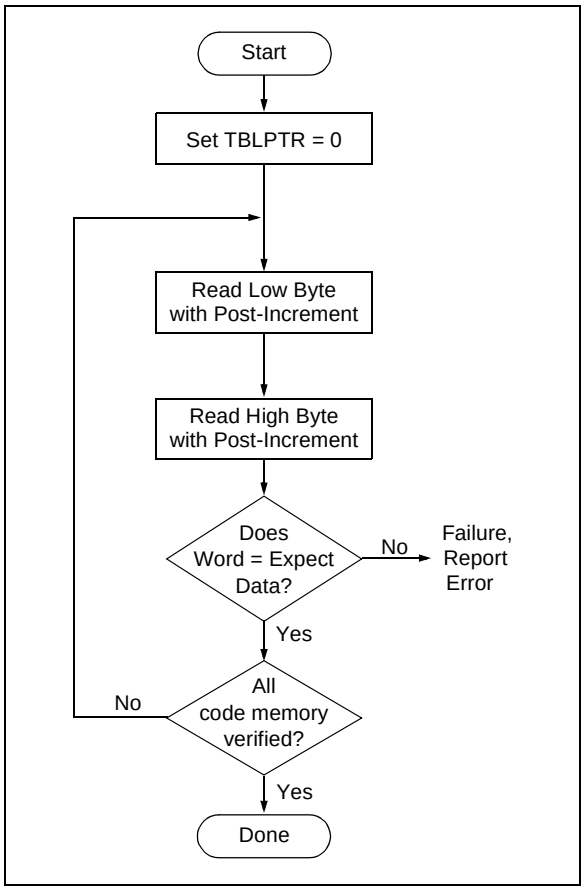
5.10 Verify Code Memory and Configuration Word

The verify step involves reading back the code memory space and comparing it against the copy held in the programmer's buffer. The Configuration registers are verified with the rest of the code.

The verify process is shown in the flowchart in Figure 5-8. Memory reads occur a single byte at a time, so two bytes must be read to compare against the word in the programmer's buffer. Refer to **Section 5.8 "Reading Code Memory"** for implementation details of reading code memory.

Note:	Because the Configuration registers include the device code protection bit, code memory should be verified immediately after writing if code protection is enabled. This is because the device will not be readable or verifiable if a device Reset occurs after the code-protect bit in the FGS Configuration register has been cleared.

FIGURE 5-8: VERIFY CODE MEMORY FLOW



5.11 Reading the Application ID Word

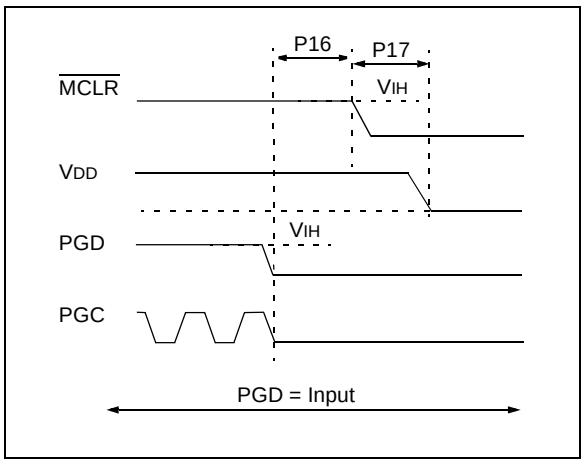
The Application ID Word is stored at address 0x8007F0 in executive code memory. To read this memory location, you must use the SIX control code to move this program memory location to the VISI register. Then, the REGOUT control code must be used to clock the contents of the VISI register out of the device. The corresponding control and instruction codes that must be serially transmitted to the device to perform this operation are shown in Table 5-11.

After the programmer has clocked out the Application ID Word, it must be inspected. If the application ID has the value 0xBB, the programming executive is resident in memory and the device can be programmed using the mechanism described in **Section 3.0 "Device Programming – Enhanced ICSP"**. However, if the application ID has any other value, the programming executive is not resident in memory; it must be loaded to memory before the device can be programmed. The procedure for loading the programming executive to memory is described in **Section 6.0 "Programming the Programming Executive to Memory"**.

5.12 Exiting ICSP Mode

Exiting Program/Verify mode is done by removing V_{IH} from $\overline{MCLR}$, as shown in Figure 5-9. The only requirement for exit is that an interval P16 should elapse between the last clock and program signals on PGC and PGD before removing V_{IH}.

FIGURE 5-9: EXITING ICSP™ MODE



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TABLE 5-11: SERIAL INSTRUCTION EXECUTION FOR READING THE APPLICATION ID WORD

Command (Binary)	Data (Hex)	Description
Step 1: Exit Reset vector.		
0000	000000	NOP
0000	000000	NOP
0000	040200	GOTO 0x200
0000	000000	NOP
Step 2: Initialize TBLPAG and the read pointer (W0) for TBLRD instruction.		
0000	200800	MOV #0x80, W0
0000	880190	MOV W0, TBLPAG
0000	205FE0	MOV #0x5BE, W0
0000	207841	MOV #VISI, W1
0000	000000	NOP
0000	BA0890	TBLRDL [W0], [W1]
0000	000000	NOP
0000	000000	NOP
Step 3: Output the VISI register using the REGOUT command.		
0001	<VISI>	Clock out contents of the VISI register

6.0 PROGRAMMING THE PROGRAMMING EXECUTIVE TO MEMORY

6.1 Overview

If it is determined that the programming executive is not present in executive memory (as described in **Section 3.2 "Confirming the Presence of the Programming Executive"**), it must be programmed into executive memory using ICSP, as described in **Section 5.0 "Device Programming – ICSP"**.

Storing the programming executive to executive memory is similar to normal programming of code memory. Namely, the executive memory must first be erased, and then the programming executive must be programmed 64 words at a time. This control flow is summarized in Table 6-1.

TABLE 6-1: PROGRAMMING THE PROGRAMMING EXECUTIVE

Command (Binary)	Data (Hex)	Description
Step 1: Exit Reset vector and erase executive memory.		
0000	000000	NOP
0000	000000	NOP
0000	040200	GOTO 0x200
0000	000000	NOP
Step 2: Initialize the NVMCON to erase a page of executive memory.		
0000	24072A	MOV #0x4042, W10
0000	883B0A	MOV W10, NVMCON
Step 3: Initiate the erase cycle, wait for erase to complete and make sure WR bit is clear.		
0000	A8E761	BSET NVMCON, #15
0000	000000	NOP
0000	000000	NOP
-	-	Externally time 'P12' msec (see **Section TABLE 8-1: "AC/DC Characteristics and Timing Requirements"**) to allow sufficient time for the Page Erase operation to complete.
0000	807600	MOV NVMCON, W0
0000	887840	MOV W0, VISI
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register. Repeat until the WR bit is clear.
Step 4: Repeat Step 3 four times to erase all four pages of executive memory.		
Step 5: Initialize the NVMCON to program 64 instruction words.		
0000	24001A	MOV #0x4001, W10
0000	883B0A	MOV W10, NVMCON
Step 6: Initialize TBLPAG and the write pointer (W7).		
0000	200800	MOV #0x80, W0
0000	880190	MOV W0, TBLPAG
0000	EB0380	CLR W7
0000	000000	NOP

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TABLE 6-1: PROGRAMMING THE PROGRAMMING EXECUTIVE (CONTINUED)

Command (Binary)	Data (Hex)	Description
Step 7: Load W0:W5 with the next 4 words of packed programming executive code and initialize W6 for programming. Programming starts from the base of executive memory (0x800000) using W6 as a read pointer and W7 as a write pointer.		
0000	2<LSW0>0	MOV #<LSW0>, W0
0000	2<MSB1:MSB0>1	MOV #<MSB1:MSB0>, W1
0000	2<LSW1>2	MOV #<LSW1>, W2
0000	2<LSW2>3	MOV #<LSW2>, W3
0000	2<MSB3:MSB2>4	MOV #<MSB3:MSB2>, W4
0000	2<LSW3>5	MOV #<LSW3>, W5
Step 8: Set the read pointer (W6) and load the (next four write) latches.		
0000	EB0300	CLR W6
0000	000000	NOP
0000	BB0BB6	TBLWTL [W6++], [W7]
0000	000000	NOP
0000	000000	NOP
0000	BBDBB6	TBLWTH.B[W6++], [W7++]
0000	000000	NOP
0000	000000	NOP
0000	BBEBB6	TBLWTH.B[W6++], [++W7]
0000	000000	NOP
0000	000000	NOP
0000	BB1BB6	TBLWTL [W6++], [W7++]
0000	000000	NOP
0000	000000	NOP
0000	BB0BB6	TBLWTL [W6++], [W7]
0000	000000	NOP
0000	000000	NOP
0000	BBDBB6	TBLWTH.B[W6++], [W7++]
0000	000000	NOP
0000	000000	NOP
0000	BBEBB6	TBLWTH.B[W6++], [++W7]
0000	000000	NOP
0000	000000	NOP
0000	BB1BB6	TBLWTL [W6++], [W7++]
0000	000000	NOP
0000	000000	NOP
Step 9: Repeat Steps 7-8 sixteen times to load the write latches for the 64 instructions.		
Step 10: Initiate the programming cycle.		
0000	A8E761	BSET NVMCON, #15
0000	000000	NOP
0000	000000	NOP
Step 11: Wait for the Row Program operation to complete.		
-	-	Externally time 'P13' msec (see **Section TABLE 8-1: "AC/DC Characteristics and Timing Requirements"**) to allow sufficient time for the Page Erase operation to complete.
0000	807600	MOV NVMCON, W0
0000	887840	MOV W0, VISI
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register. Repeat until the WR bit is clear.

TABLE 6-1: PROGRAMMING THE PROGRAMMING EXECUTIVE (CONTINUED)

Command (Binary)	Data (Hex)	Description
Step 12: Reset the device internal PC.		
0000	040200	GOTO 0x200
0000	000000	NOP
Step 13: Repeat Steps 7-12 until all 32 rows of executive memory have been programmed.		

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6.2 Programming Verification

After the programming executive has been programmed to executive memory using ICSP, it must be verified. Verification is performed by reading out the contents of executive memory and comparing it with the image of the programming executive stored in the programmer.

Reading the contents of executive memory can be performed using the same technique described in **Section 5.8 "Reading Code Memory"**. A procedure for reading executive memory is shown in Table 6-2. Note that in Step 2, the TBLPAG register is set to 0x80, such that executive memory may be read.

TABLE 6-2: READING EXECUTIVE MEMORY

Command (Binary)	Data (Hex)	Description
Step 1: Exit the Reset vector.		
0000	000000	NOP
0000	000000	NOP
0000	040200	GOTO 0x200
0000	000000	NOP
Step 2: Initialize TBLPAG and the read pointer (W6) for TBLRD instruction.		
0000	200800	MOV #0x80, W0
0000	880190	MOV W0, TBLPAG
0000	EB0300	CLR W6
Step 3: Initialize the write pointer (W7) to point to the VISI register.		
0000	207847	MOV #VISI, W7
Step 4: Read and clock out the contents of the next two locations of executive memory through the VISI register using the REGOUT command.		
0000	000000	NOP
0000	BA1B96	TBLRDL [W6], [W7]
0000	000000	NOP
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register
0000	BA9BB6	TBLRDH [W6++], [W7]
0000	000000	NOP
0000	000000	NOP
0001	<VISI>	Clock out contents of VISI register
Step 5: Reset the device internal PC.		
0000	040200	GOTO 0x200
0000	000000	NOP
Step 6: Repeat Steps 4-5 until all 2048 instruction words of executive memory are read.		

7.0 DEVICE ID

The device ID region of memory can be used to determine mask, variant and manufacturing information about the chip. The device ID region is 2 x 16-bits and it can be read using the READC command. This region of memory is read-only and can also be read when code protection is enabled.

Table 7-1 shows the device ID for each device, Table 7-2 shows the Device ID registers and Table 7-3 describes the bit field of each register.

TABLE 7-1: DEVICE IDs

Device	DEVID	DEVREV
dsPIC33FJ64GP206	0xC1	0x3000
dsPIC33FJ64GP306	0xCD	0x3000
dsPIC33FJ64GP310	0xCF	0x3000
dsPIC33FJ64GP706	0xD5	0x3000
dsPIC33FJ64GP708	0xD6	0x3000
dsPIC33FJ64GP710	0xD7	0x3000
dsPIC33FJ128GP206	0xD9	0x3000
dsPIC33FJ128GP306	0xE5	0x3000
dsPIC33FJ128GP310	0xE7	0x3000
dsPIC33FJ128GP706	0xED	0x3000
dsPIC33FJ128GP708	0xEE	0x3000
dsPIC33FJ128GP710	0xEF	0x3000
dsPIC33FJ256GP506	0xF5	0x3000
dsPIC33FJ256GP510	0xF7	0x3000
dsPIC33FJ256GP710	0xFF	0x3000
dsPIC33FJ64MC506	0x89	0x3000
dsPIC33FJ64MC508	0x8A	0x3000
dsPIC33FJ64MC510	0x8B	0x3000
dsPIC33FJ64MC706	0x91	0x3000
dsPIC33FJ64MC710	0x97	0x3000
dsPIC33FJ128MC506	0xA1	0x3000
dsPIC33FJ128MC510	0xA3	0x3000
dsPIC33FJ128MC706	0xA9	0x3000
dsPIC33FJ128MC708	0xAE	0x3000
dsPIC33FJ128MC710	0xAF	0x3000
dsPIC33FJ256MC510	0xB7	0x3000
dsPIC33FJ256MC710	0xBF	0x3000
PIC24HJ64GP206	0x41	0x3000
PIC24HJ64GP210	0x47	0x3000
PIC24HJ64GP506	0x49	0x3000
PIC24HJ64GP510	0x4B	0x3000
PIC24HJ128GP206	0x5D	0x3000
PIC24HJ128GP210	0x5F	0x3000
PIC24HJ128GP306	0x65	0x3000
PIC24HJ128GP310	0x67	0x3000
PIC24HJ128GP506	0x61	0x3000
PIC24HJ128GP510	0x63	0x3000
PIC24HJ256GP206	0x71	0x3000
PIC24HJ256GP210	0x73	0x3000
PIC24HJ256GP610	0x7B	0x3000
dsPIC33FJ12GP201	0x802	0x3000
dsPIC33FJ12GP202	0x803	0x3000
dsPIC33FJ12MC201	0x800	0x3000
dsPIC33FJ12MC202	0x801	0x3000
PIC24HJ12GP201	0x80A	0x3000
PIC24HJ12GP202	0x80B	0x3000

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TABLE 7-2: dsPIC33F/PIC24H PROGRAMMING SPECIFICATION DEVICE ID REGISTERS

Address	Name	Bit															
		15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0xFF0000	DEVID	MASK<9:0>										VARIANT<5:0>					
0xFF0002	DEVREV	PROC<3:0>				REV<5:0>						DOT<5:0>					

TABLE 7-3: DEVICE ID BITS DESCRIPTION

Bit Field	Register	Description
MASK<9:0>	DEVID	Encodes the MASKSET ID of the device.
VARIANT<5:0>	DEVID	Encodes the VARIANT derived from MASKSET of the device.
PROC<3:0>	DEVREV	Encodes the process of the device.
REV<5:0>	DEVREV	Encodes the major revision number of the device.
DOT<5:0>	DEVREV	Encodes the minor revision number of the device.

8.0 AC/DC CHARACTERISTICS AND TIMING REQUIREMENTS

Table 8-1 lists AC/DC characteristics and timing requirements.

TABLE 8-1: AC/DC CHARACTERISTICS AND TIMING REQUIREMENTS

Standard Operating Conditions Operating Temperature: –40°C-85°C. Programming at 25°C is recommended.						
Param No.	**Symbol**	**Characteristic**	**Min**	**Max**	**Units**	**Conditions**
D111	VDD	Supply Voltage During Programming	VDDCORE	3.60	V	Normal programming(1)
D112	IPP	Programming Current on $\overline{MCLR}$	—	5	µA	
D113	IDDP	Supply Current During Programming	—	2	mA	
D031	VIL	Input Low Voltage	VSS	0.2 VDD	V	
D041	VIH	Input High Voltage	0.8 VDD	VDD	V	
D080	VOL	Output Low Voltage	—	0.6	V	IOL = 8.5 mA @ 3.6V
D090	VOH	Output High Voltage	VDD – 0.7	—	V	IOH = -3.0 mA @ 3.6V
D012	CIO	Capacitive Loading on I/O pin (PGD)	—	50	pF	To meet AC specifications
D013	CF	Filter Capacitor Value on VCAP	1	10	µF	Required for controller core
P1	TPGC	Serial Clock (PGC) Period	136	—	ns	
P1A	TPGCL	Serial Clock (PGC) Low Time	40	—	ns	
P1B	TPGCH	Serial Clock (PGC) High Time	40	—	ns	
P2	TSET1	Input Data Setup Time to Serial Clock ↓	15	—	ns	
P3	THLD1	Input Data Hold Time from PGC ↓	15	—	ns	
P4	TDLY1	Delay between 4-bit Command and Command Operand	40	—	ns	
P4A	TDLY1A	Delay between Command Operand and Next 4-bit Command	40	—	ns	
P5	TDLY2	Delay between Last PGC ↓ of Command to First PGC ↑ of Read of Data Word	20	—	ns	
P6	TSET2	VDD ↑ Setup Time to $\overline{MCLR}$ ↑	100	—	ns	
P7	THLD2	Input Data Hold Time from $\overline{MCLR}$ ↑	25	—	ms	
P8	TDLY3	Delay between Last PGC ↓ of Command Byte to PGD ↑ by Programming Executive	12	—	µs	
P9a	TDLY4	Programming Executive Command Processing Time	10	—	µs	
P9b	TDLY5	Delay between PGD ↓ by Programming Executive to PGD Released by Programming Executive	15	23	µs	
P10	TDLY6	PGC Low Time After Programming	400	—	ns	
P11	TDLY7	Bulk Erase Time	200	—	ms	
P12	TDLY8	Page Erase Time	20	—	ms	
P13	TDLY9	Row Programming Time	1.5	—	ms	
P14	TR	$\overline{MCLR}$ Rise Time to Enter ICSP mode	—	1.0	µs	
P15	TVALID	Data Out Valid from PGC ↑	10	—	ns	
P16	TDLY10	Delay between Last PGC ↓ and $\overline{MCLR}$ ↓	0	—	s	
P17	THLD3	$\overline{MCLR}$ ↓ to VDD ↓	—	100	ns	

Note 1: VDD must also be supplied to the AVDD pins during programming. AVDD and AVSS should always be within ±0.3V of VDD and VSS, respectively.

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TABLE 8-1: AC/DC CHARACTERISTICS AND TIMING REQUIREMENTS (CONTINUED)

Standard Operating Conditions Operating Temperature: –40°C-85°C. Programming at 25°C is recommended.						
Param No.	**Symbol**	**Characteristic**	**Min**	**Max**	**Units**	**Conditions**
P18	TKEY1	Delay from First $\overline{\text{MCLR}}$ ↓ to First PGC ↑ for Key Sequence on PGD	40	—	ns	
P19	TKEY2	Delay from Last PGC ↓ for Key Sequence on PGD to Second $\overline{\text{MCLR}}$ ↑	25	—	ns	
P20	TDLY11	Maximum Wait Time for Configuration Register Programming		25	ms	

Note 1: VDD must also be supplied to the AVDD pins during programming. AVDD and AVSS should always be within ±0.3V of VDD and VSS, respectively.

NOTES:

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APPENDIX A: REVISION HISTORY

Revision C (June 2006)

- Added code protection Configuration register descriptions
- Added information about Unit ID
- Added `ERASES`, `ERASEG` and `ERASEC` programming executive commands
- Added checksum computation equation

Revision D (March 2007)

- Added information specific to the dsPIC33FJ12GP201/202, dsPIC33FJ12MC201/202 and PIC24HJ12GP201/202 devices in several sections, including pinout diagrams, program memory sizes and Device ID values
- Added specific checksum computations for all dsPIC33F and PIC24H devices
- Updated ICSP bulk/page erase and row/byte program code examples to show externally timed operation (waiting for specific delay periods)
- Added the P20 timing characteristic
- Updated timing characteristics and references to the timing characteristics
- Updated the ICSP code examples

NOTES:

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Netherlands - Drunen
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Fax: 31-416-690340

Spain - Madrid
Tel: 34-91-708-08-90
Fax: 34-91-708-08-91

UK - Wokingham
Tel: 44-118-921-5869
Fax: 44-118-921-5820

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